ITSPA0-33

Formative Assessment

EDUV12936584



SECURITY+ ASSIGNMENT

EDUV12936584

ABSTRACT

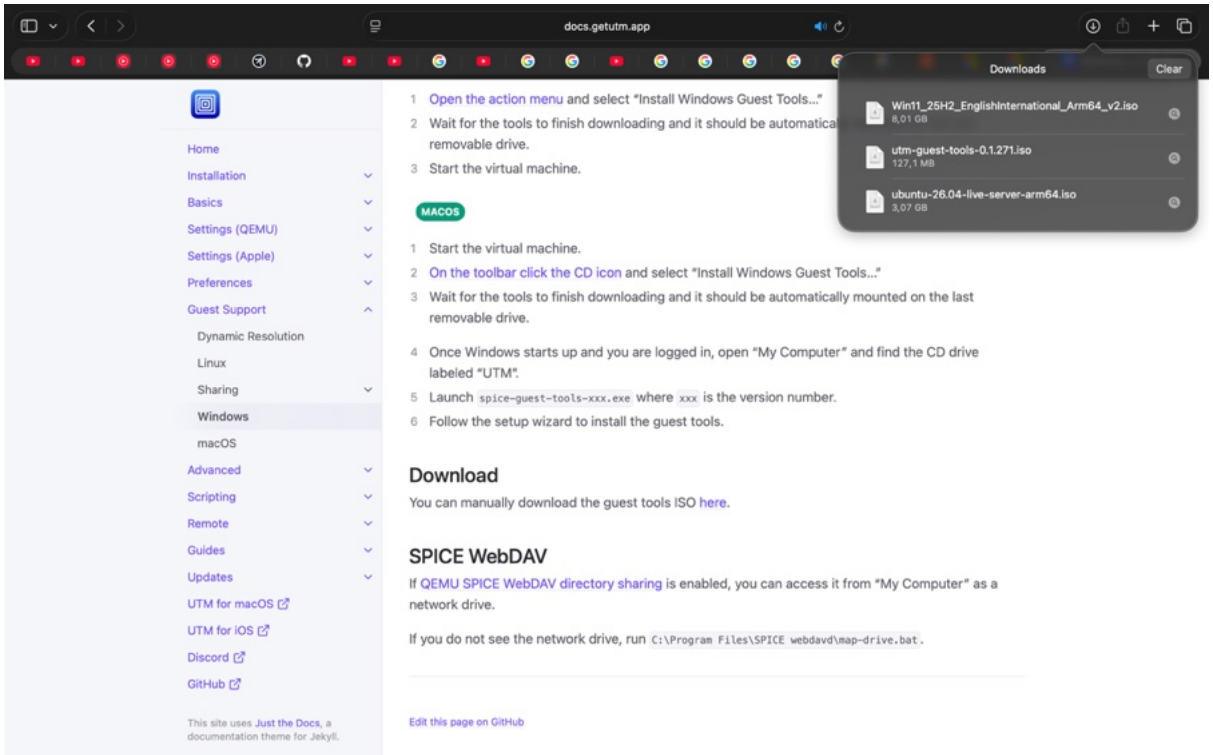
Improve security of workstations by applying hardening techniques, configuring user accounts and IAM controls, and analysing network traffic with Wireshark. Implement and document security controls in a virtual lab.

Branny Aurthur Branfield

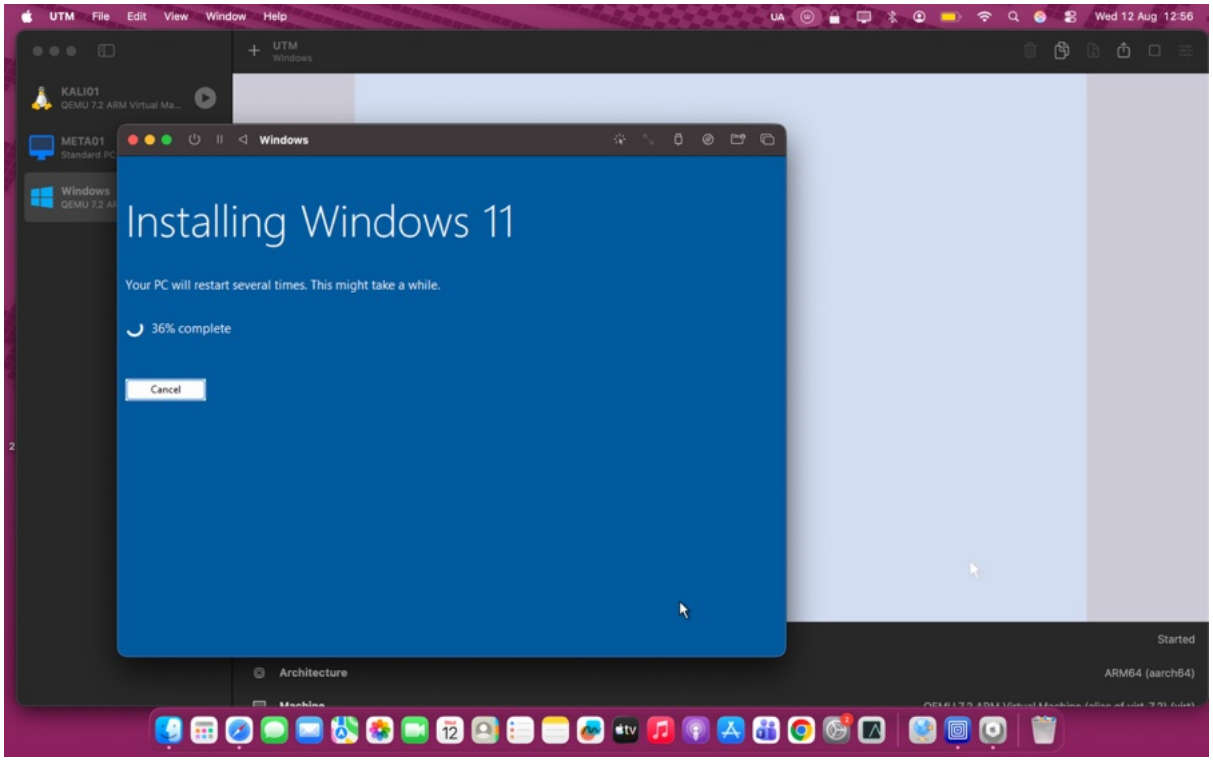
Securing System

NOTE: I'll be using a different Virtual Machine software called UTM instead of VirtualBox or VMware. Because those VM machines are not fully functional in Apples ARM64 silicon M-Chip.

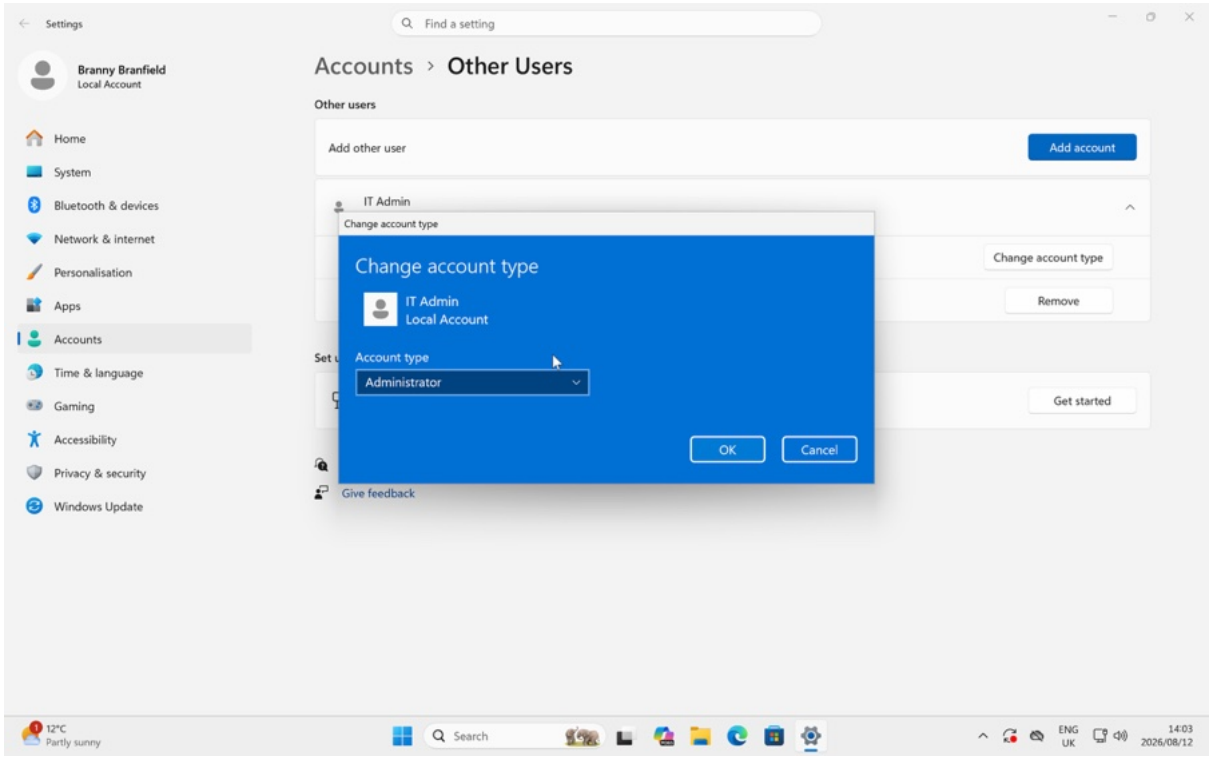
Question 1



Installing Windows 11 iso for ARM64



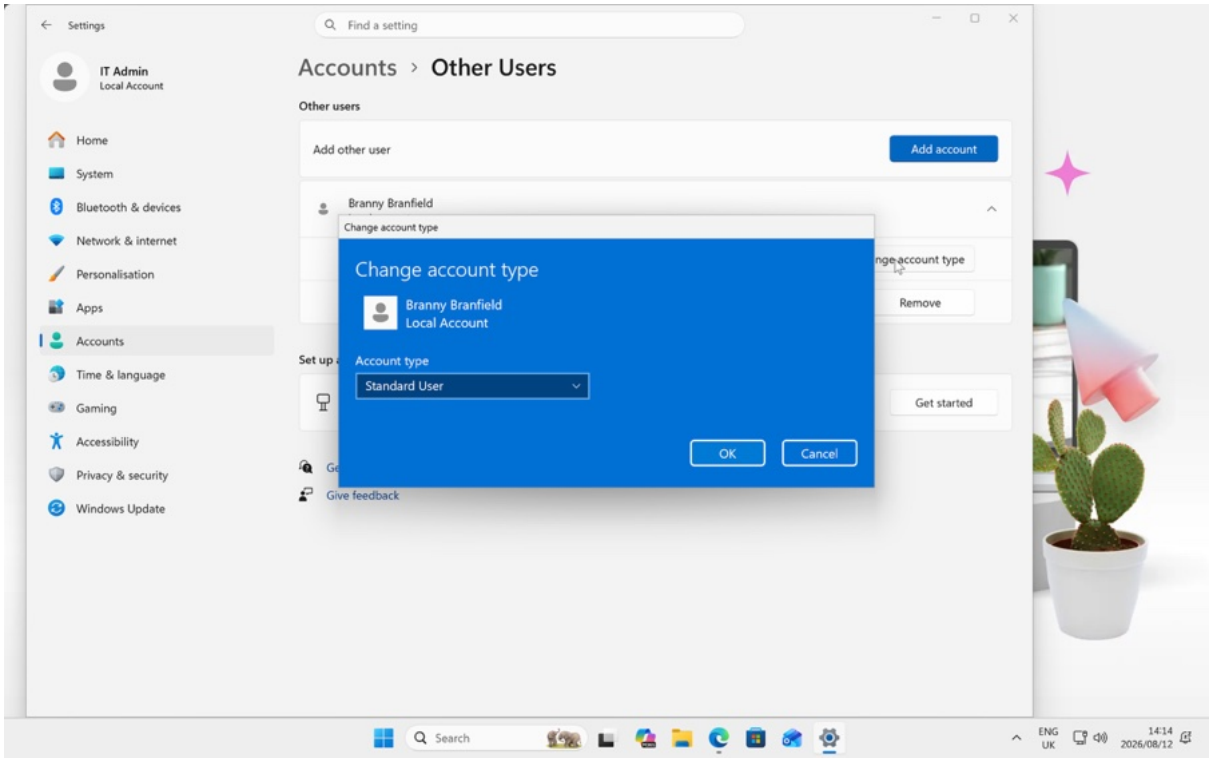
Set up and installation of Windows OS in UTM Virtual Machine with 2 core CPU , 64 GB Storage, 4 GB ram and Shared Network configuration.



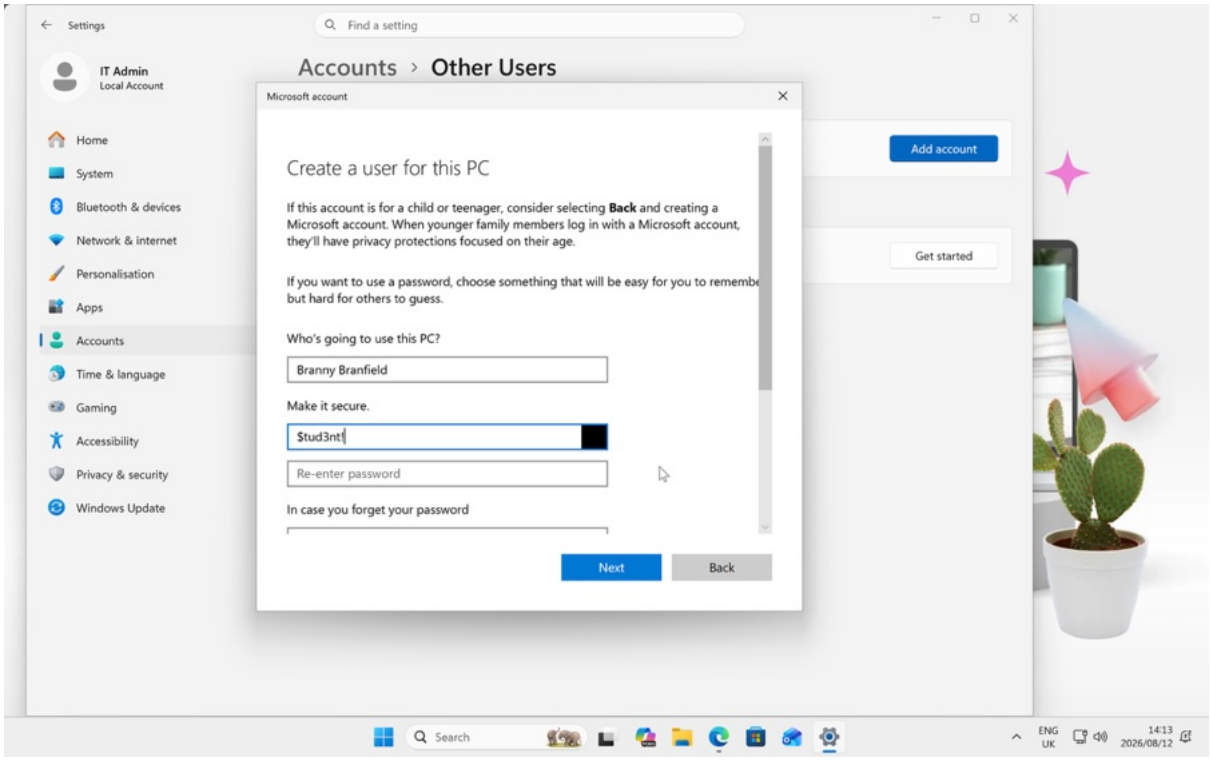
Creating an IT Admin account with Administrator privileges.



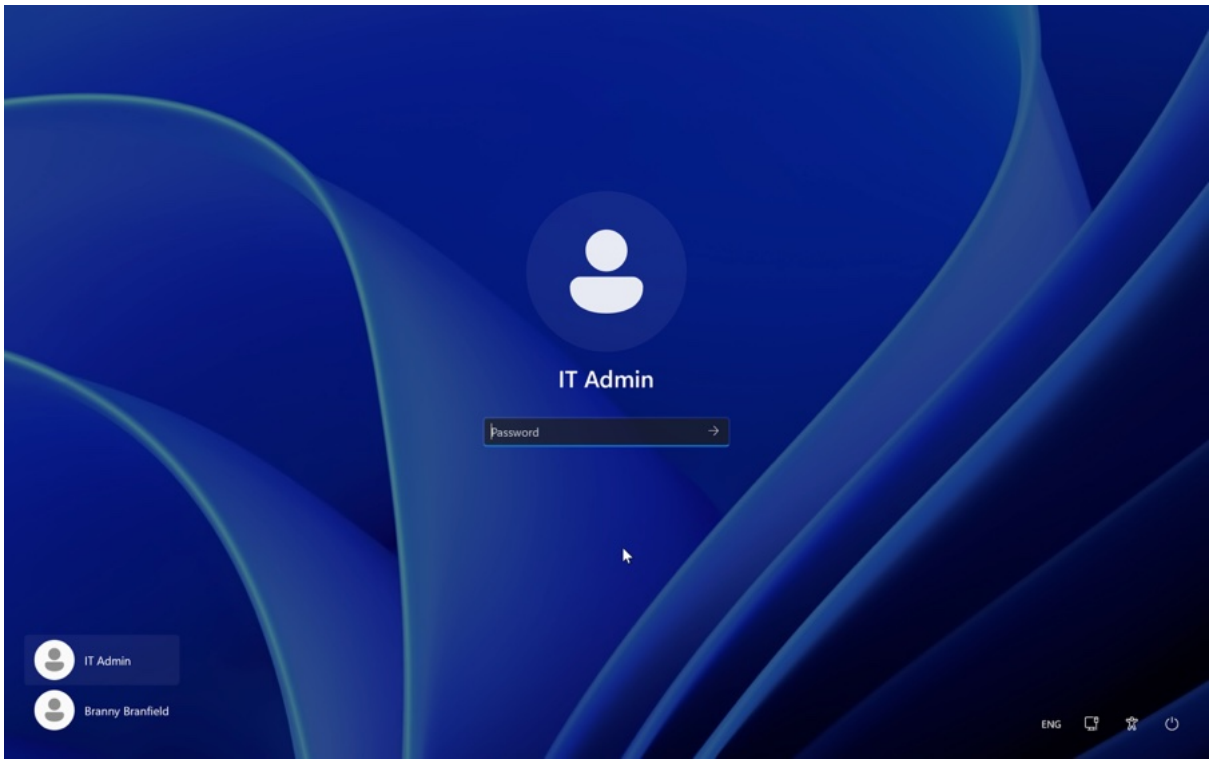
IT Admin account with the password : P@55w0rd



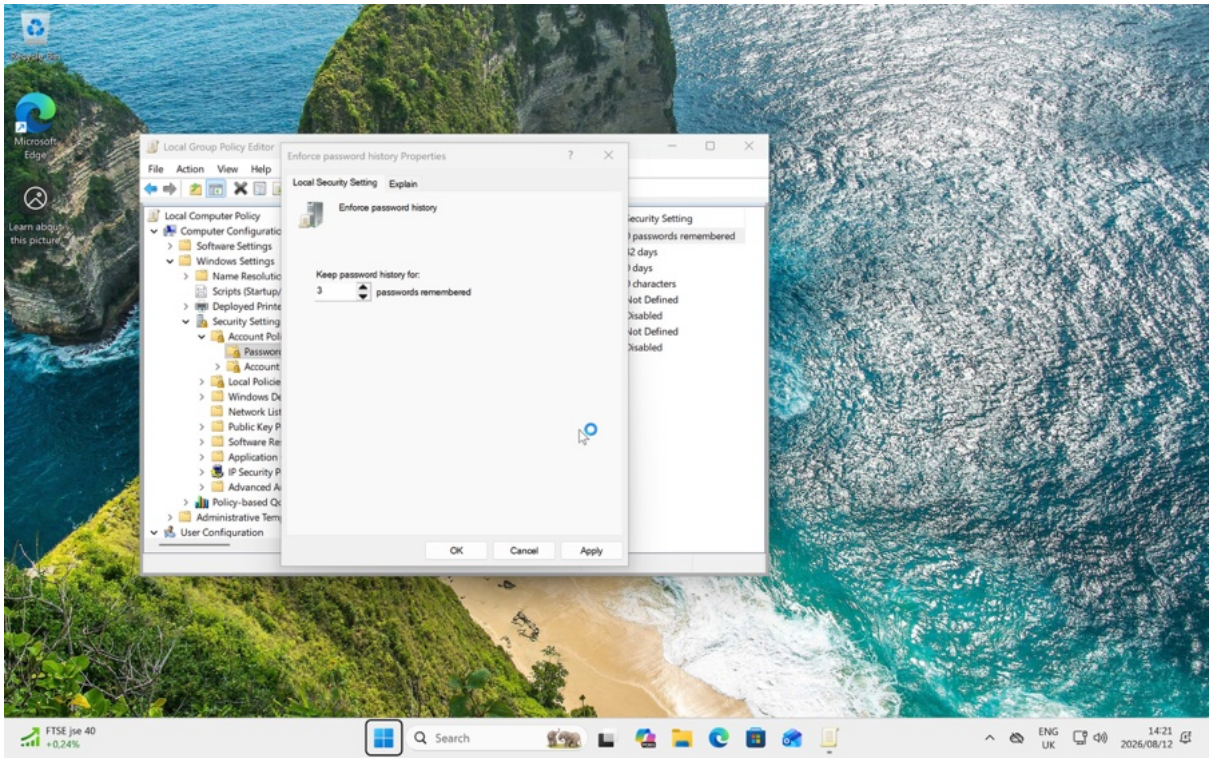
Creating a Standard user account with student name



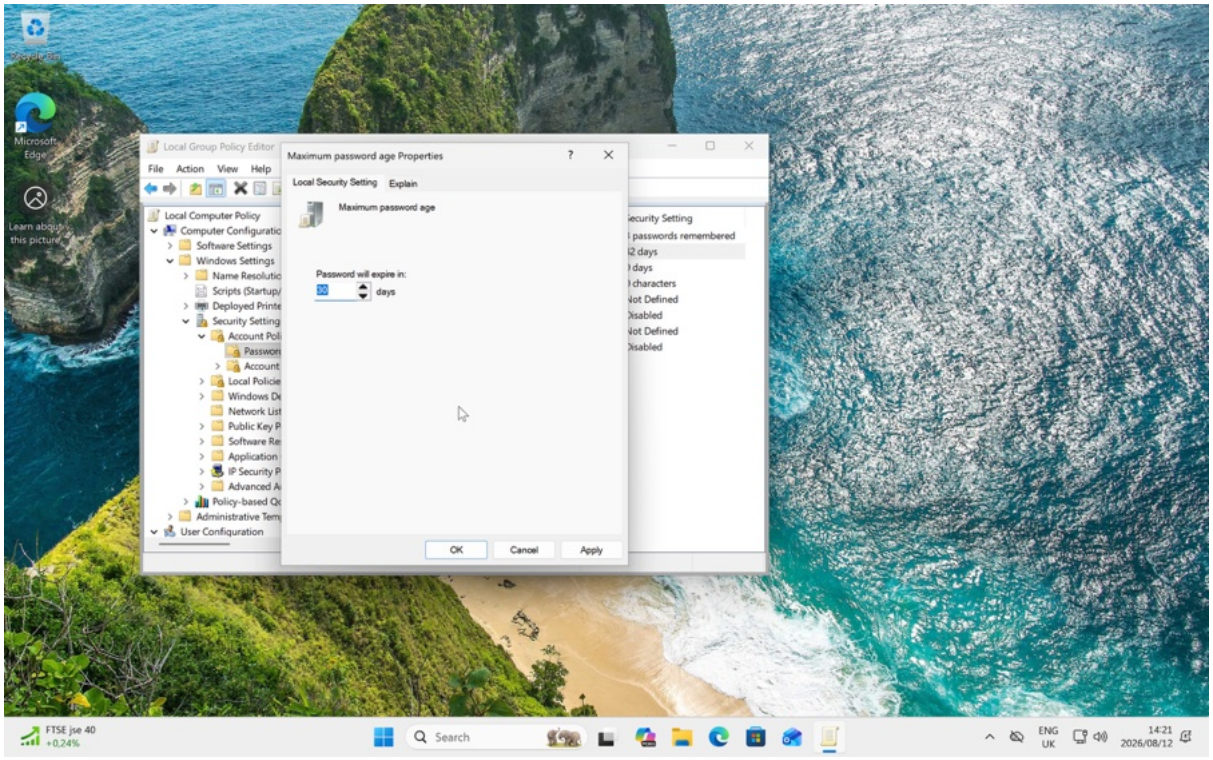
Student account with password : $tud3nt!



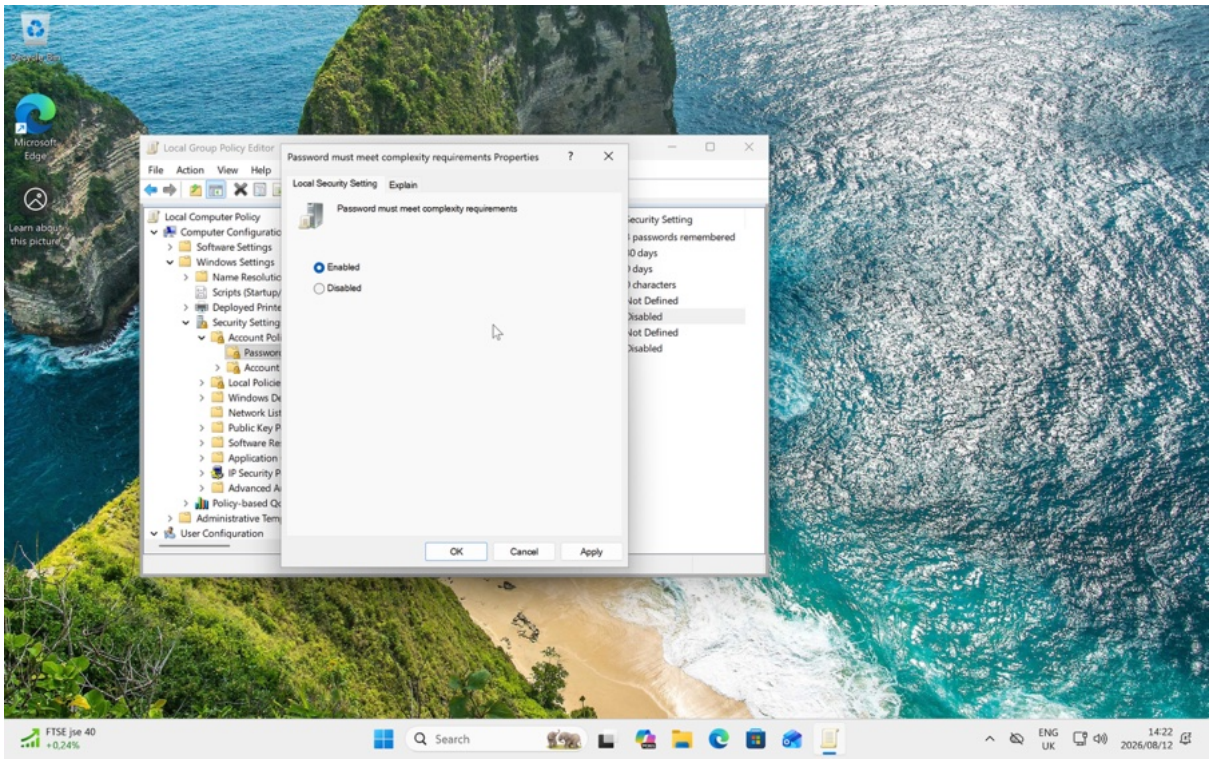
Two Account IT Admin and Student Account ready to use.



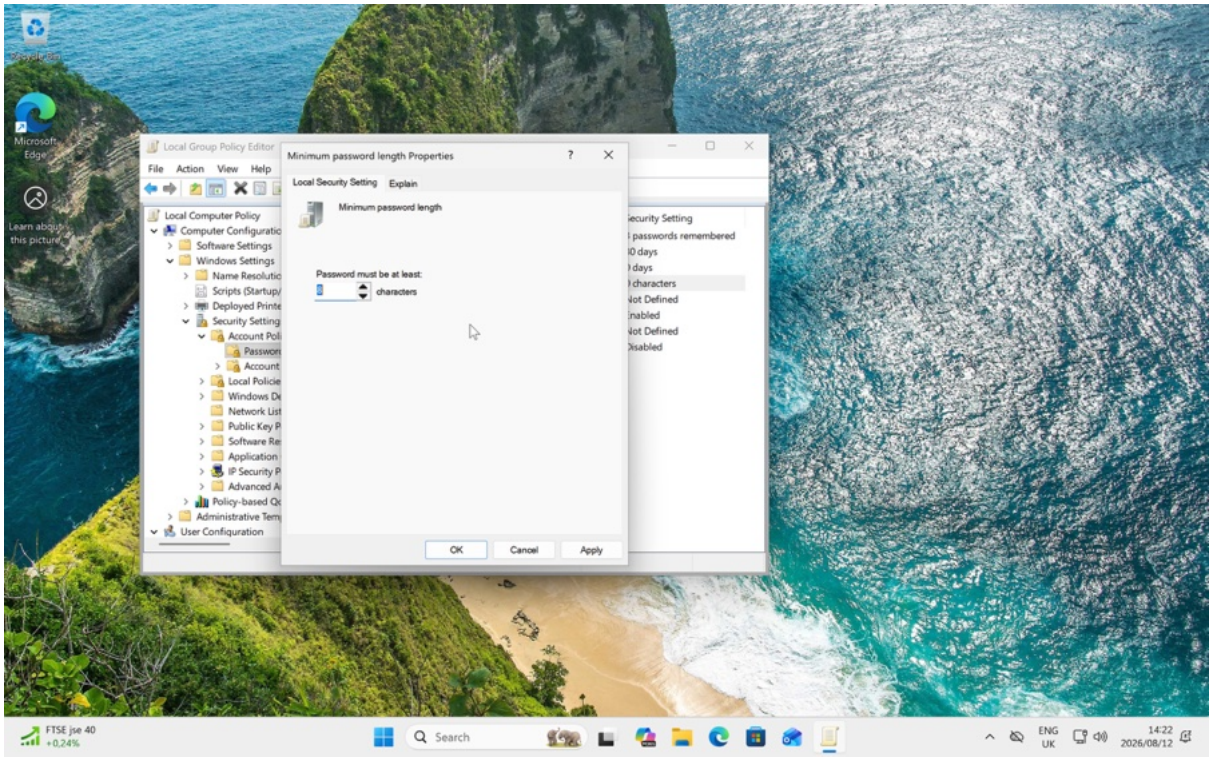
Enforcing password history of 3 last passwords



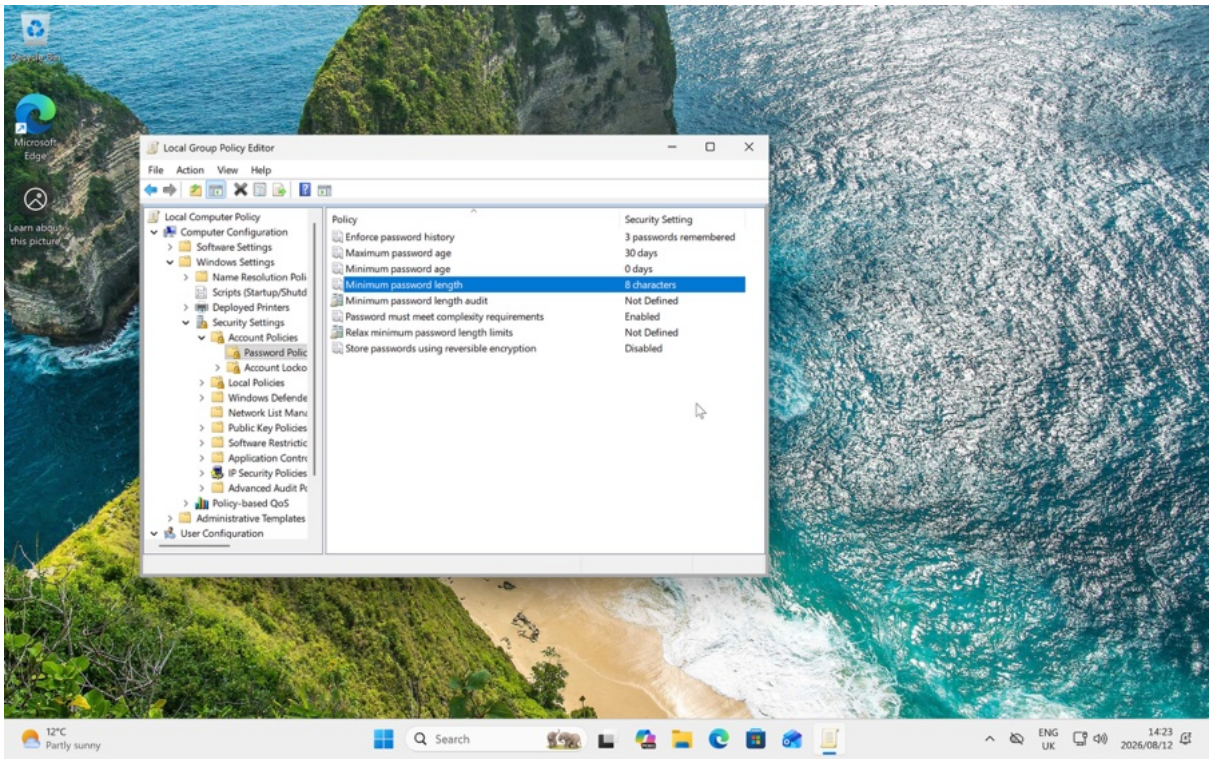
Maximising age of password to 30 days



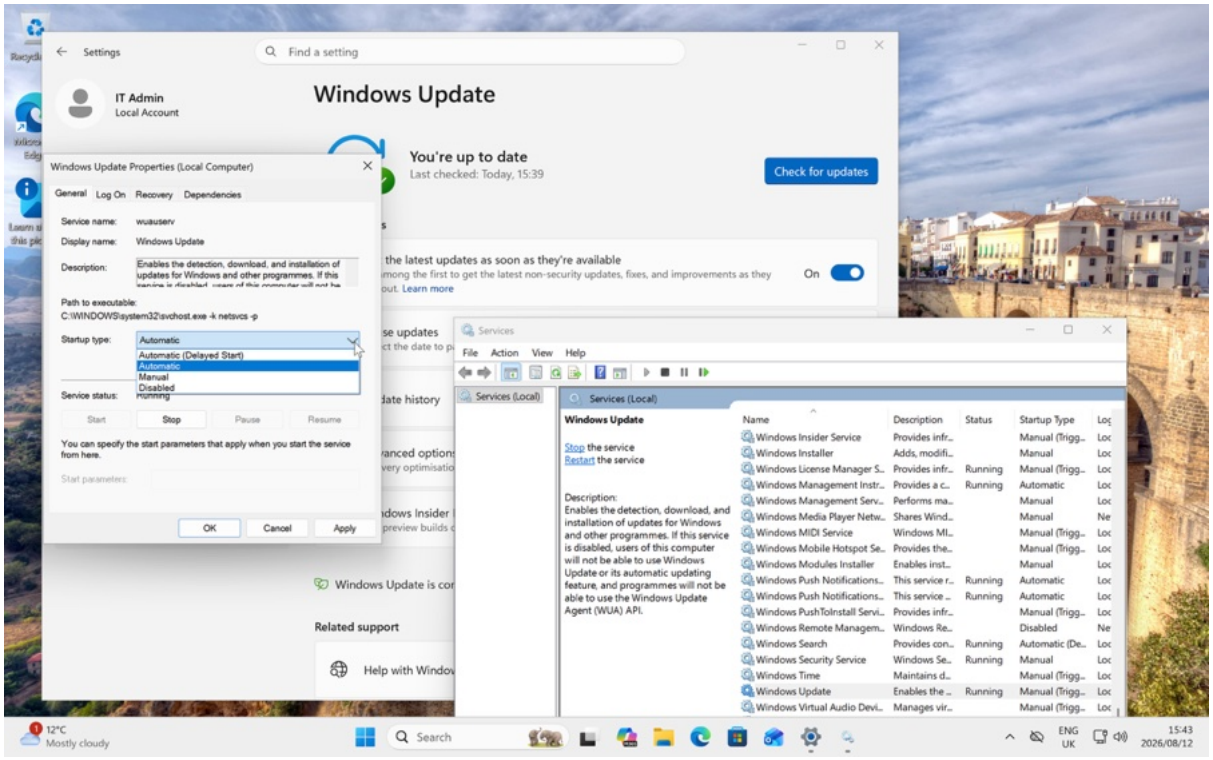
Enabling passwords to meet complexity requirements



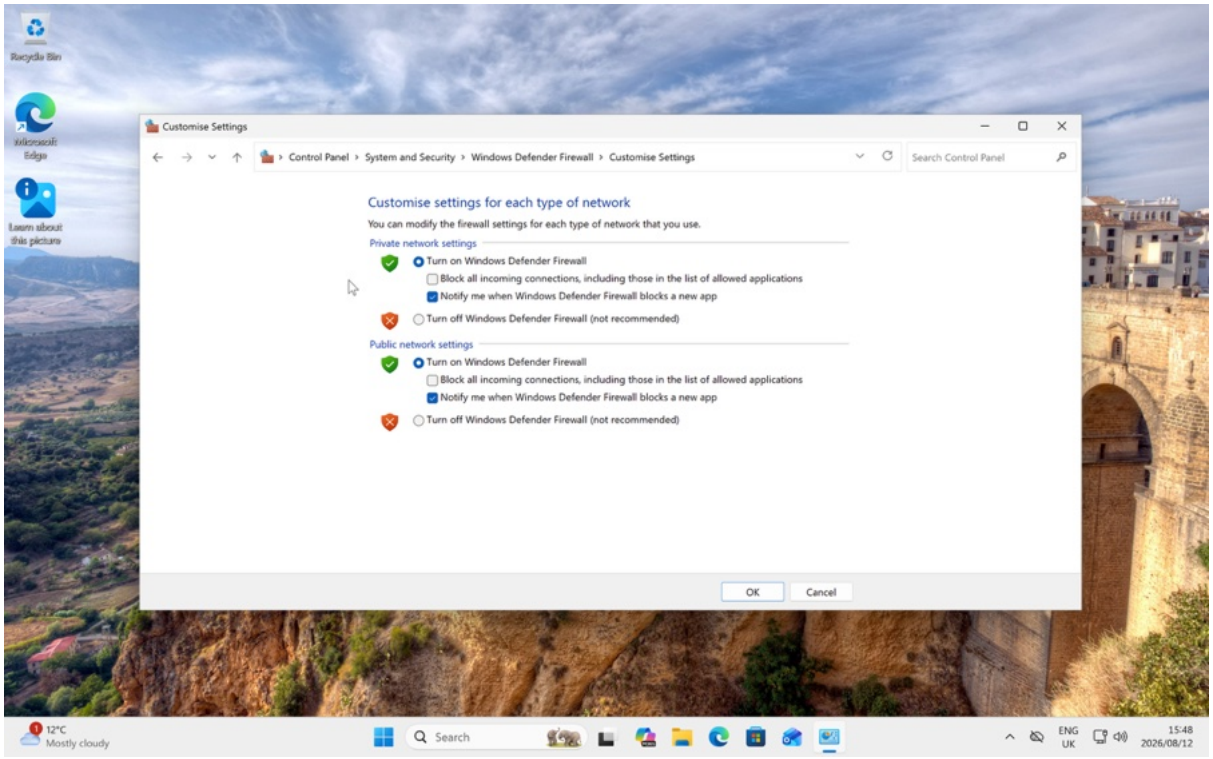
Set up minimum password length of 8 characters



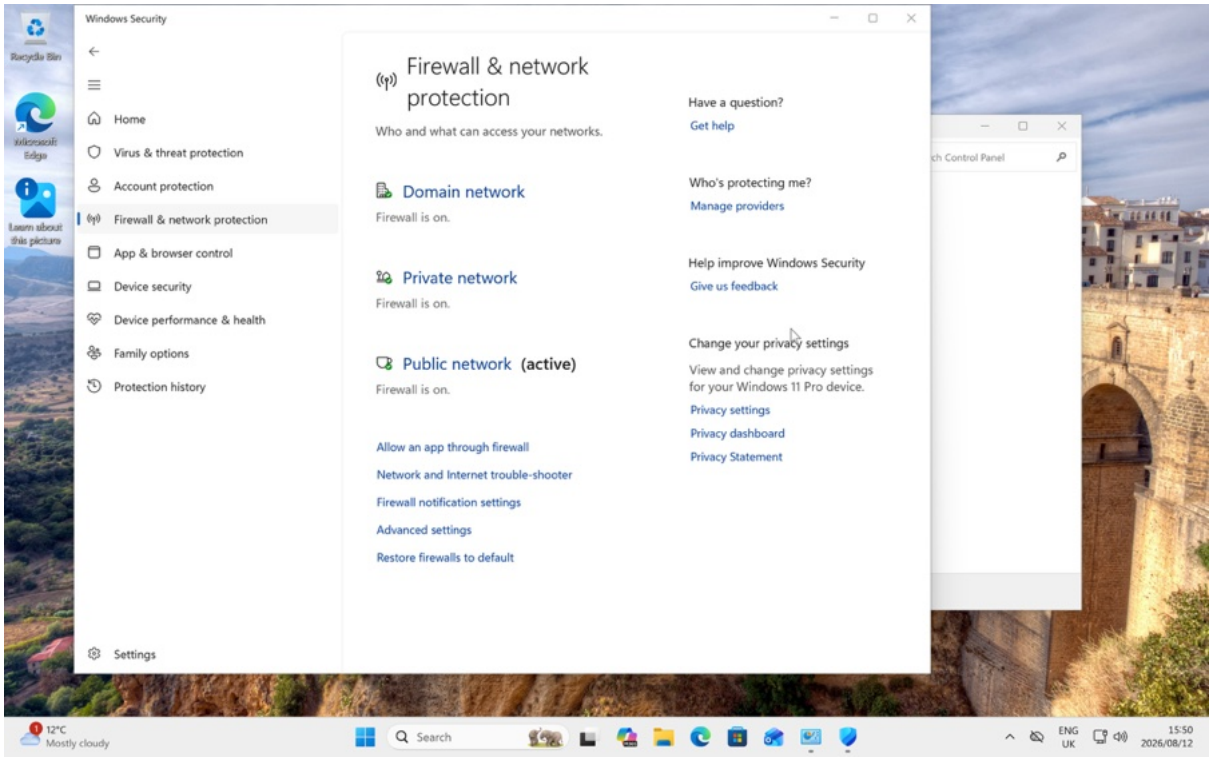
Confirming password policy is set accordingly.



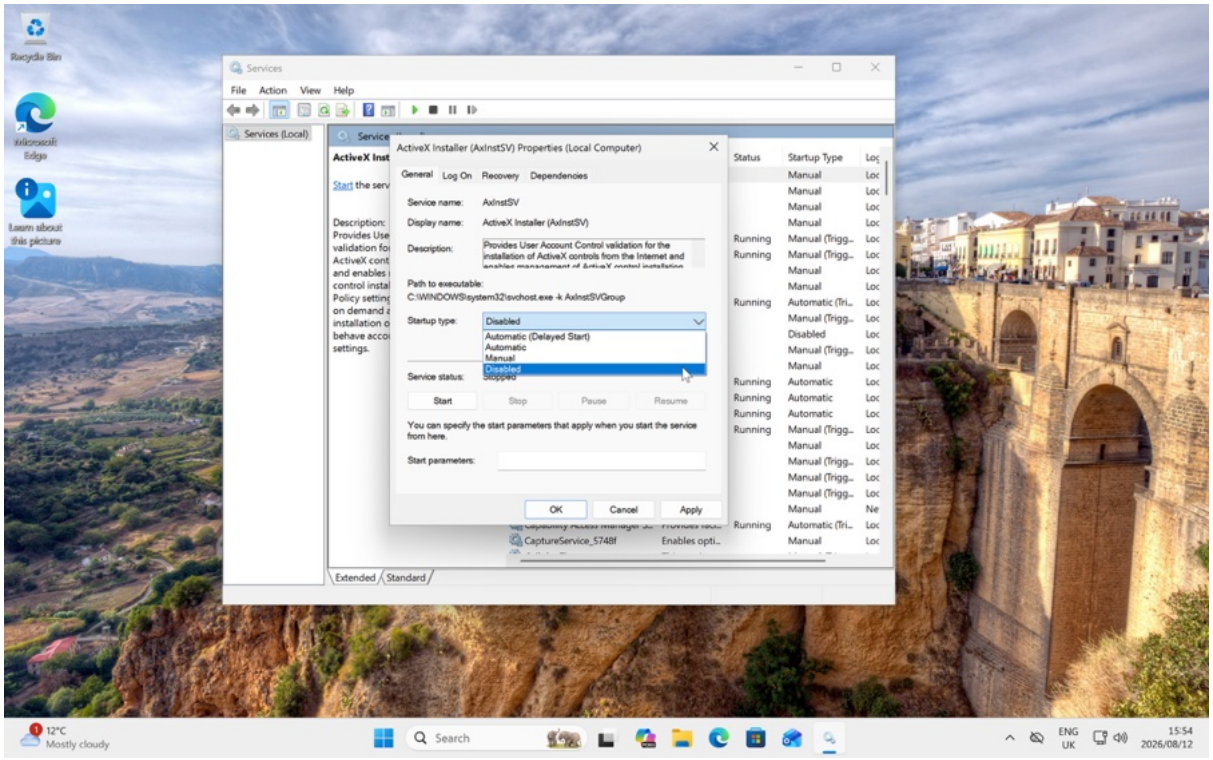
Insuring Up To Date Operation System and enabling Automatic OS updates as well as confirming OS update configuration to get the latest fixes and patches prevent vulnerabilities or exploits like zero-day.



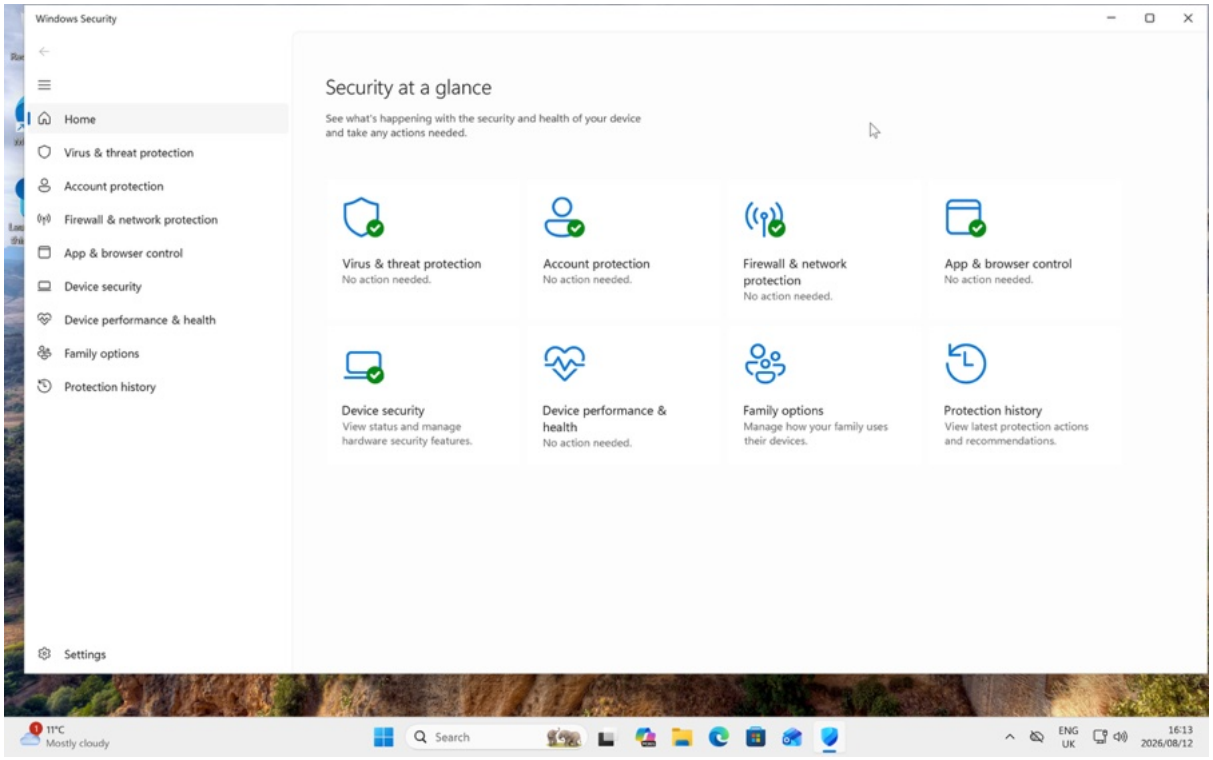
Enabling Host firewall on windows defender for both private and public network to protect the host in different layer environments e.g. layer 2 and 3.



Confirming that firewall configurations are active and enabled to insure the network remains secure.

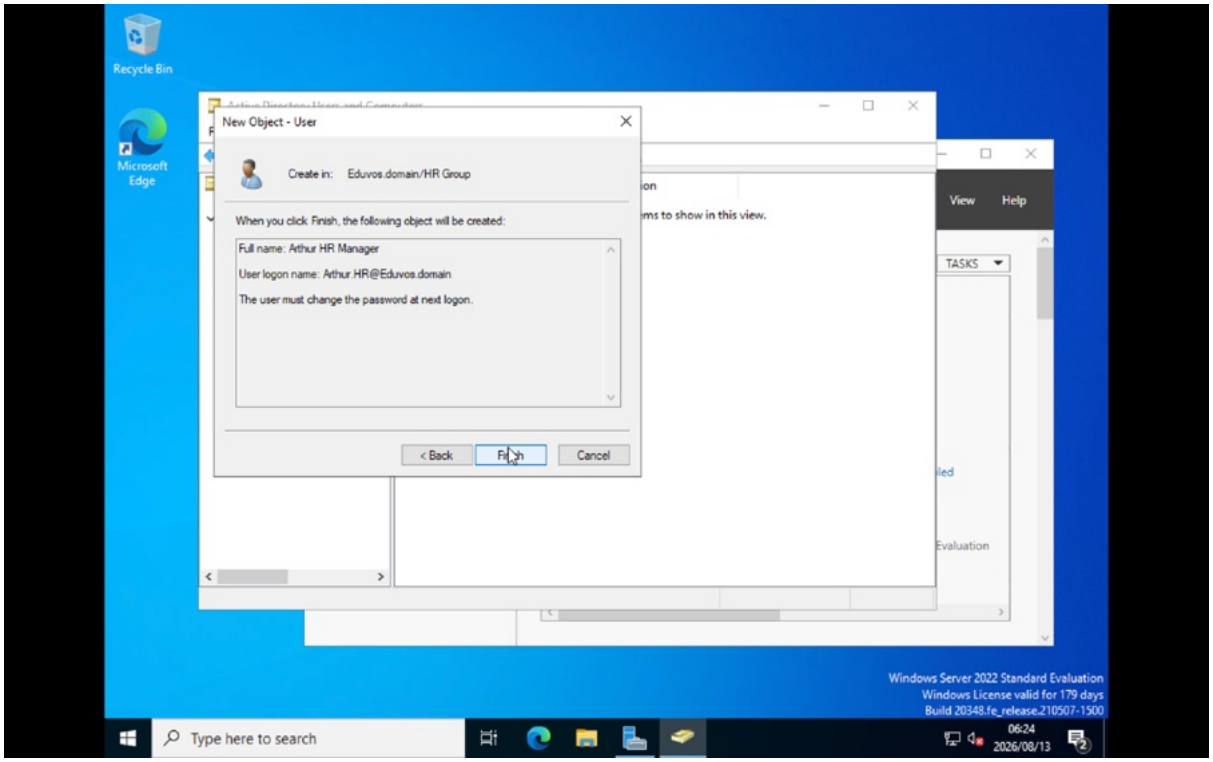


Disable an unnecessary service “Activex Installer” to free up system resources, stop data collection and prevent software conflicts.

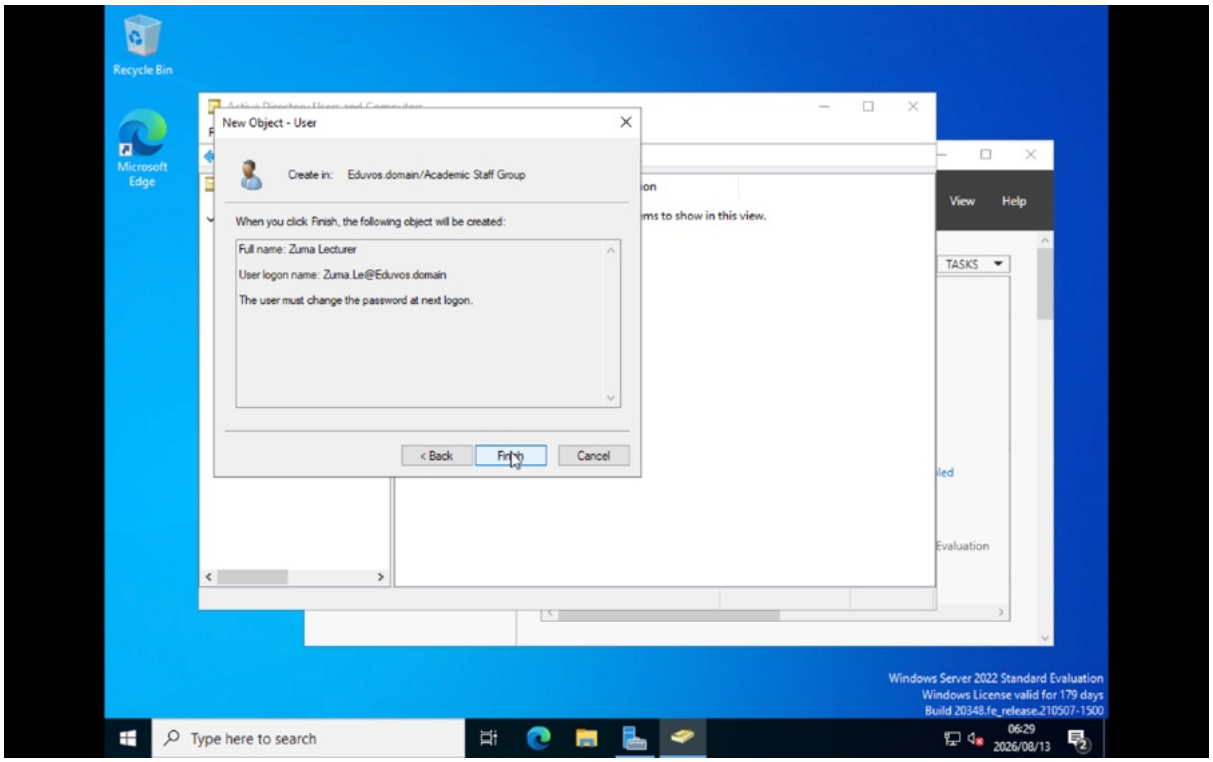


Verifying all security settings to insure the device remain safe and secure.

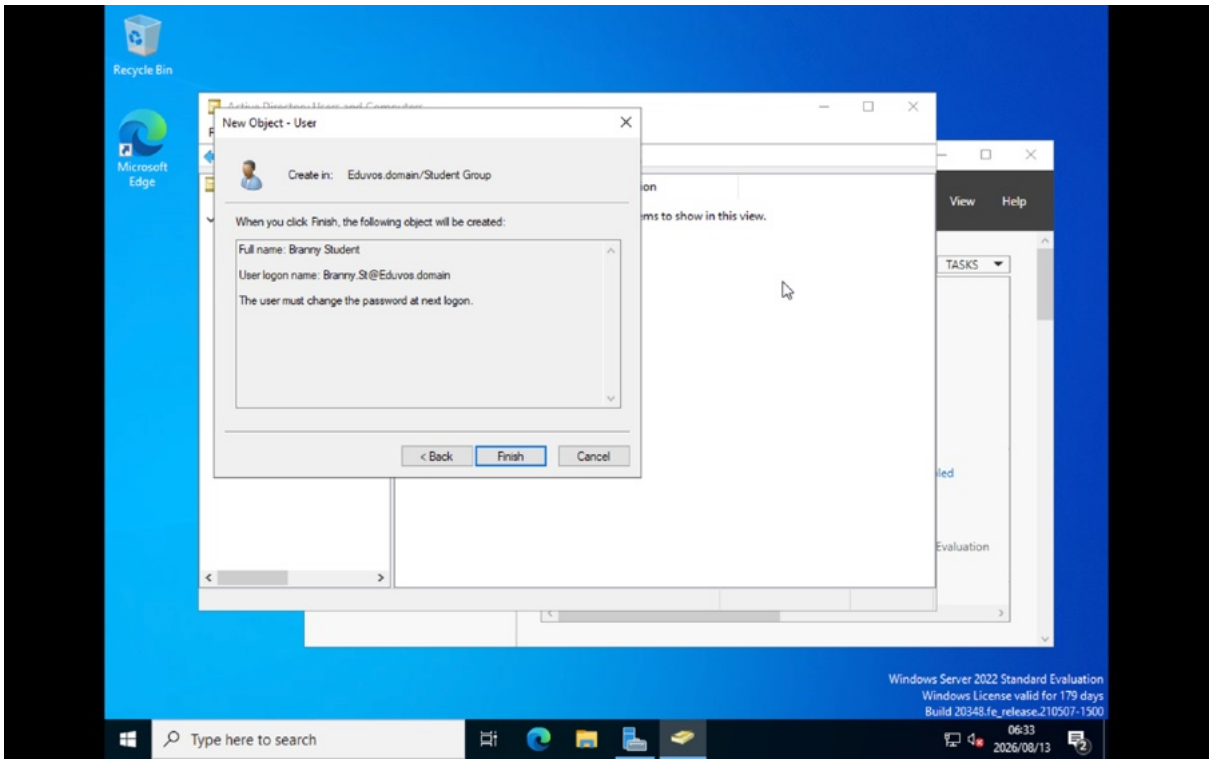
Question 2



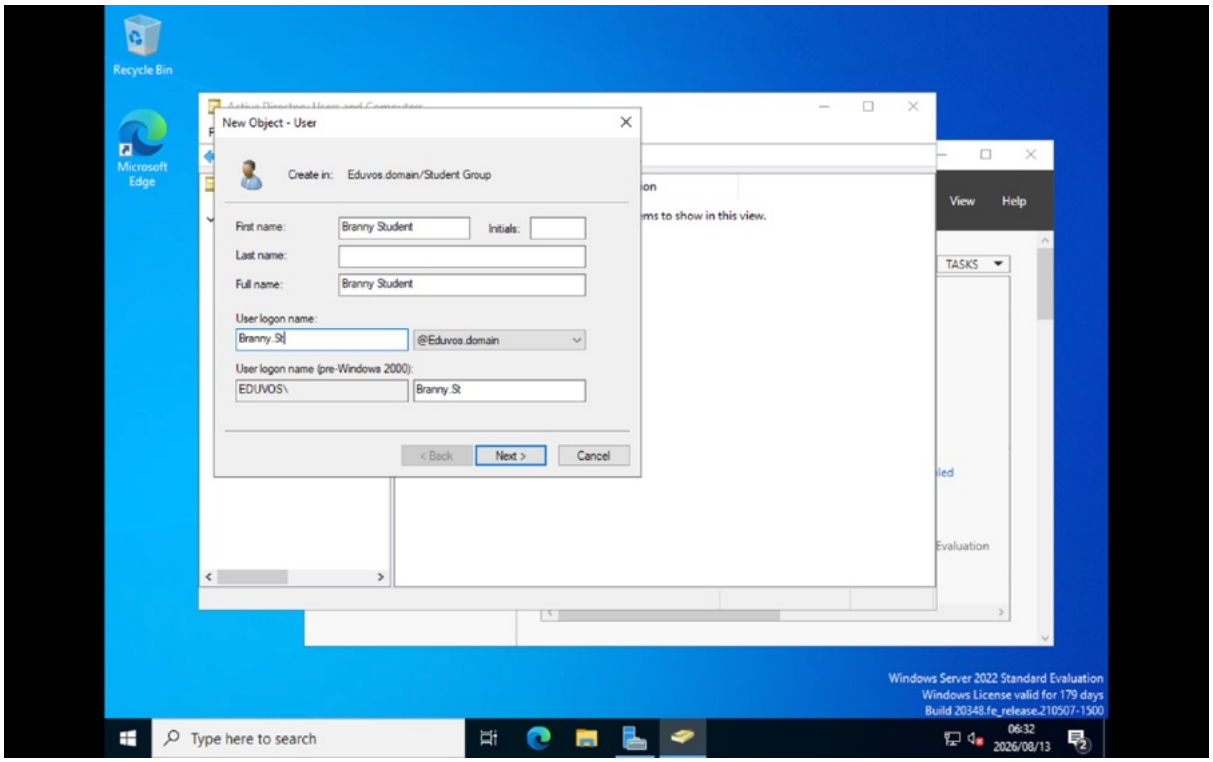
Using Admin account to create an account representing HR Manager



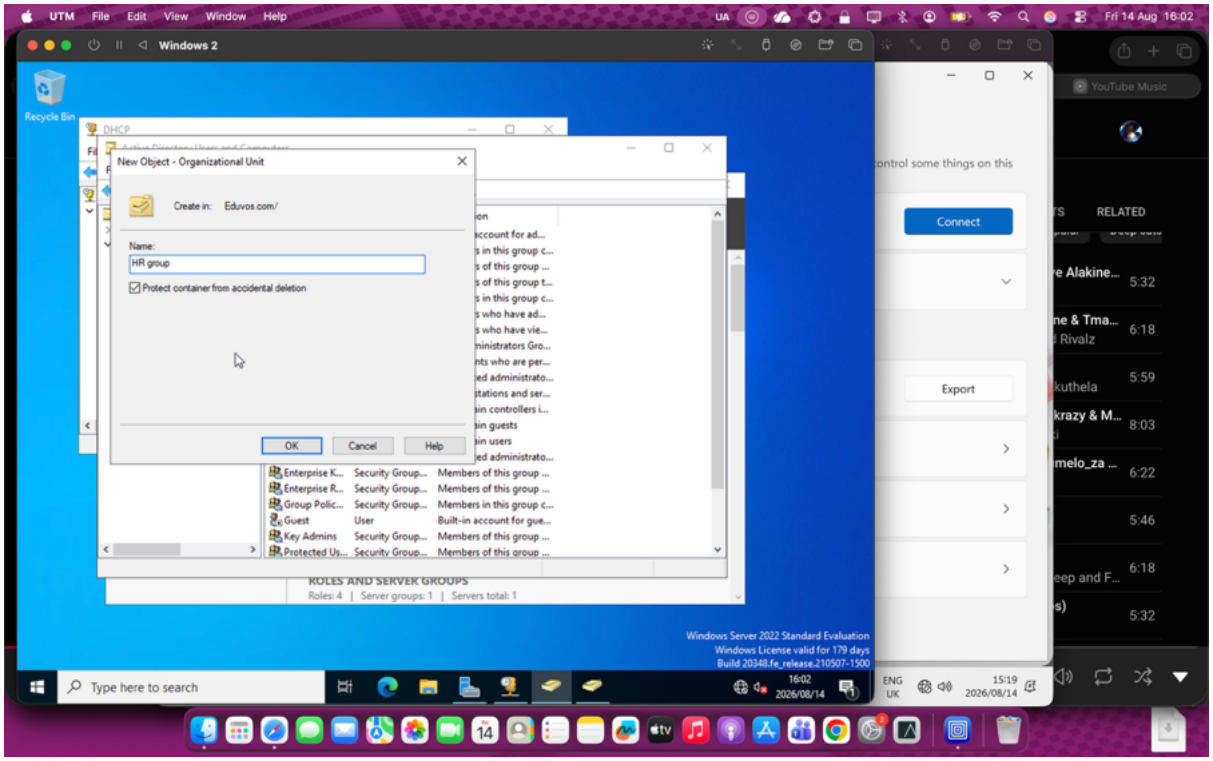
Using Admin account to create an account representing Lecturer



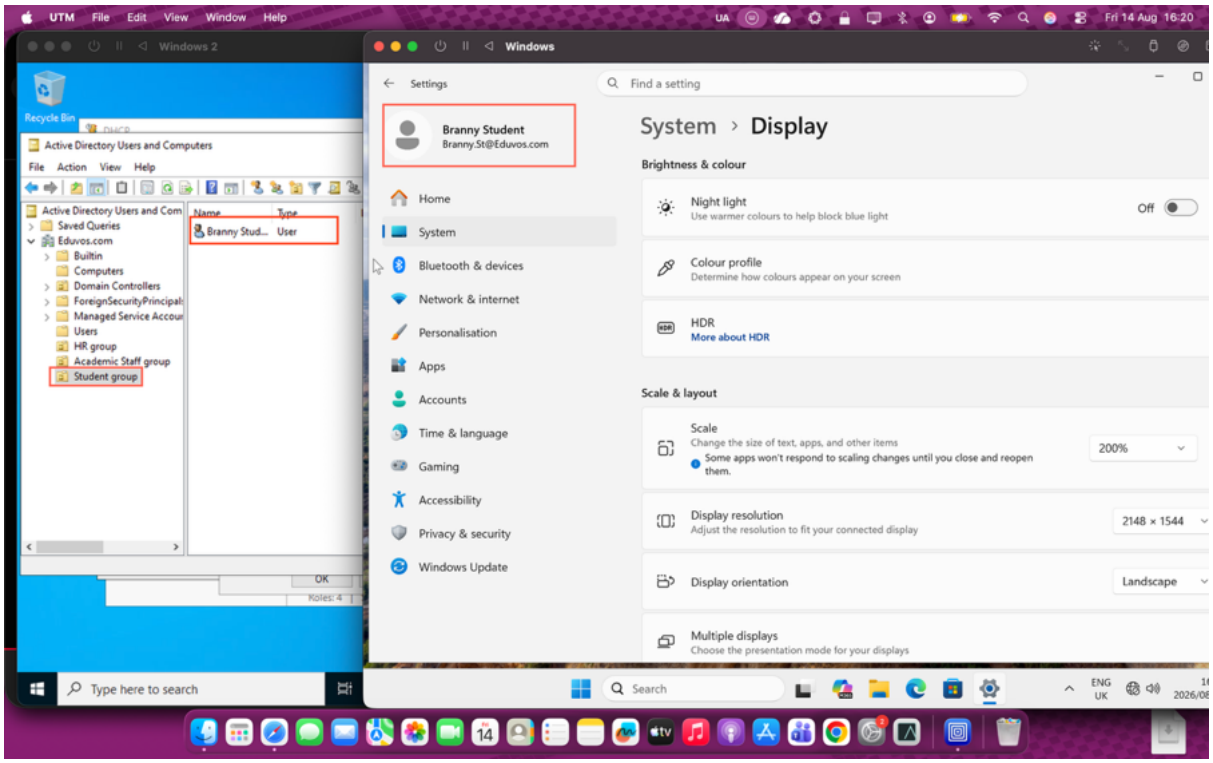
Using Admin account to create an account representing Student



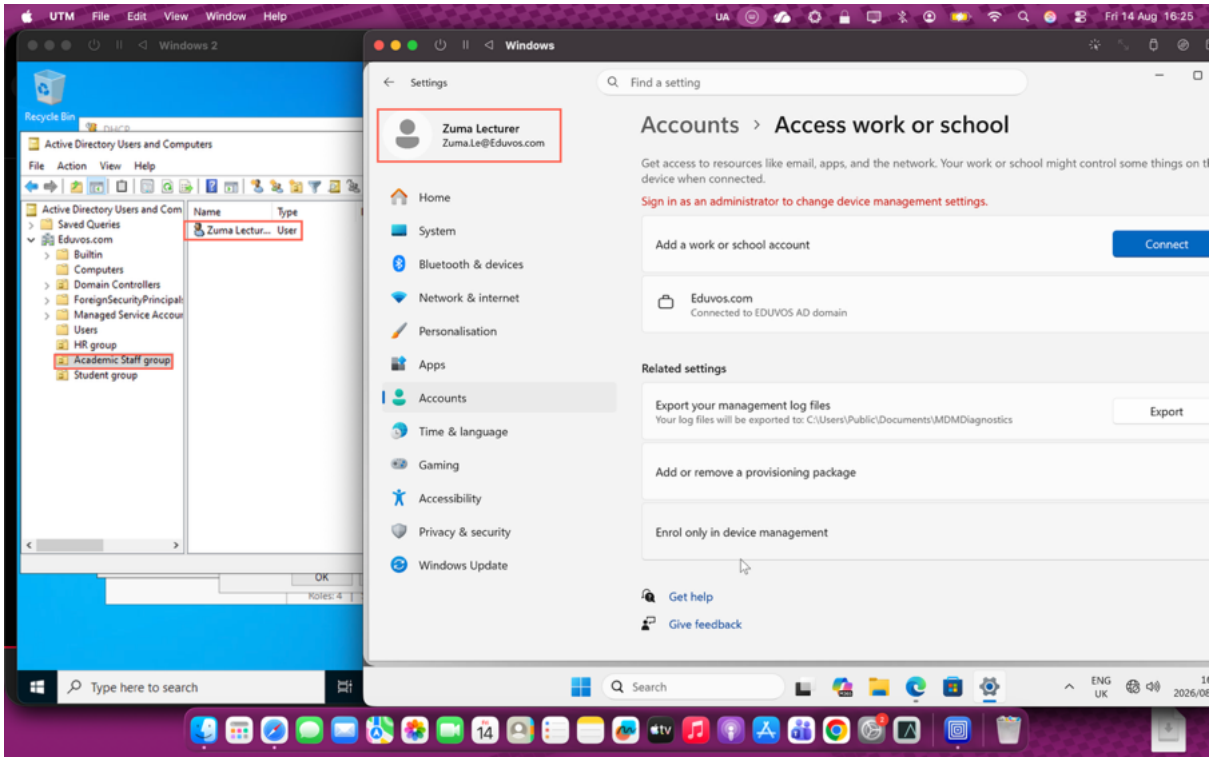
Assigning accounts to security groups. E.g. Student Account to Student Group.



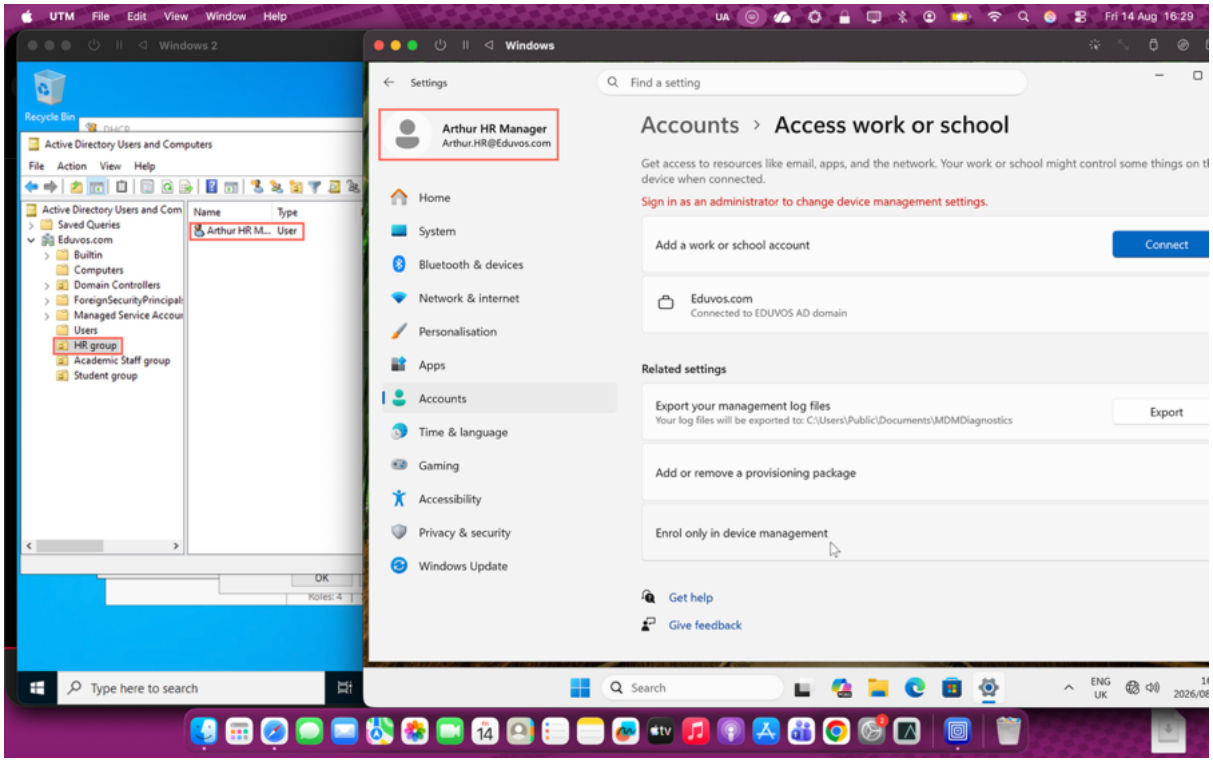
Example of another security group HR Group in which HR manager will be assigned too.



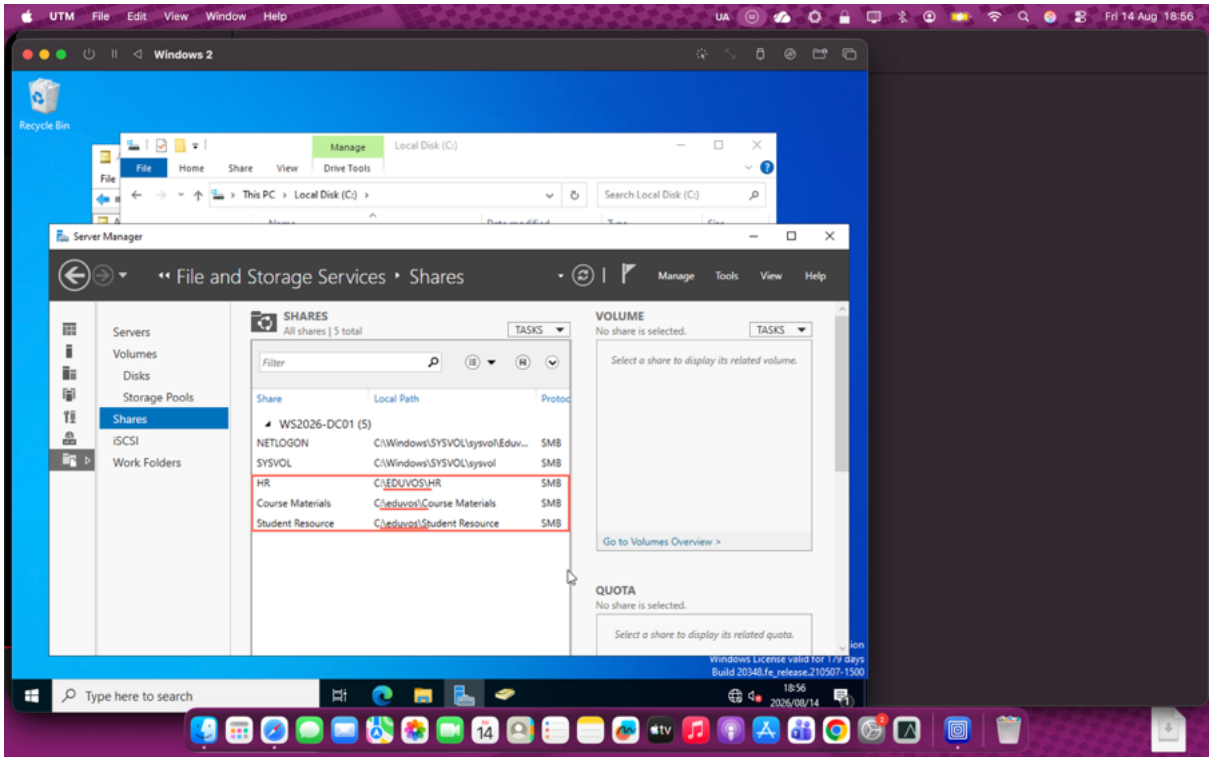
Student account assigned to Student Group.



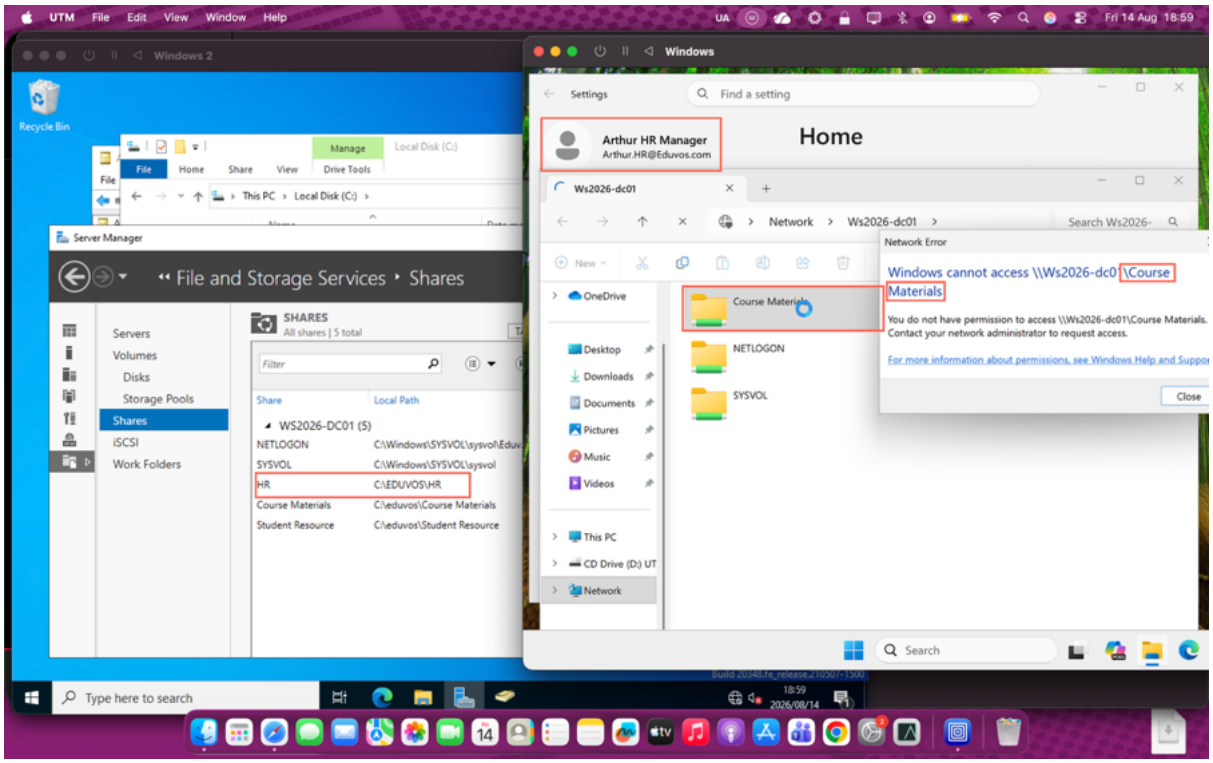
Lecturer account assigned to Academic Staff Group.



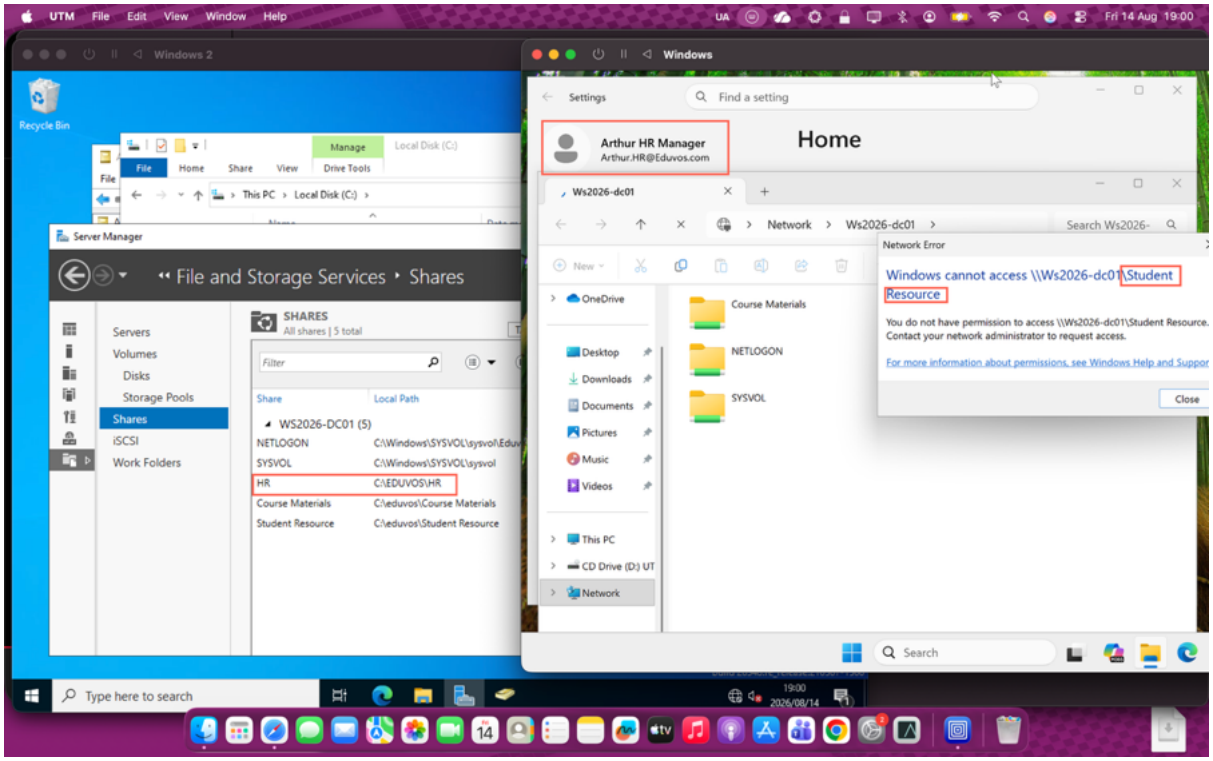
HR account assigned to HR Group.



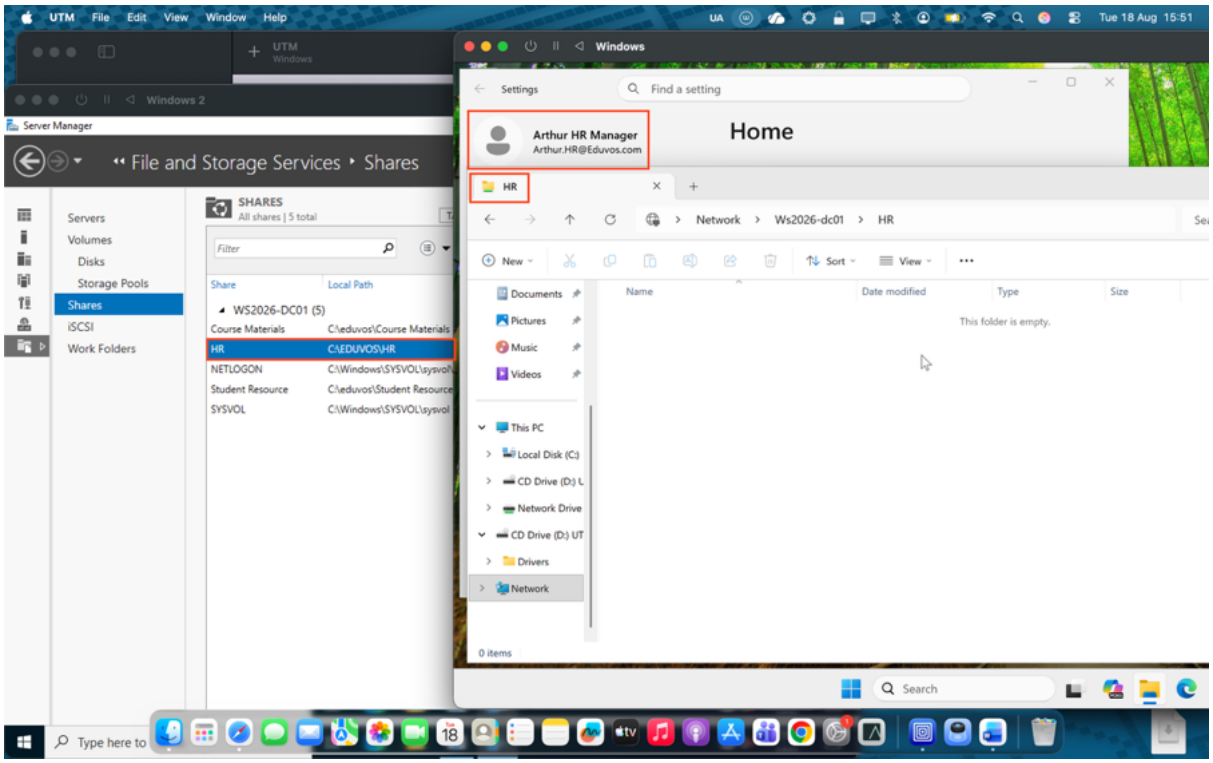
3 Created subfolders : HR, Course Materials and Student Resources inside a primary folder : Eduvos



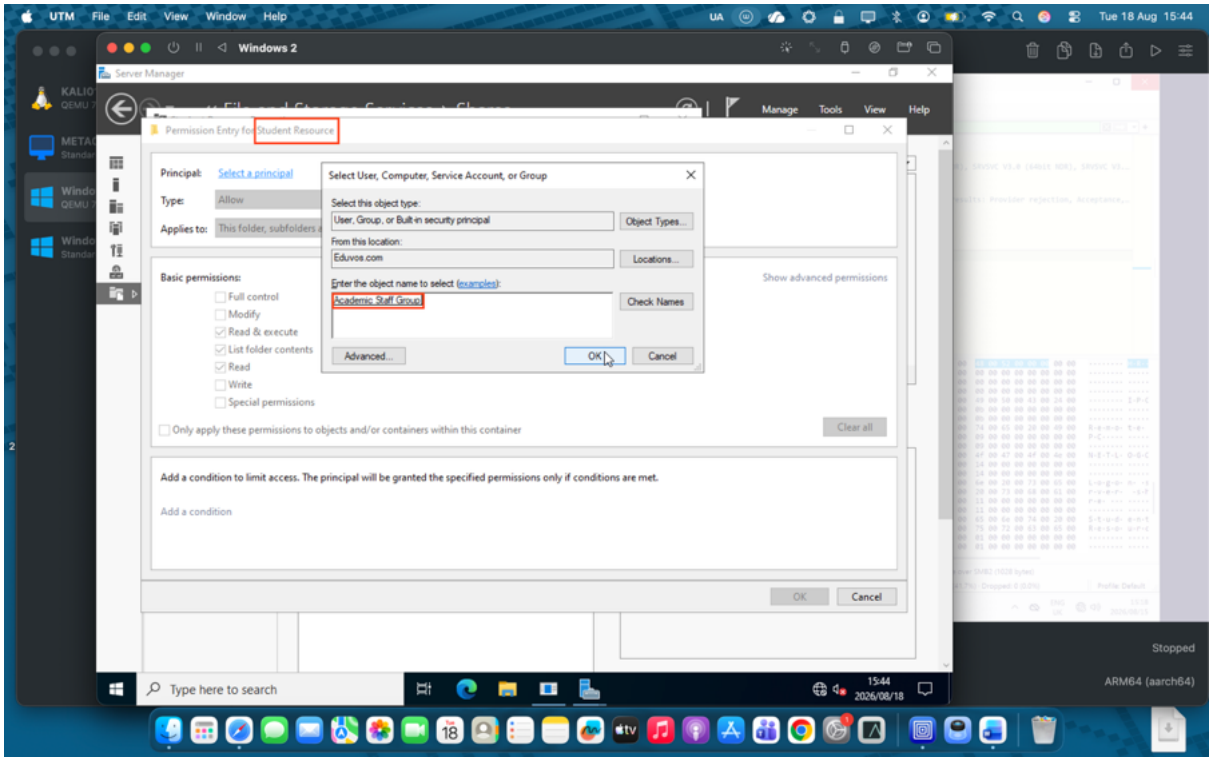
HR user from HR Group denied access to Course Materials



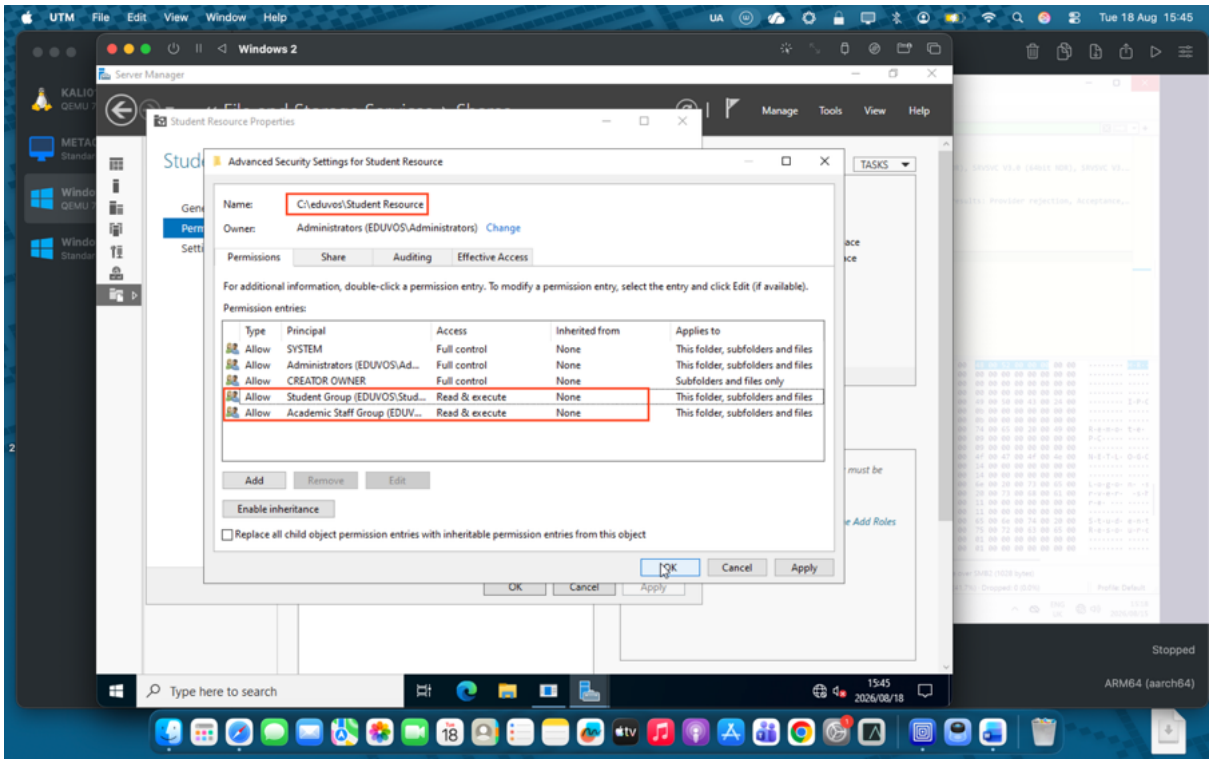
HR user from HR Group denied access to Student resources



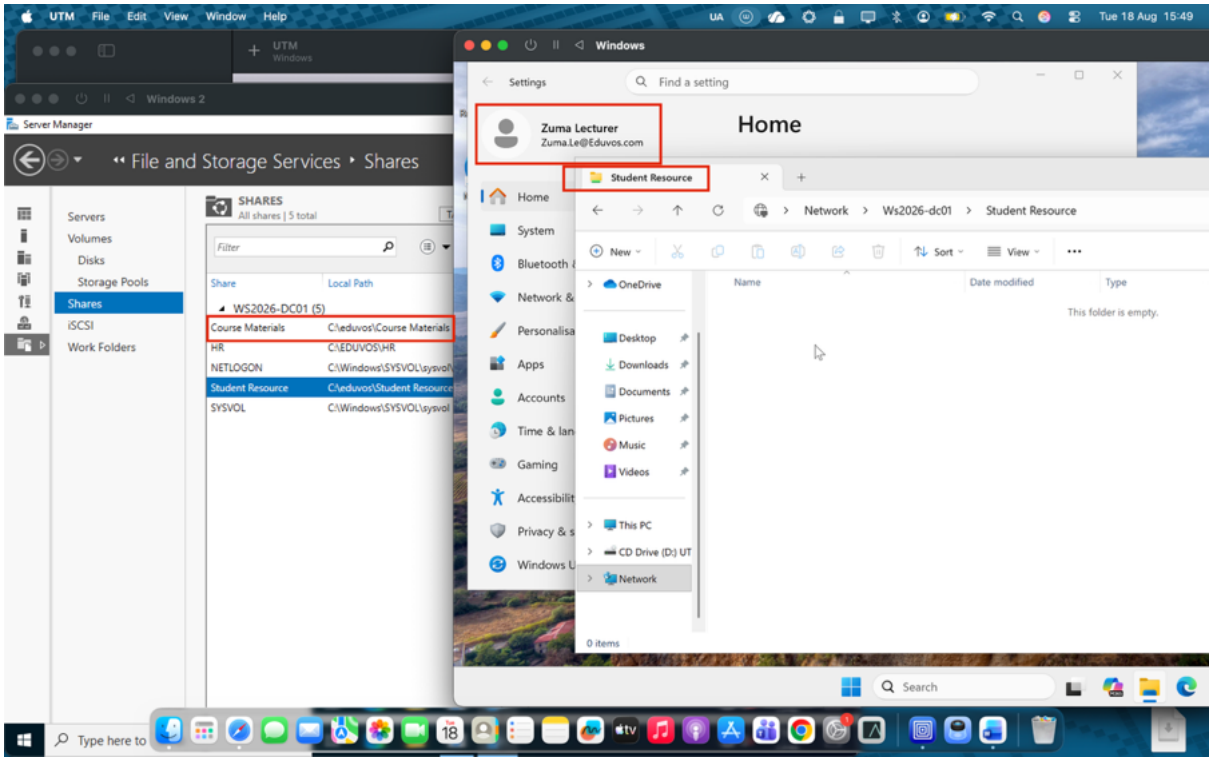
HR user from HR Group granted access to HR folder



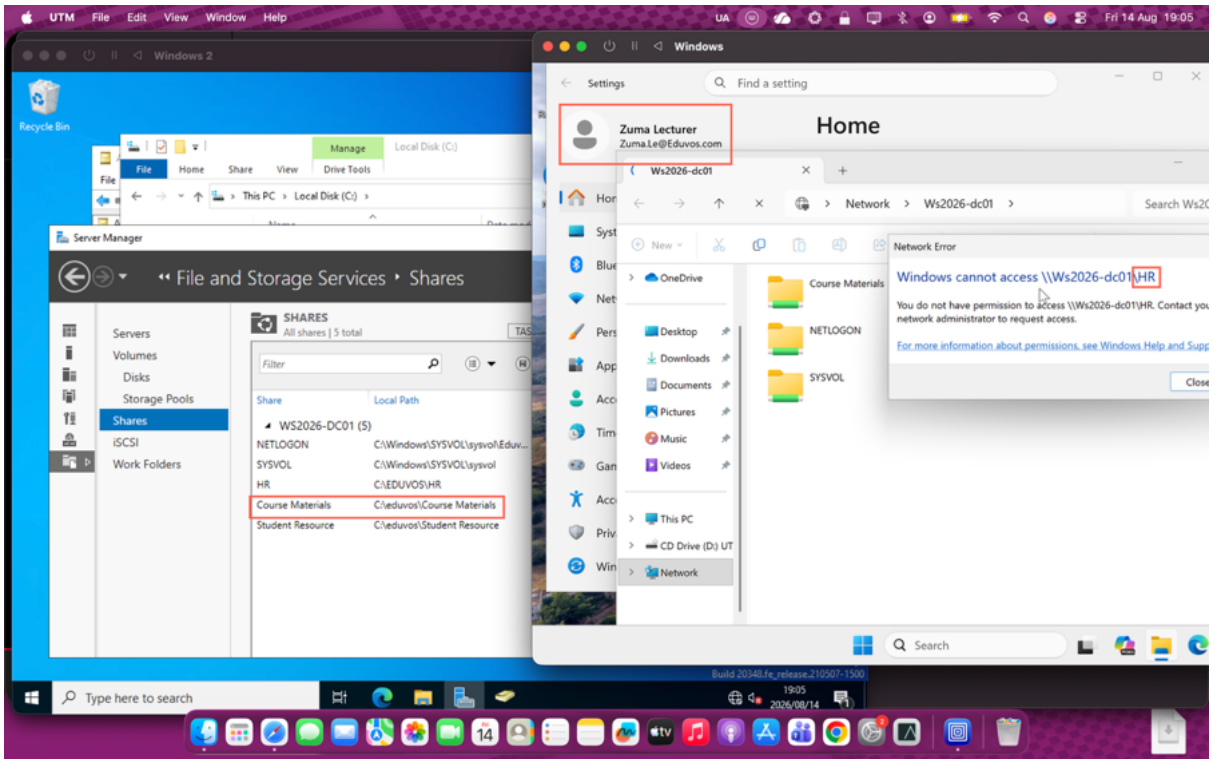
Adding Academic Staff Group with Lecturers access to Student Resources.



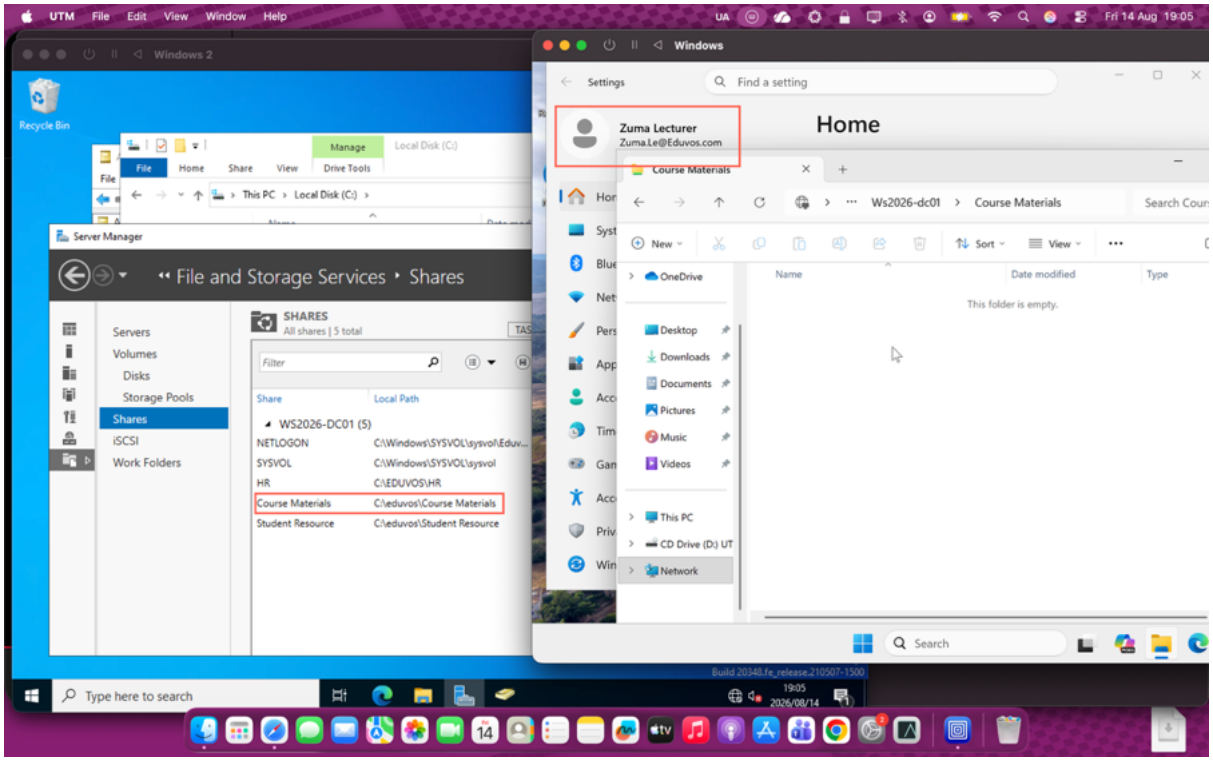
Student Group and Academic Staff Group now have access to Student Resources.



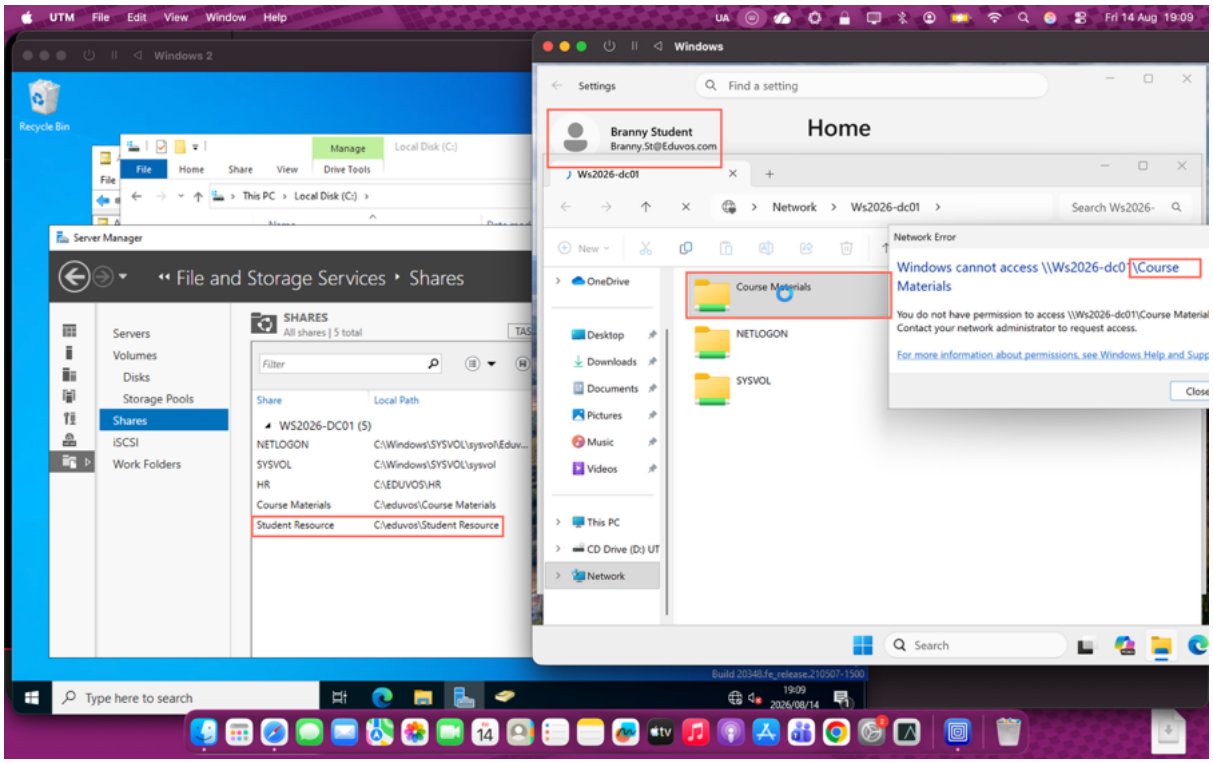
Lecturer user from Academic Staff Group granted access to Student Resources.



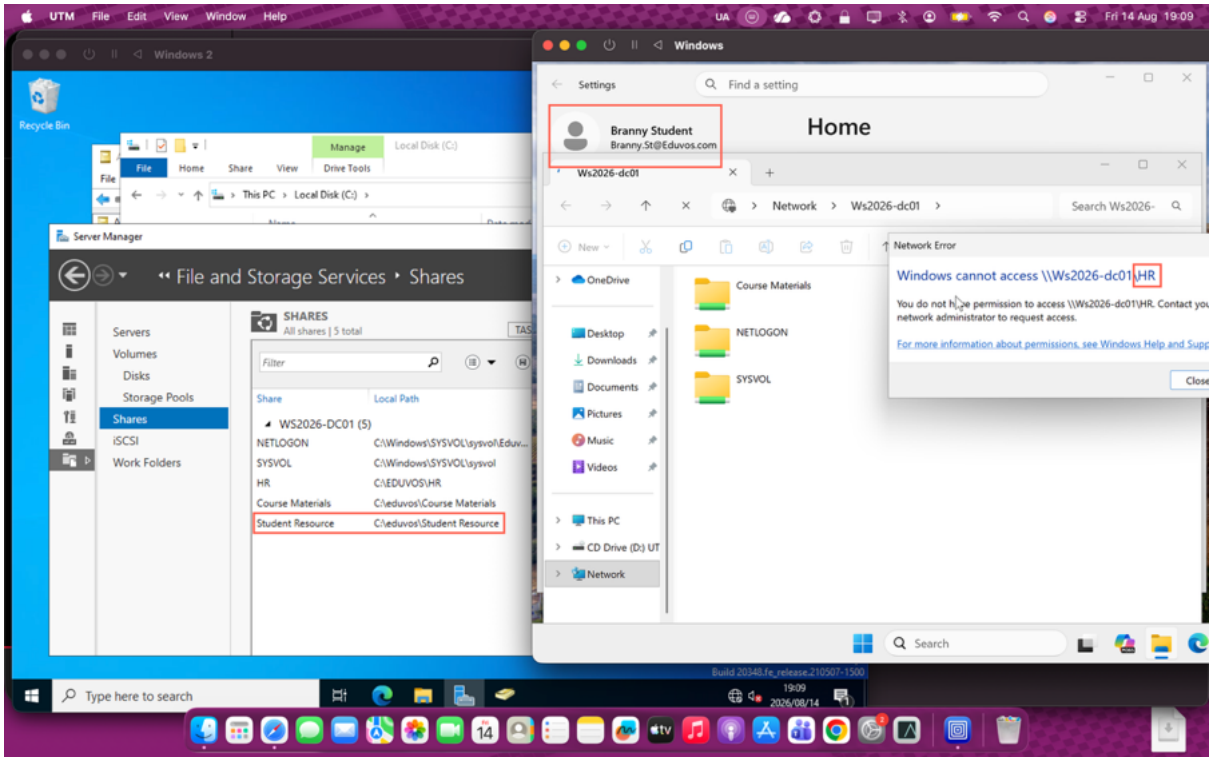
Lecturer user from Academic Staff Group denied access to HR folder



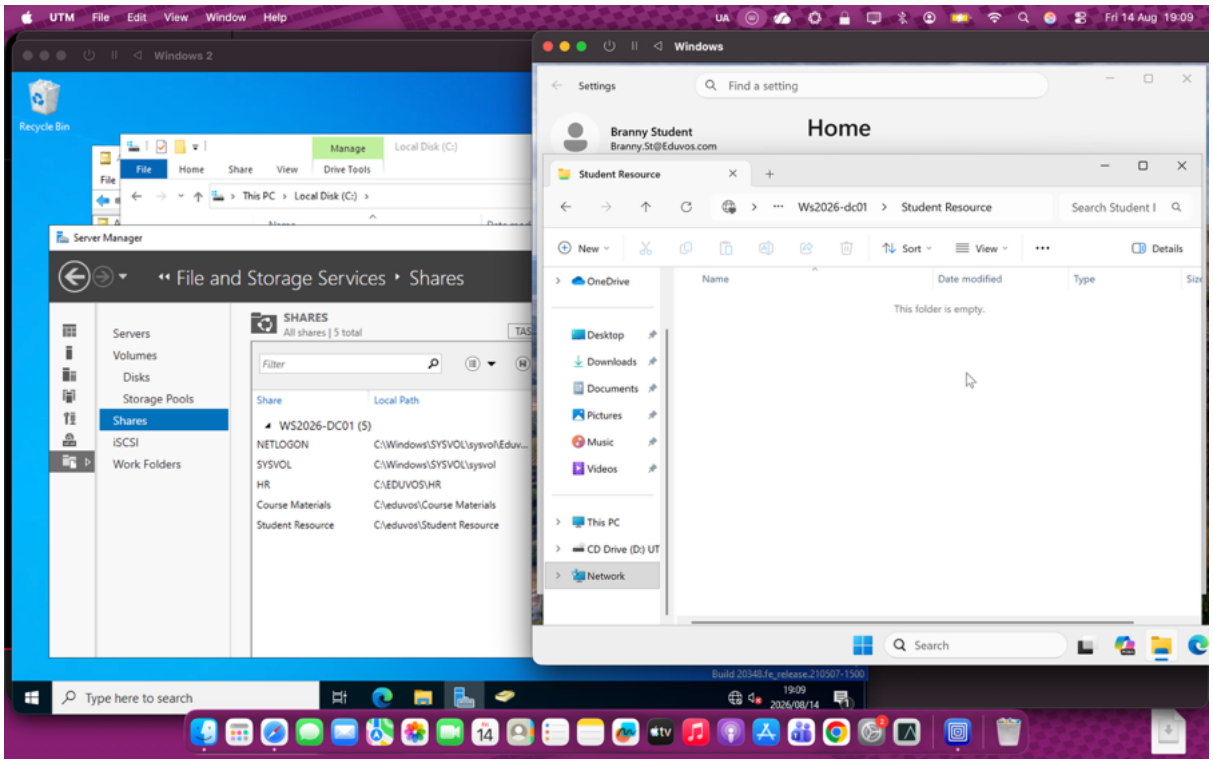
Lecturer user from Academic Staff Group granted access to Course Materials



Student user from Student Group denied access to Course Material.

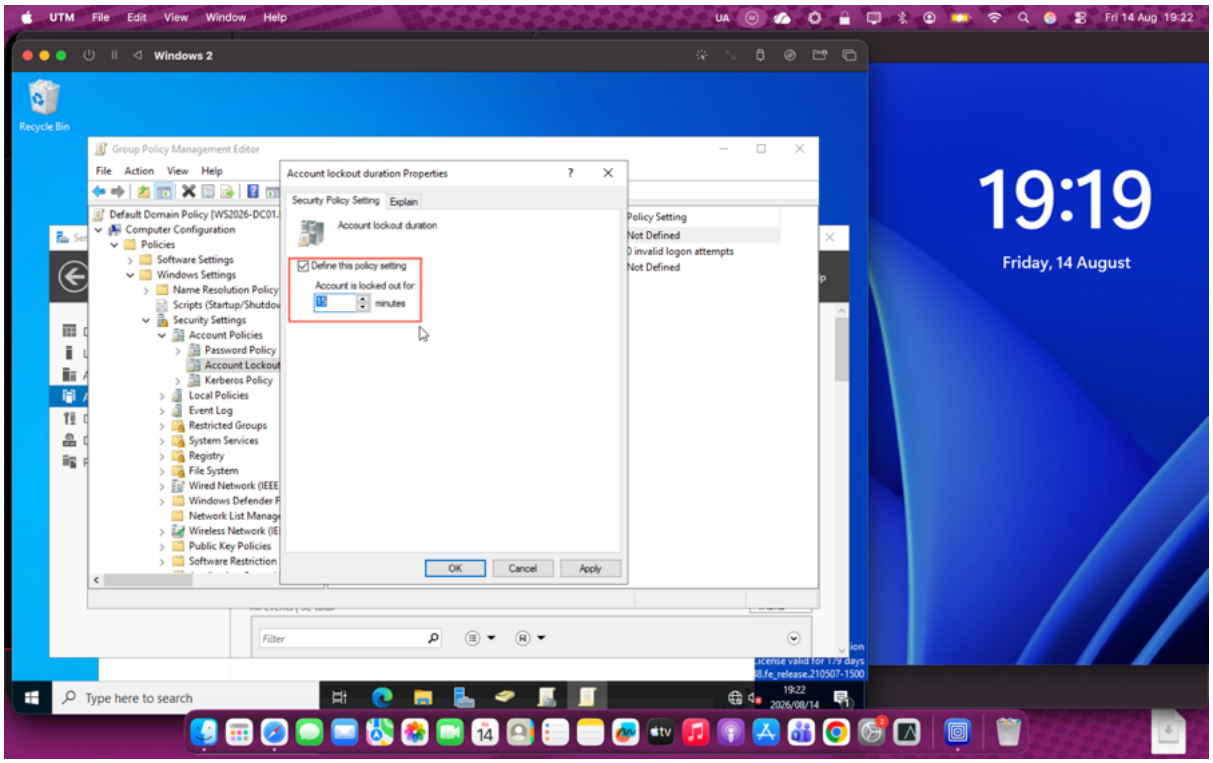


Student user from Student Group denied access to HR folder

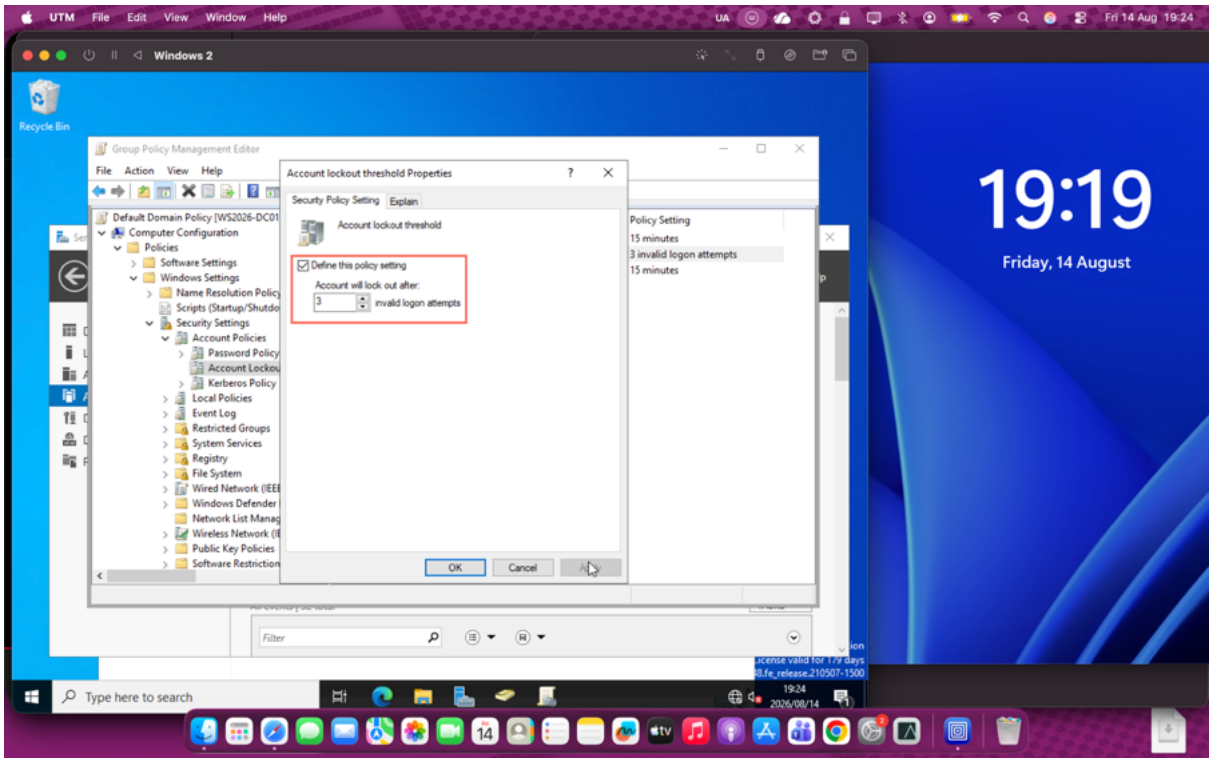


Student user from Student Group granted access to Student Resource.

With all this verifying that cross-department access is blocked.



Using Admin account to define lockout duration to 15 minutes



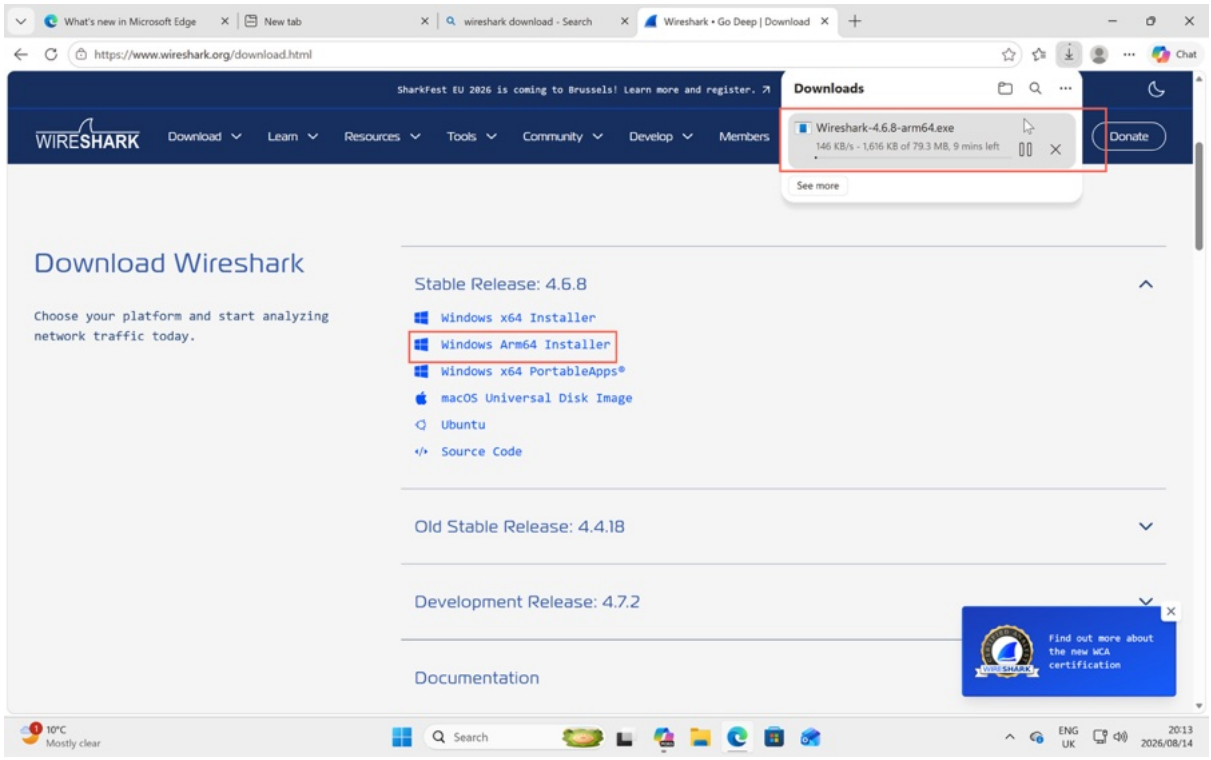
Using Admin account to set lockout threshold to 3 invalid logon attempts.

Lockout duration and lockout threshold prevents brute-force password attacks by minimising the number of failed sign in attempts and defining how long a account stays blocked after hitting the limit.

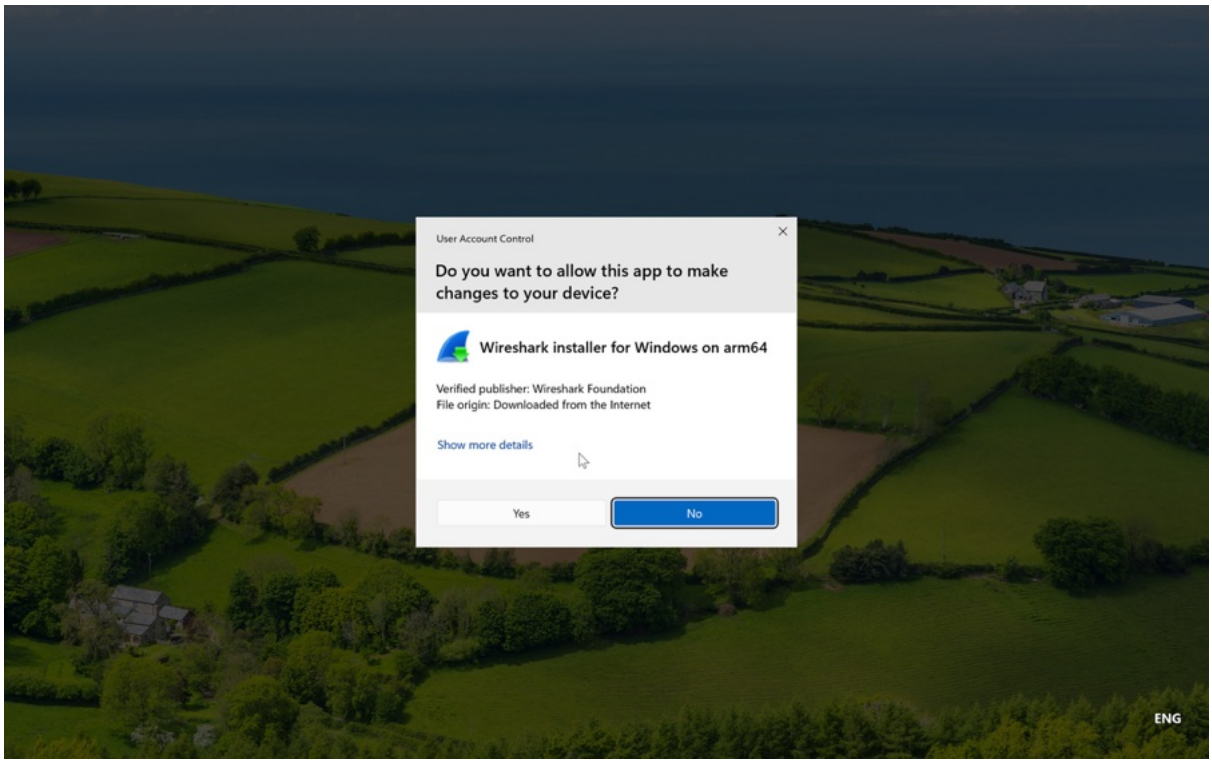
With Threshold counting mistakes, Triggering defence and stop instant guessing

While Duration forces timeout, slow attack crawl and reopen access safety.

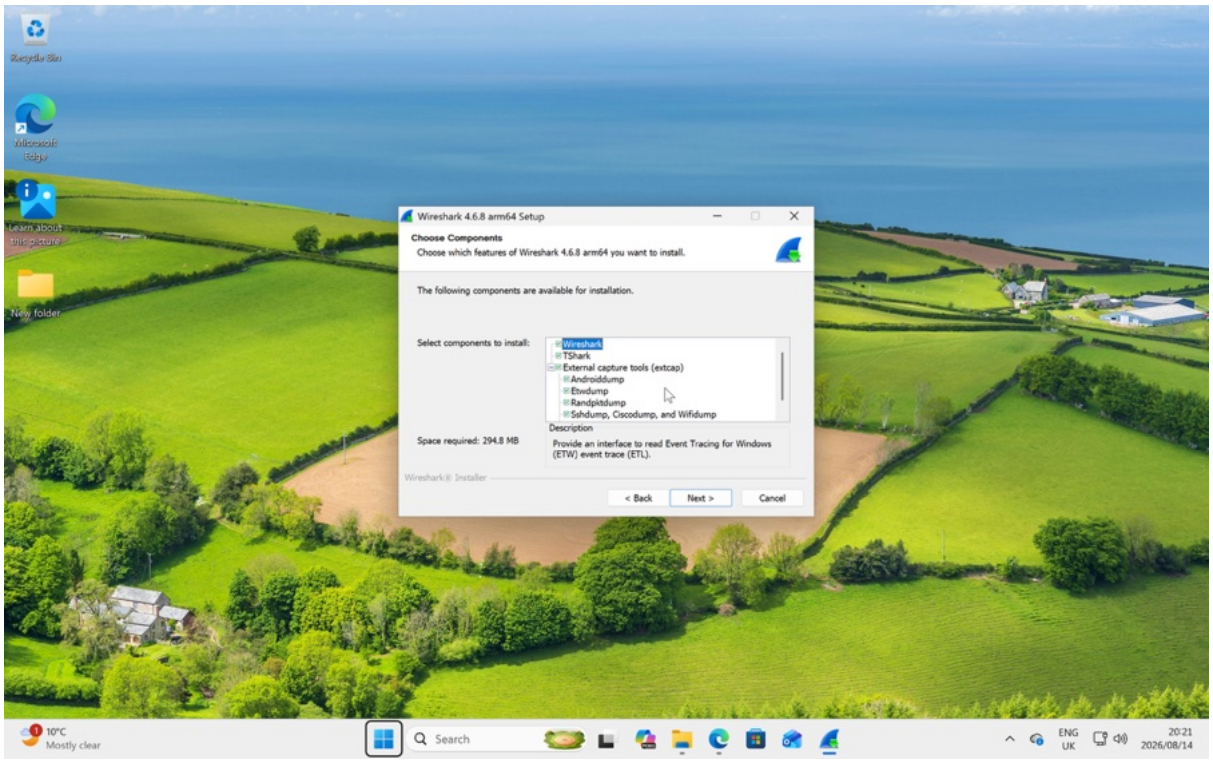
Question 3



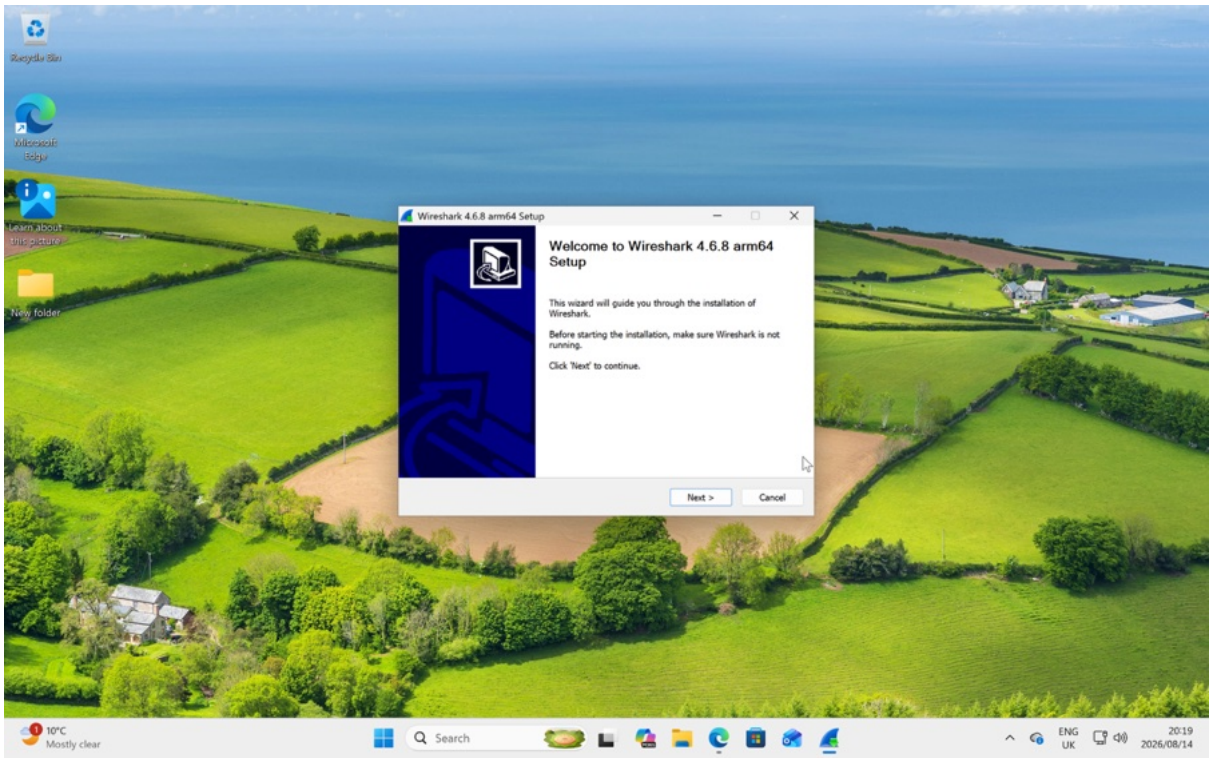
Installing Wireshark installer for ARM64 windows.



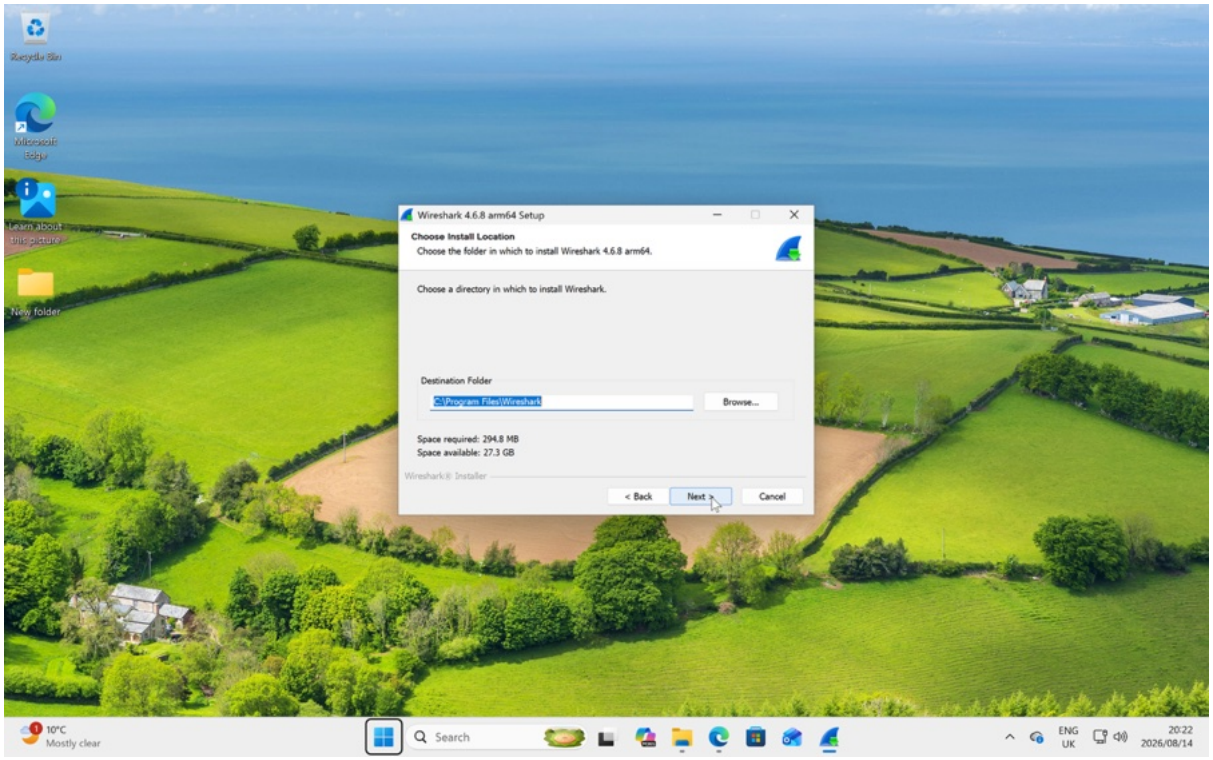
Allowing Wireshark to make changes on the device so that it can be fully functional.



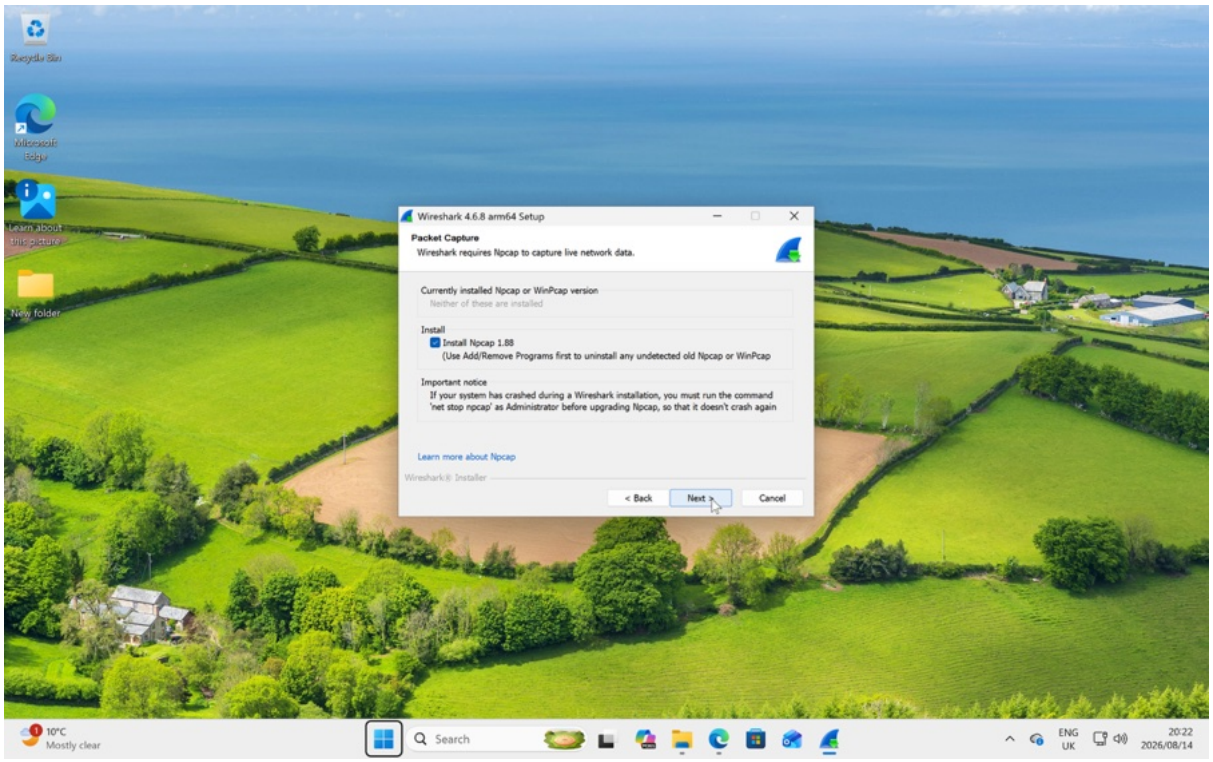
Installing all Wireshark components.



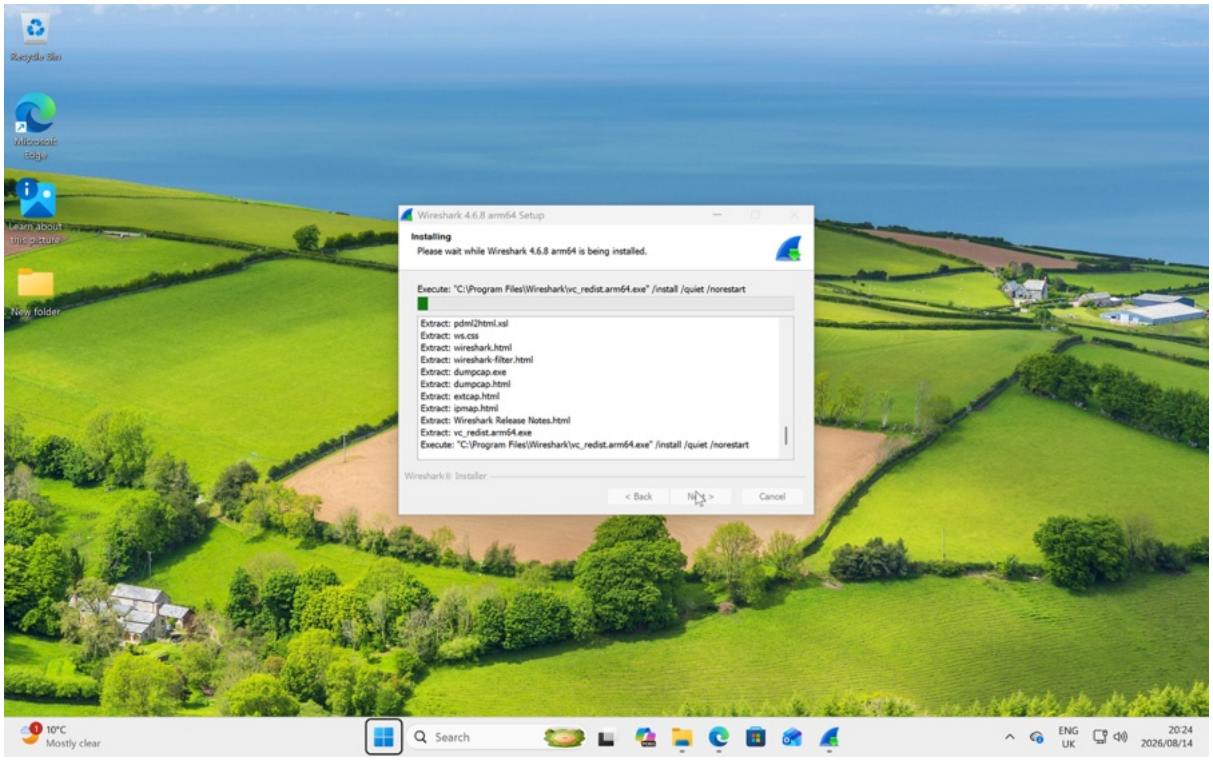
Setting up Wireshark wizard.



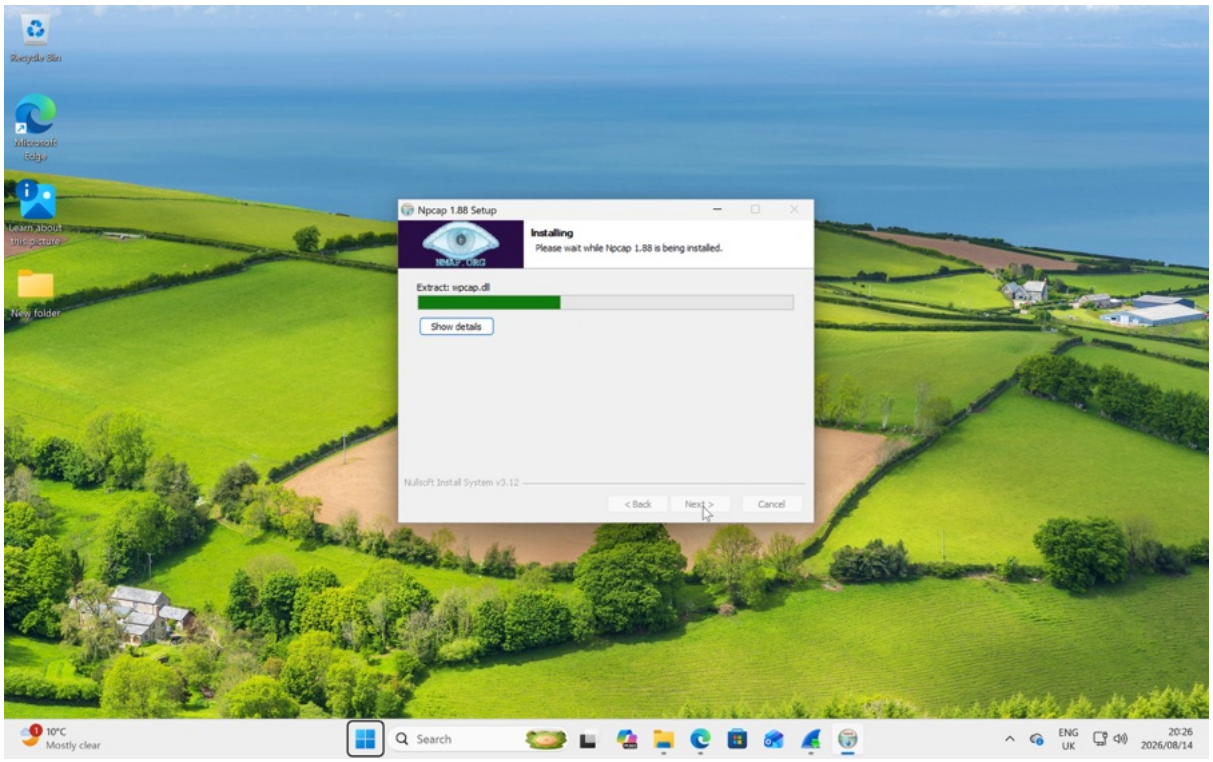
Setting up location or destination for Wireshark program.



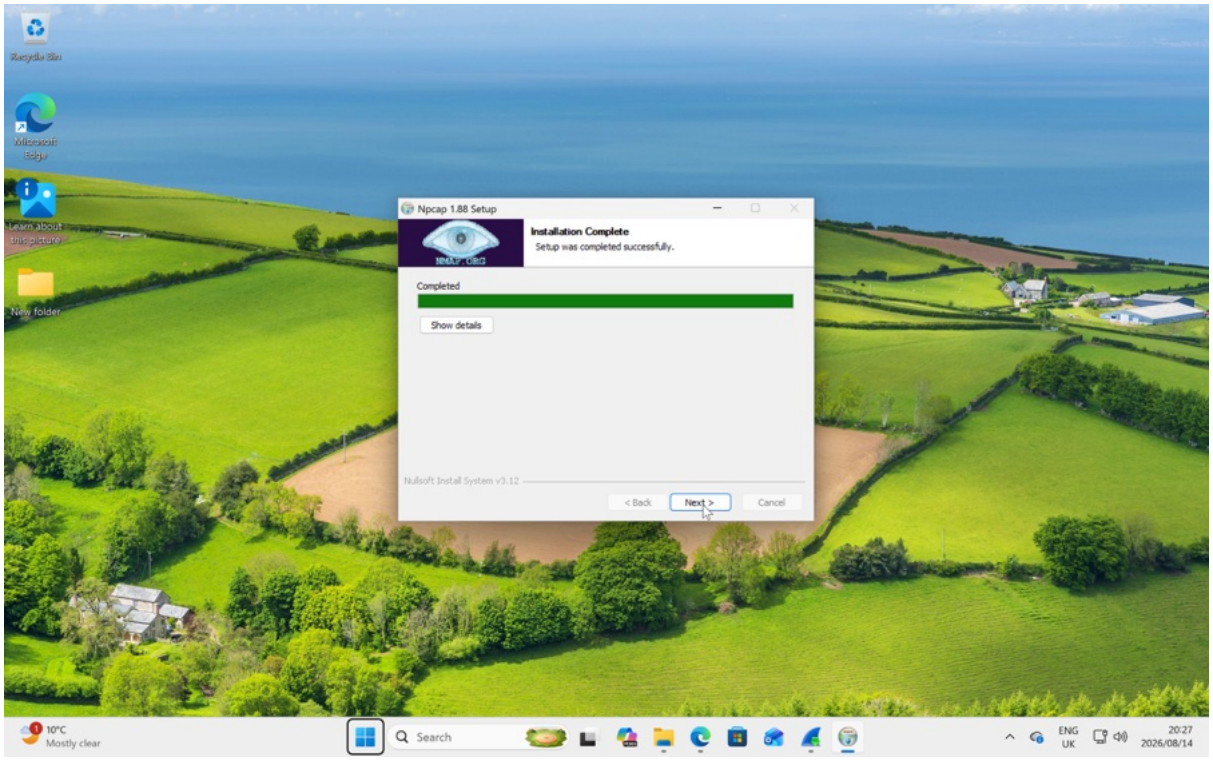
Installing Npcap on Wireshark.



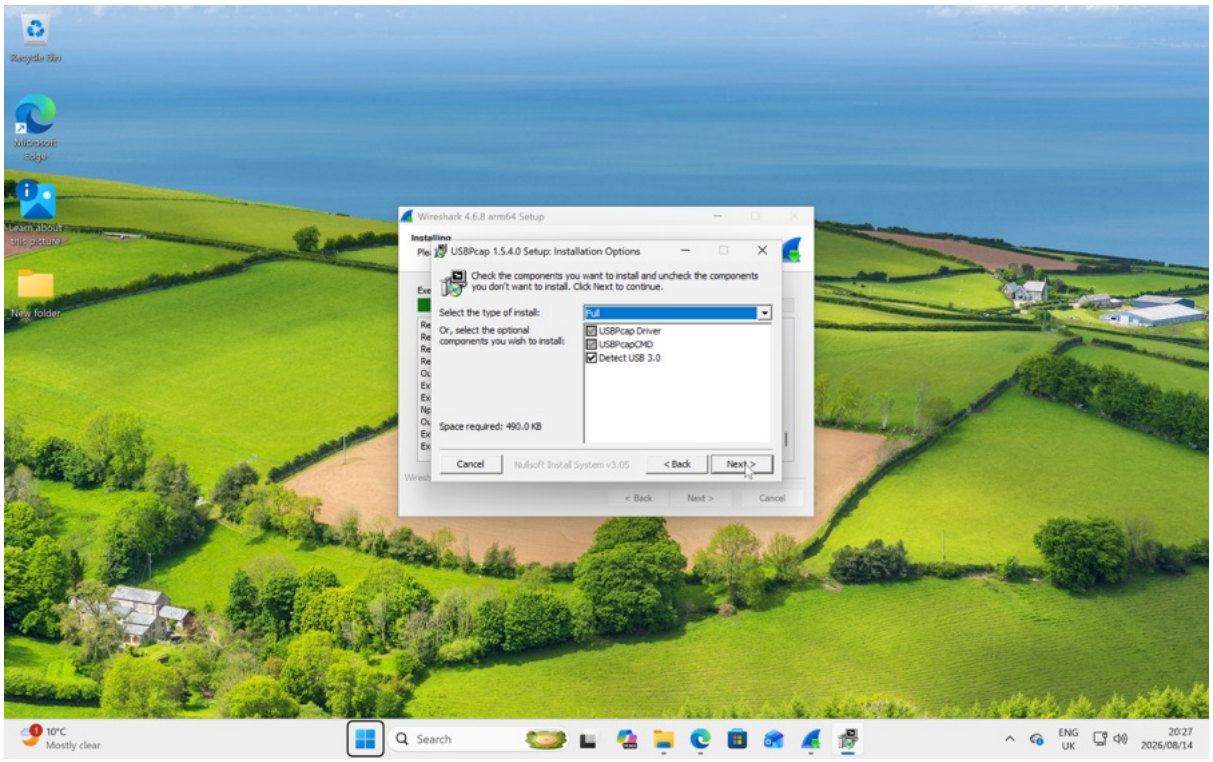
Wireshark installing.



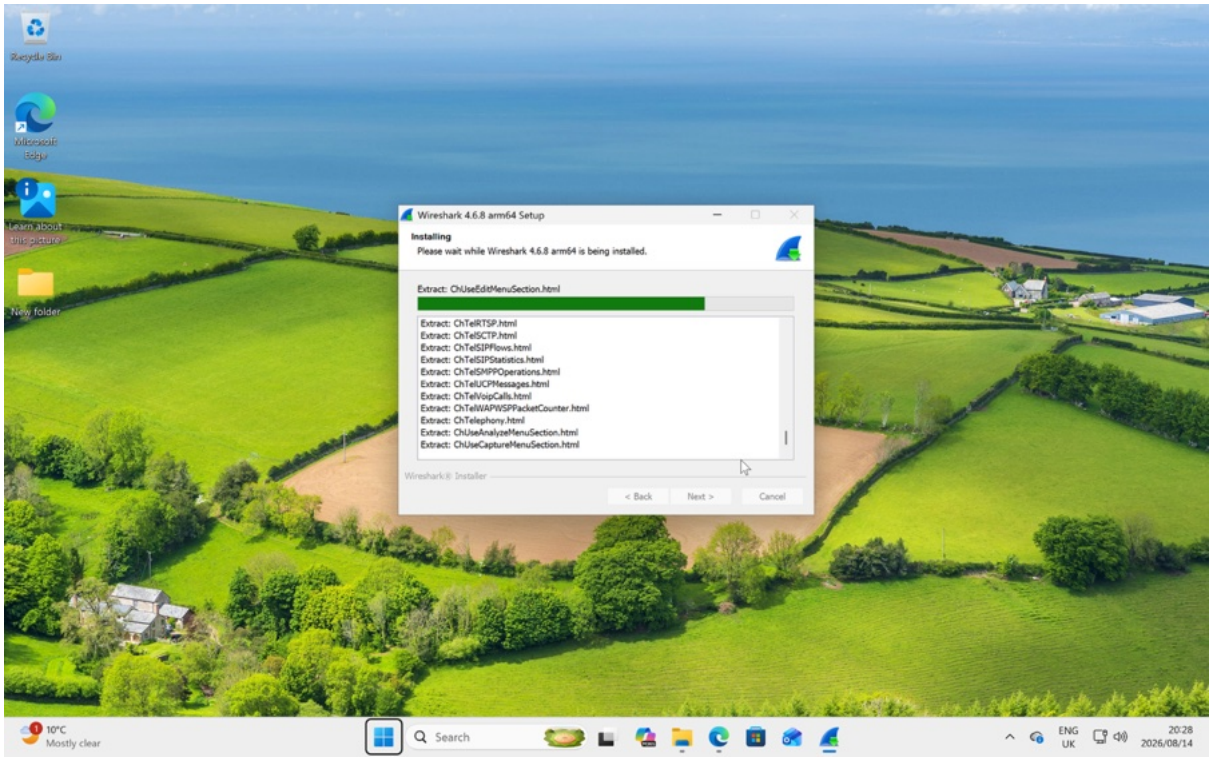
Npcap installing.



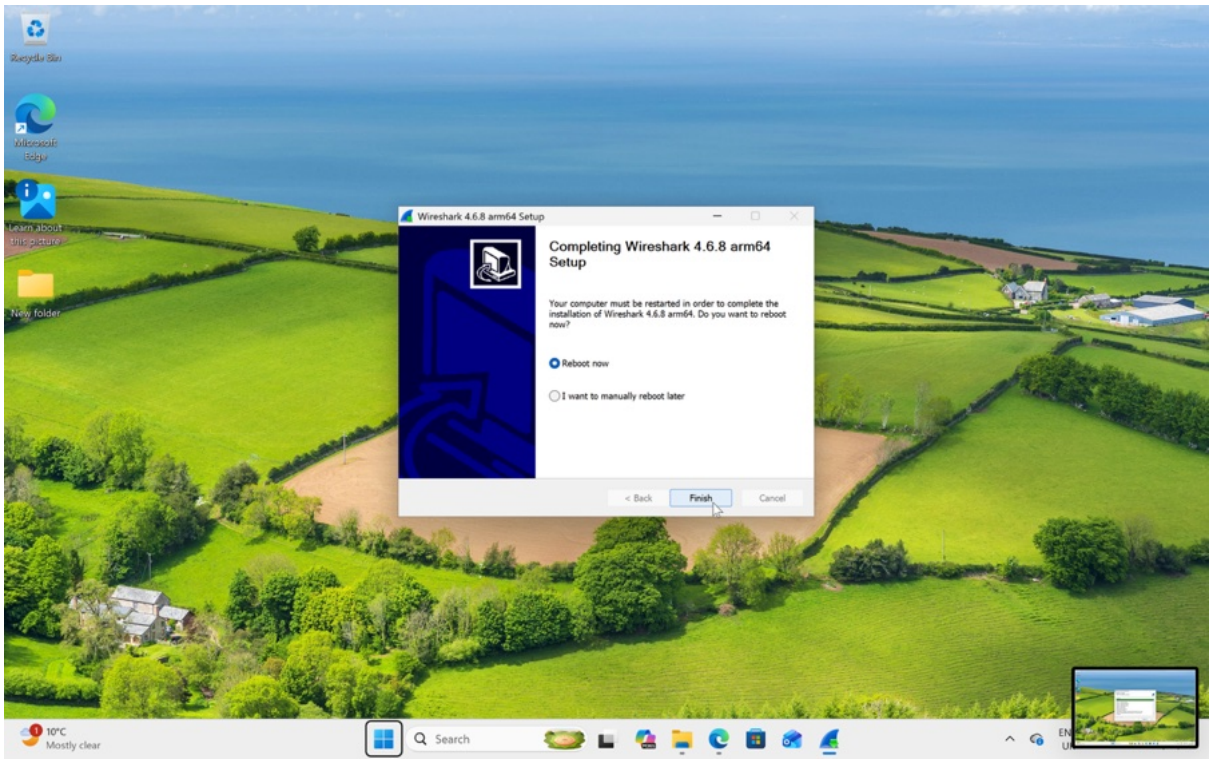
Complete installation



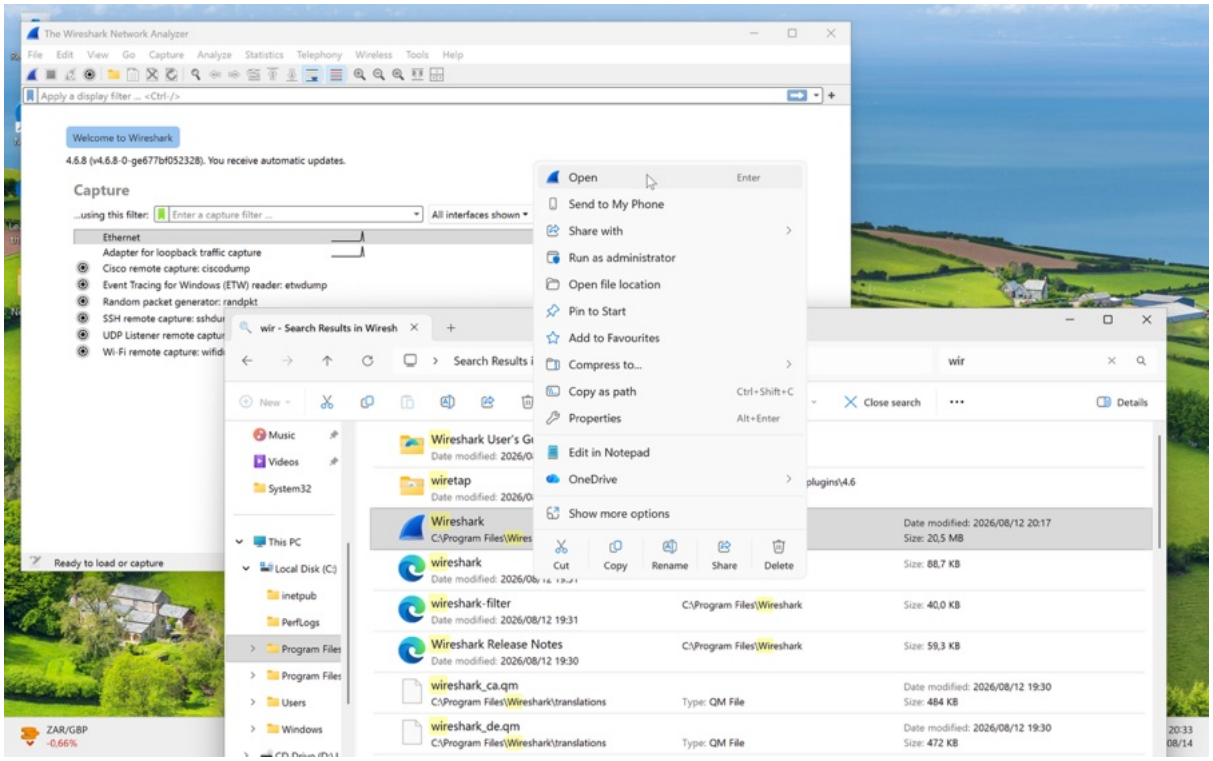
Installing all components.



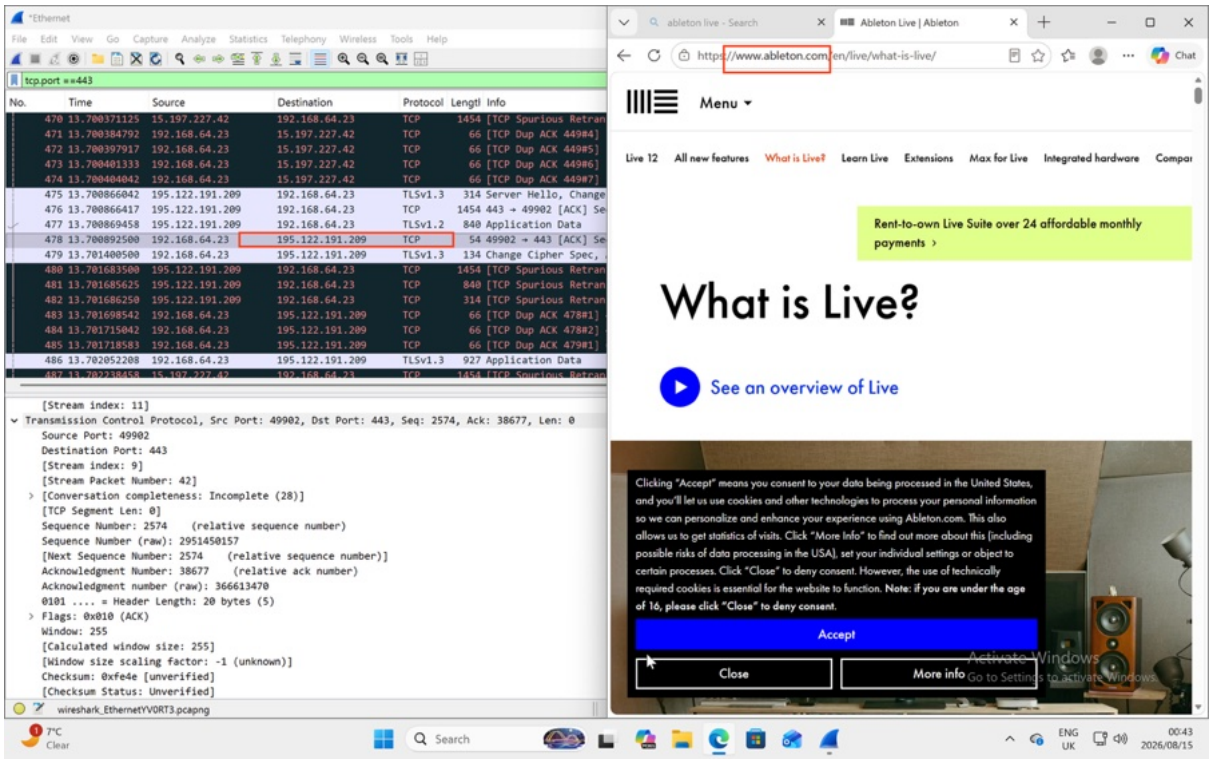
Wireshark continuing installation



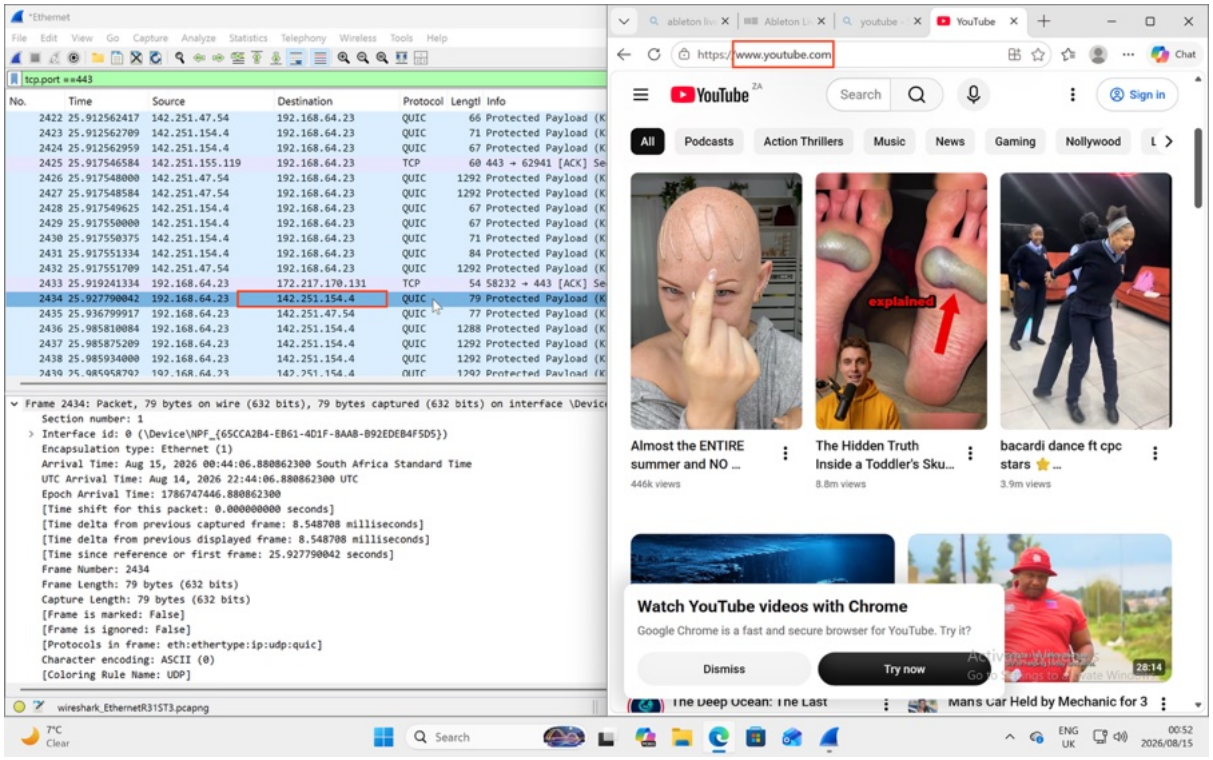
Wireshark installation complete and ready to use after reboot.



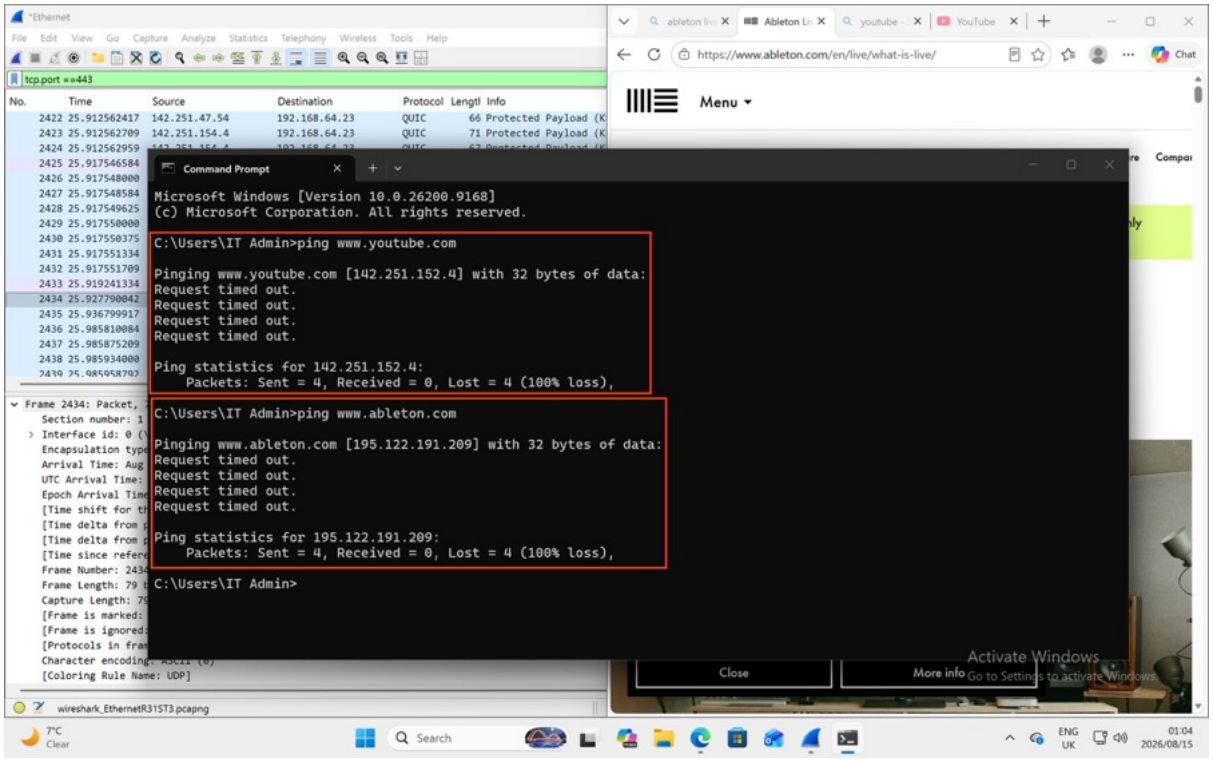
Wireshark now running and Successfully installed.



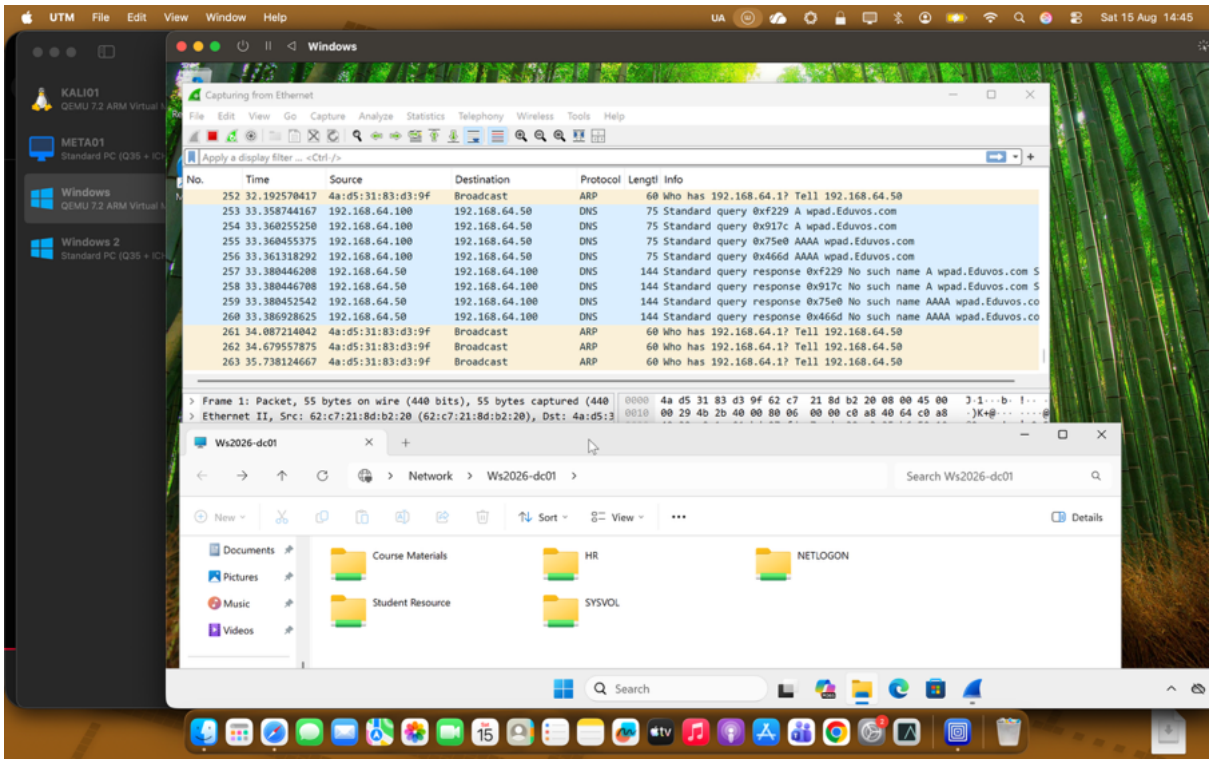
Generating Traffic by browsing the website www.ableton.com with the IP address of 195.122.191.209 and using filter : tcp.port ==443 to find https.



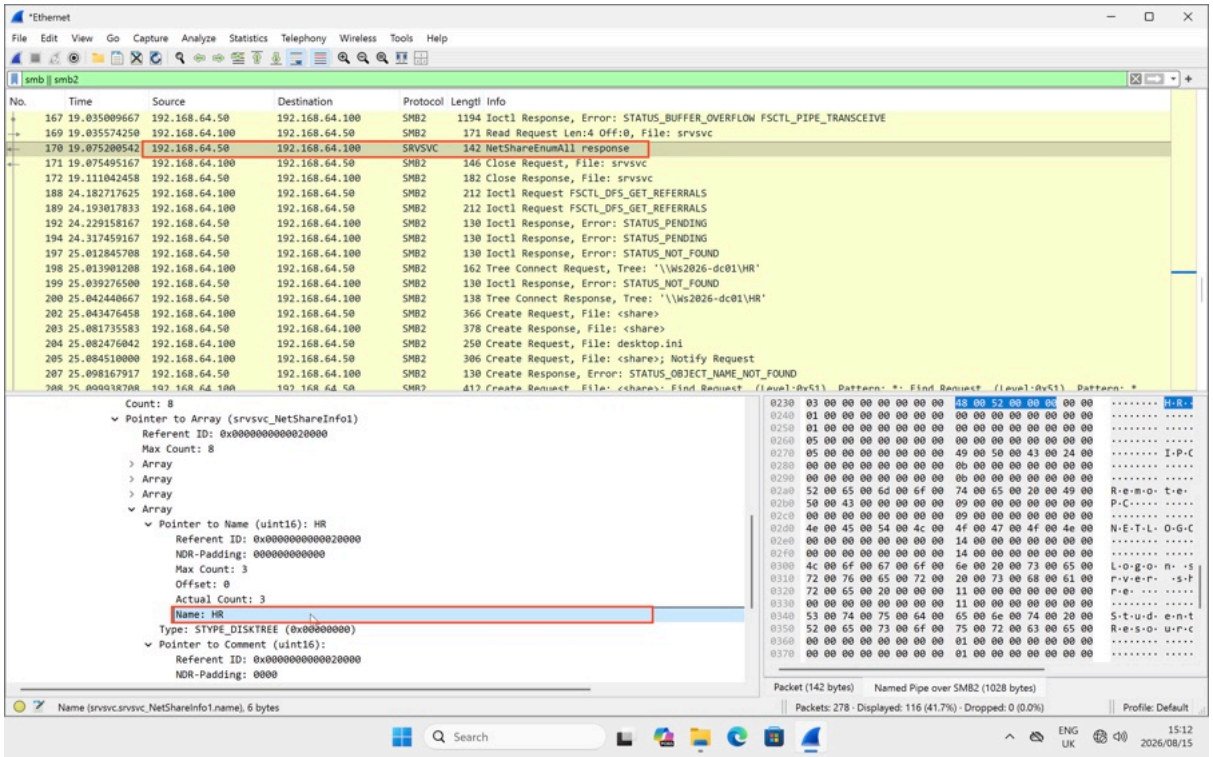
Generating Traffic by browsing the website www.youtube.com with the IP address of 142.251.154.4 and using filter : tcp.port ==443 to find https.



Pinging the two websites www.youtube.com and www.ableton.com with the IP Addresses 142.251.152.4 and 195.122.191.209

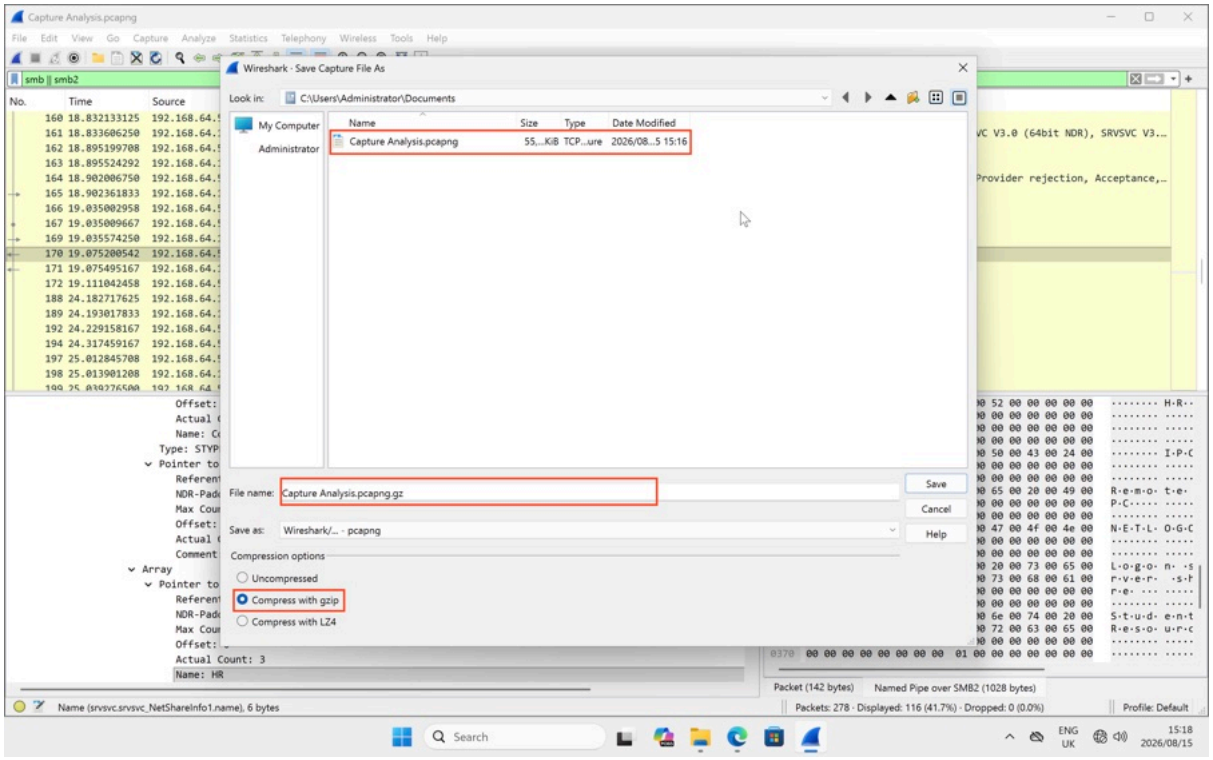


Generating traffic by accessing file shares from shared folder.



Simulating file sharing access to shared folders with Wireshark. Using the filter : smb || smb2

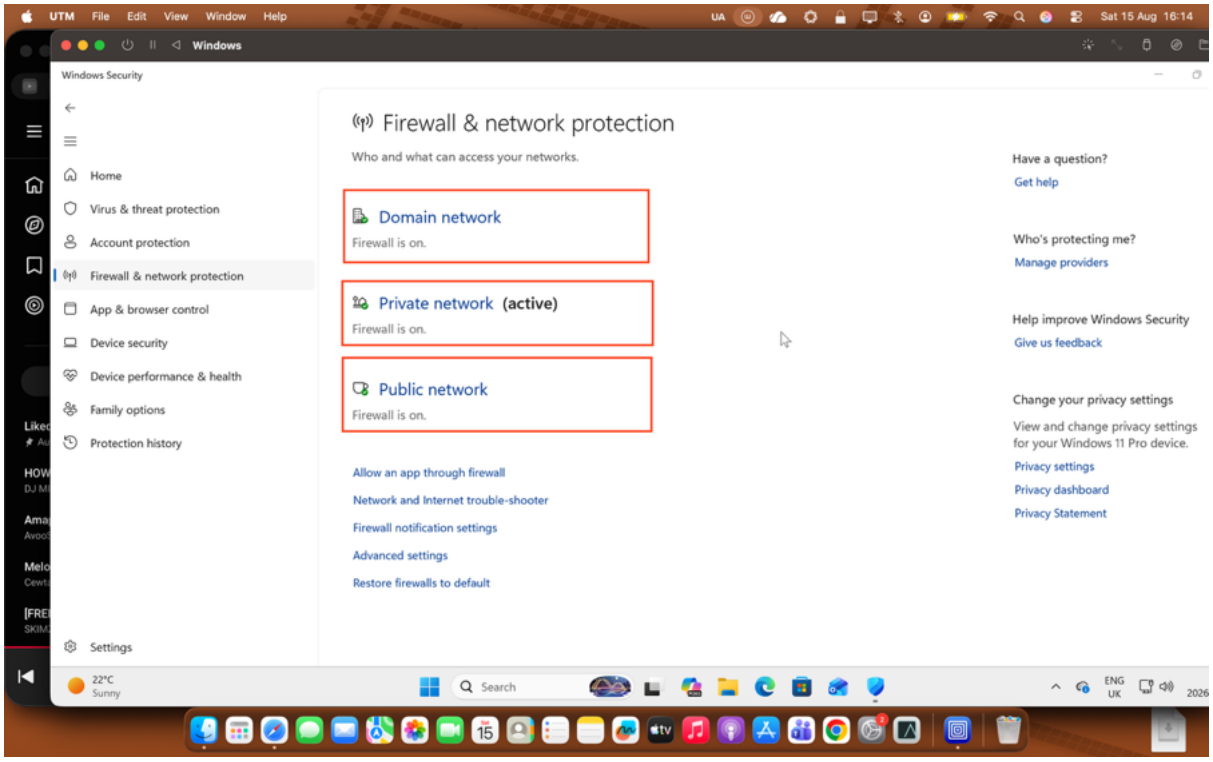
Protocols	PURPOSE
SMB	Purpose is to enable network files, printer, and serial port sharing using a request-response model.
TCP	Purpose is to enable reliable, ordered, and error-checked data delivery between network applications.
TLS	Purpose is to enable secure communication over a public network like the internet.
DNS	Purpose is to enable a human-readable translated website names from computer-readable numerical IP addresses.
ARP	Purpose is to map logical IP address to physical MAC address on a local network.



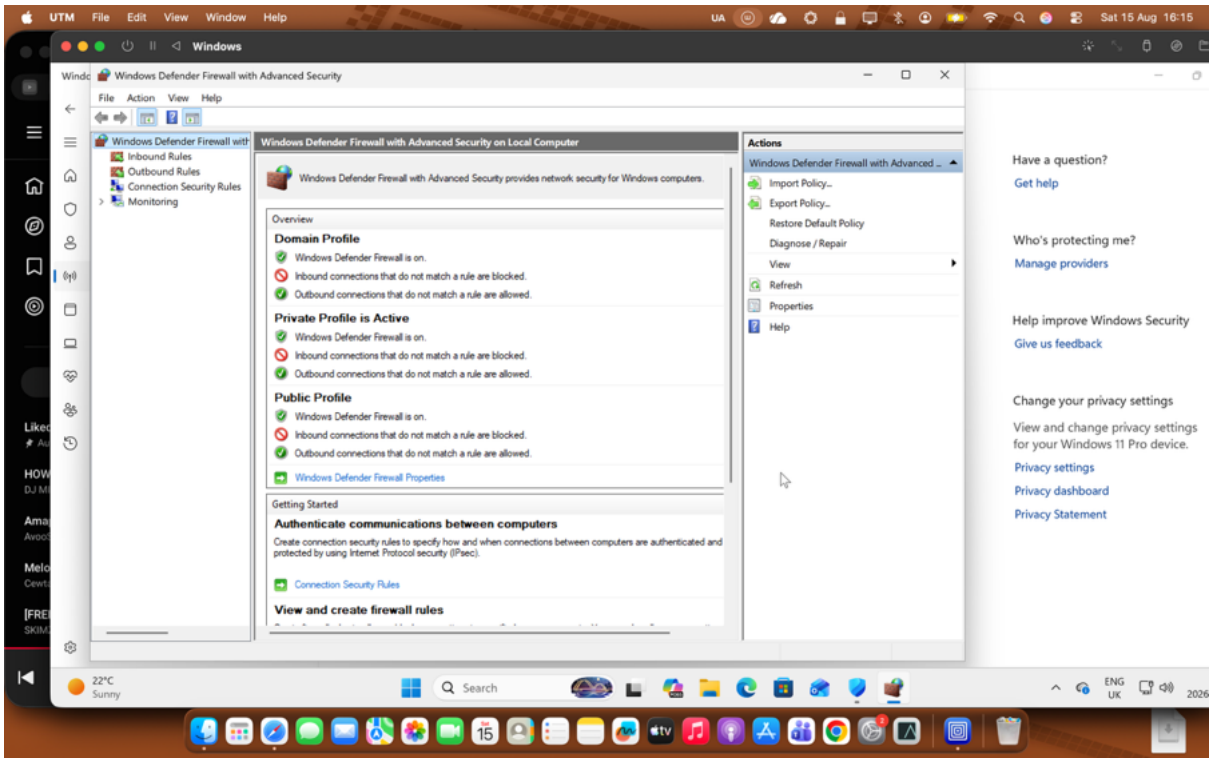
Screenshot of the filters taken and saved capture into a file named Capture Analysis.

File upload of Capture Analysis.

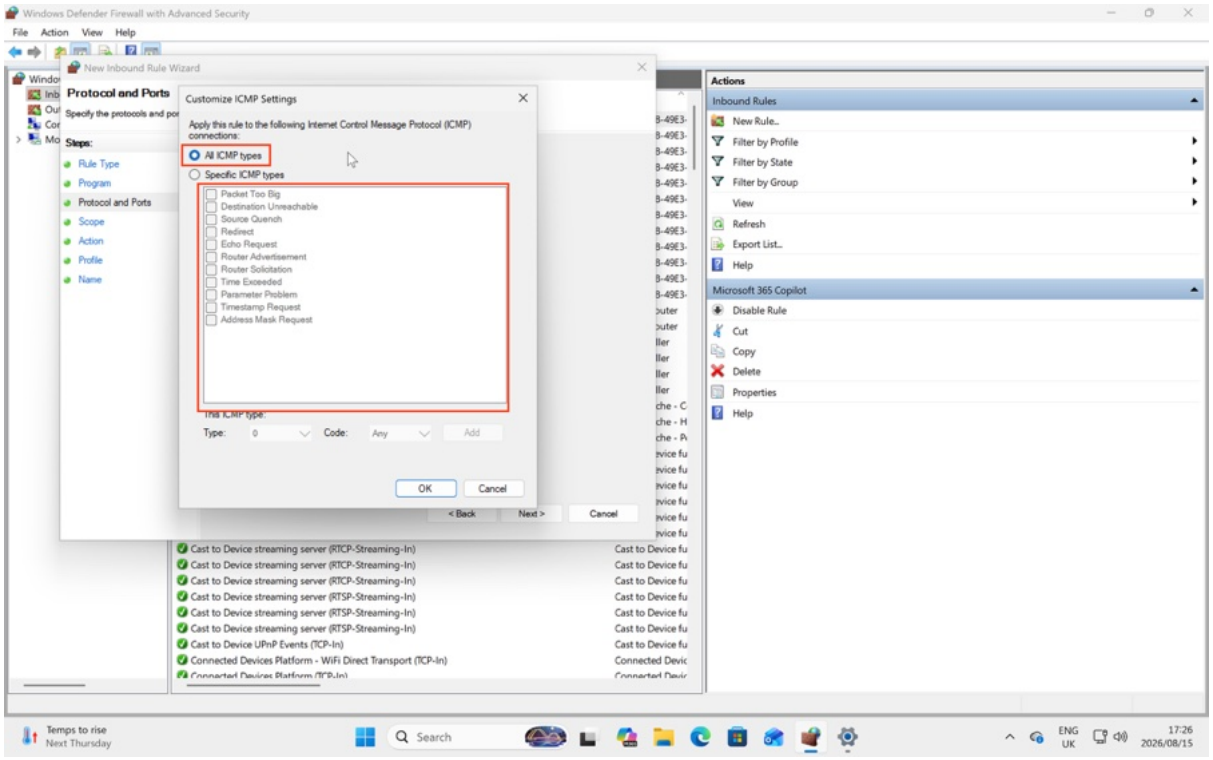
Question 4



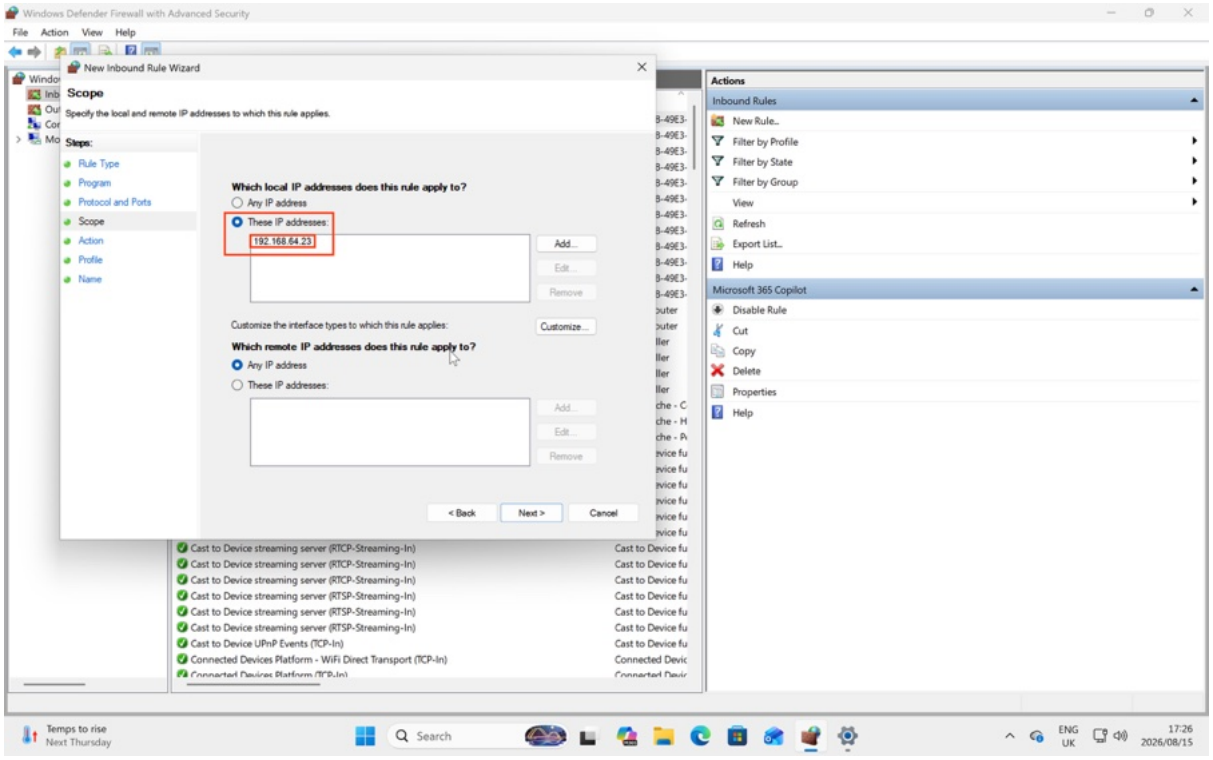
Verifying Firewall status with enable firewall protection for Domain, Private and Public.



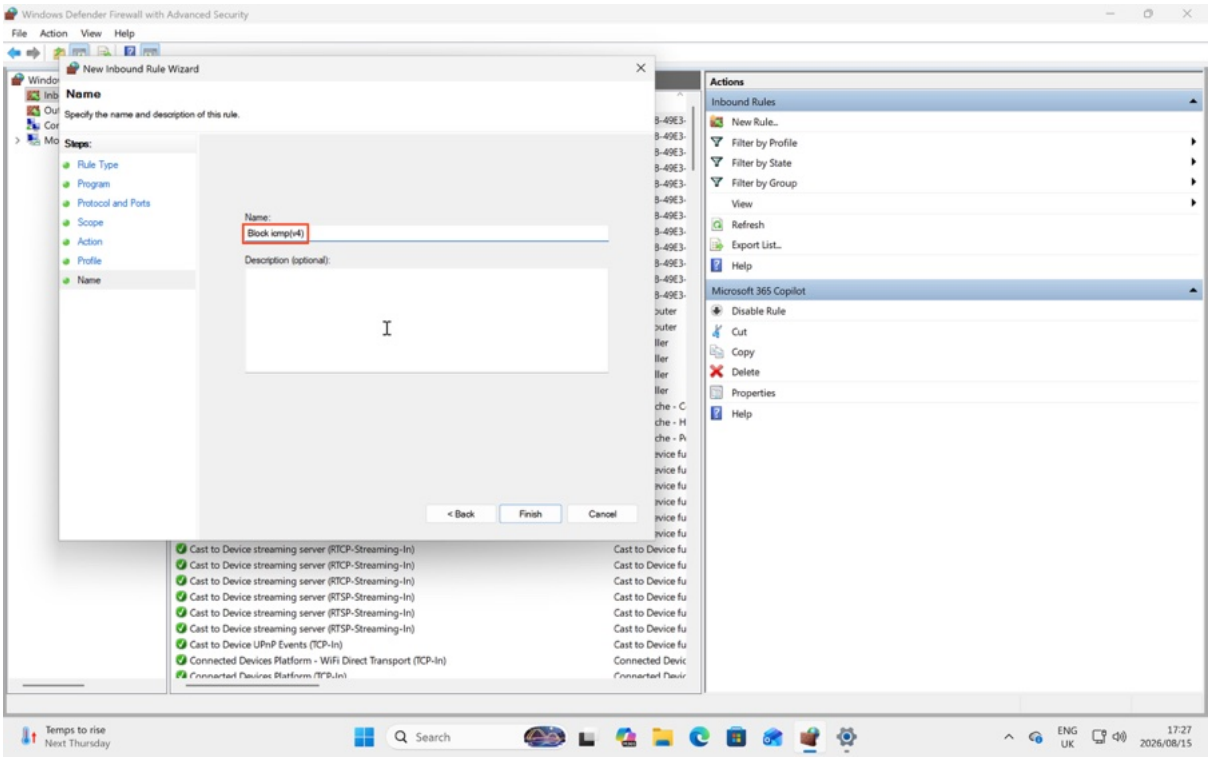
Opening Windows Defender firewall with Advanced Security and verifying that firewall is enabled for Domain, Private and Public Profile.



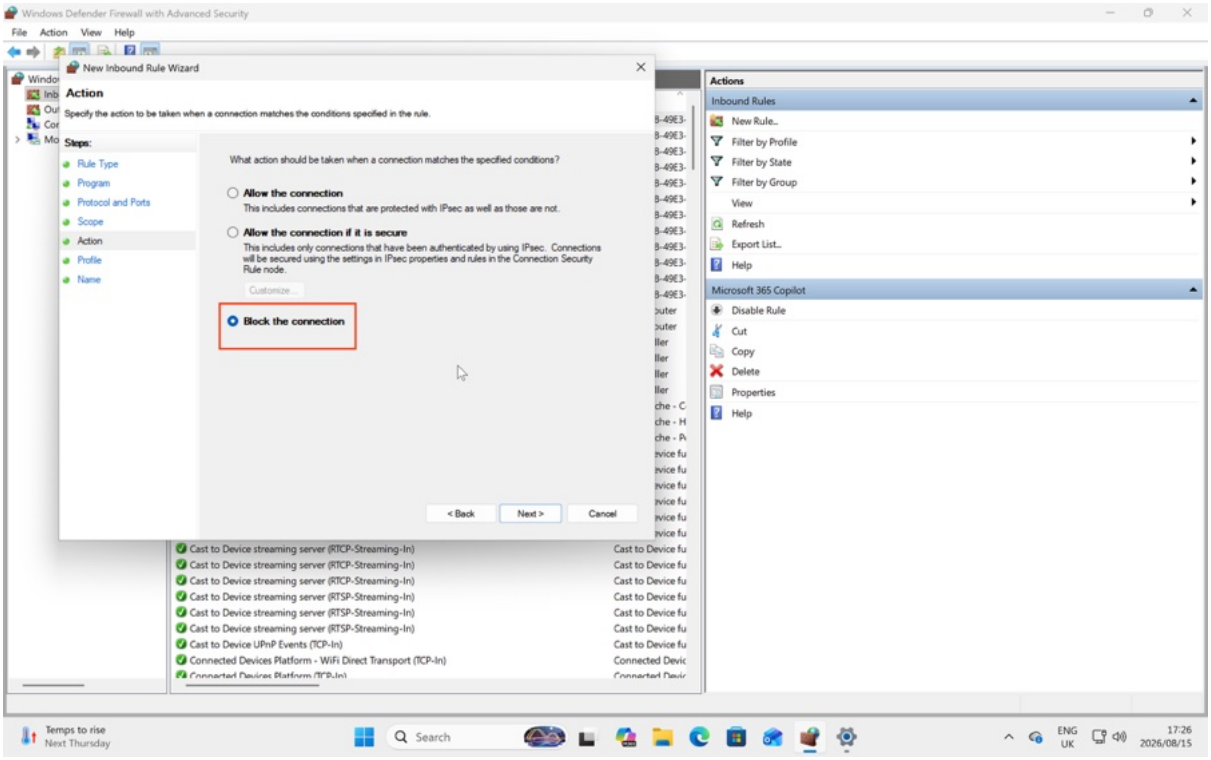
Creating an inbound rule for ICMP(Ping) Traffic.



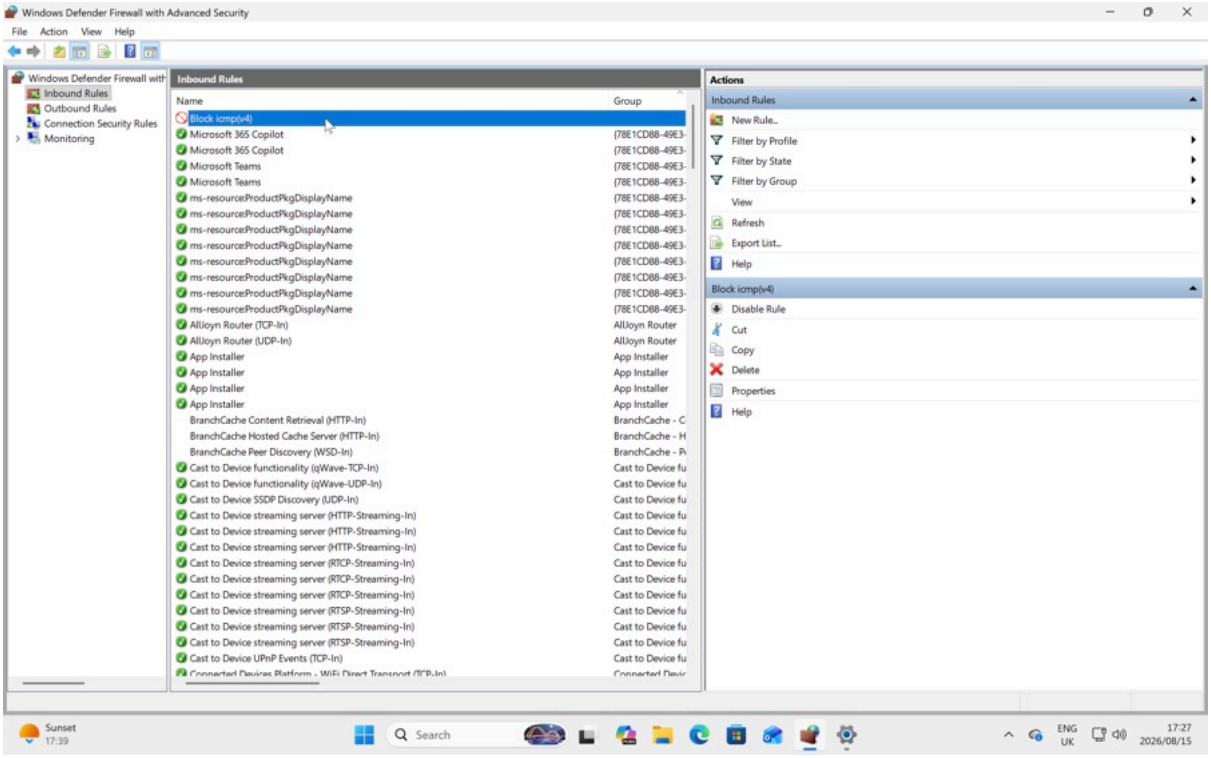
Setting Scope rule for ICMP



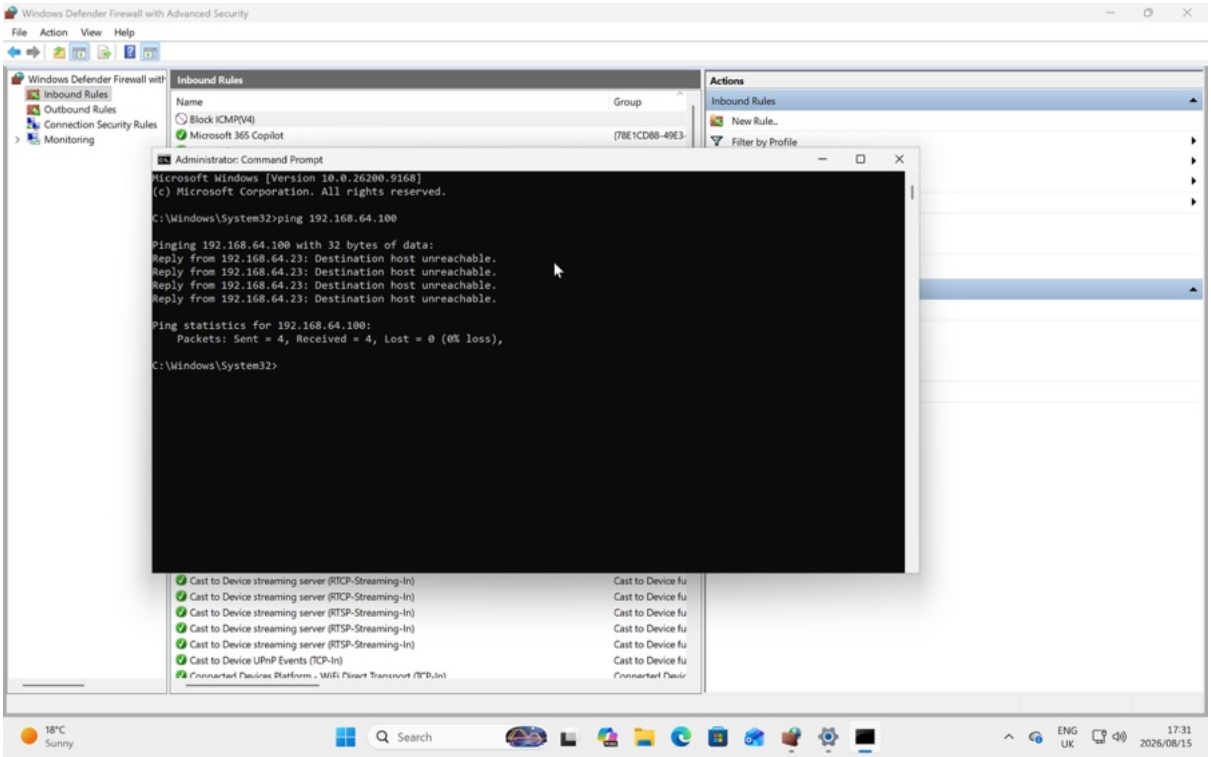
Name the ICMP Traffic rule to Block icmp(v4)



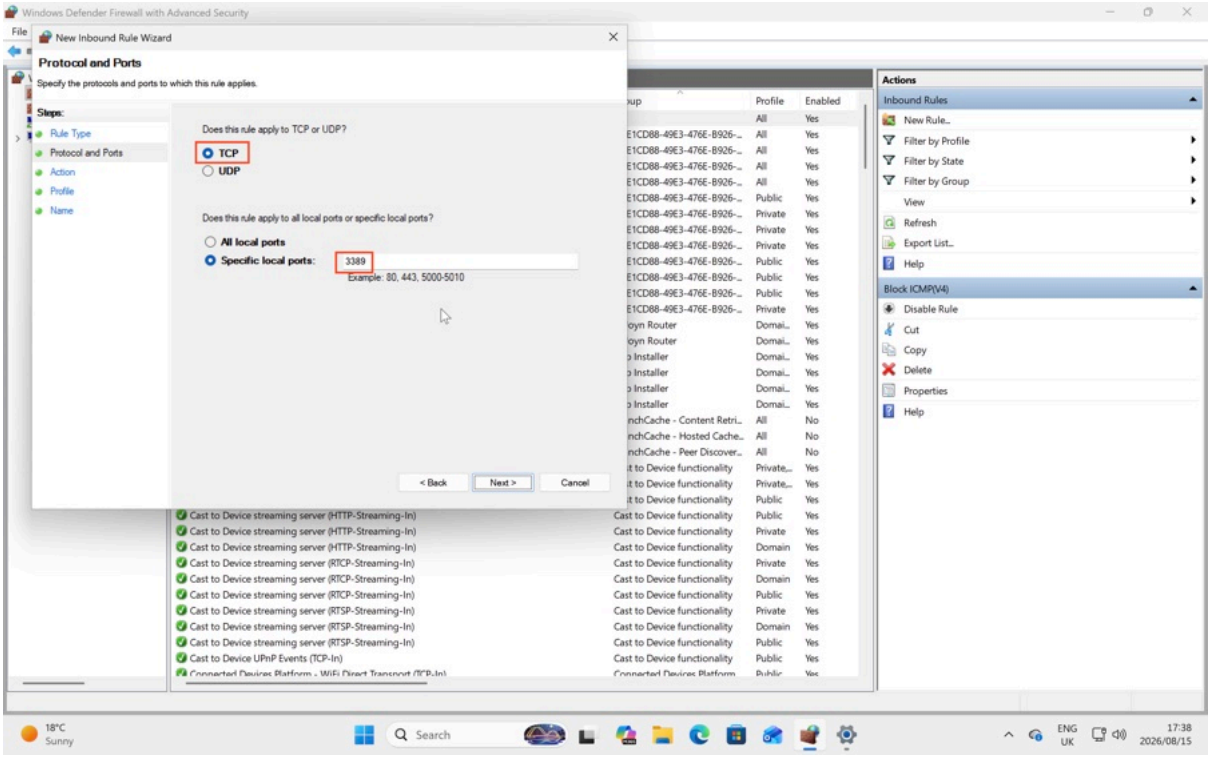
Blocking ICMP Traffic connection.



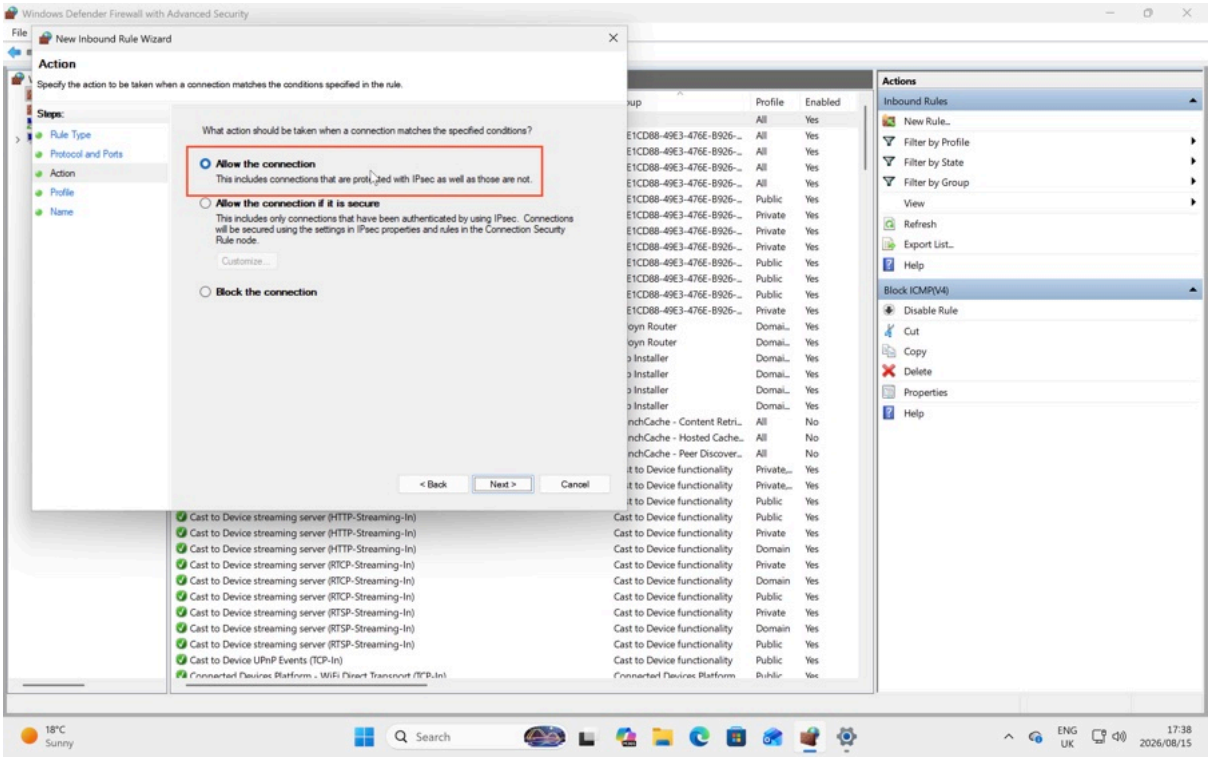
ICMP Block rule now appears in the Inbound Rules.



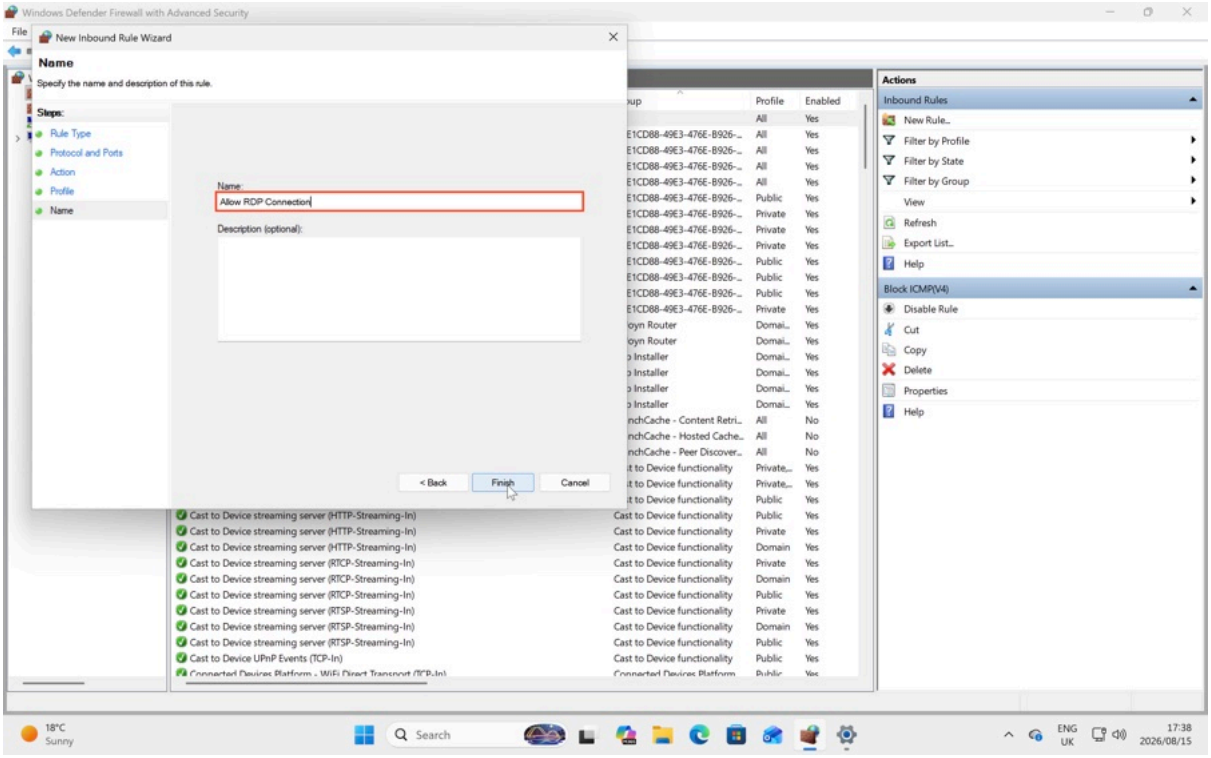
Test that the Computer no longer responses to ping. ICMP destination host unreachable.



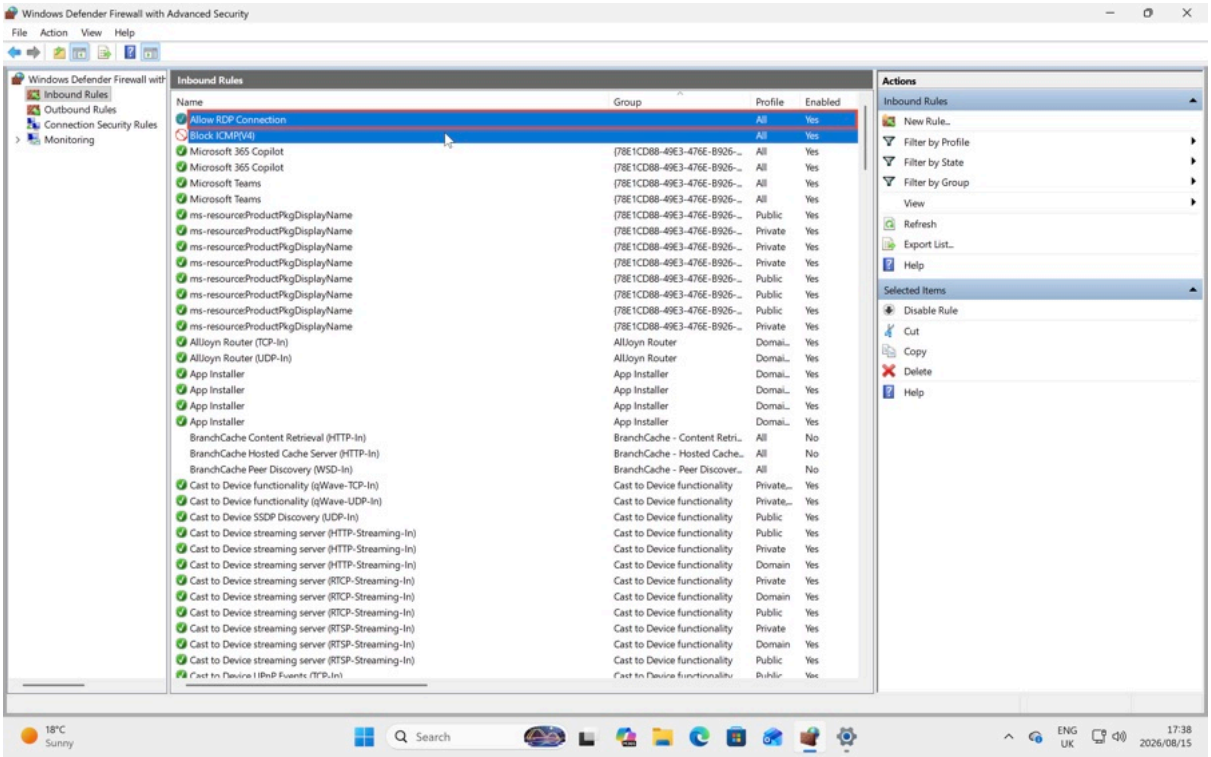
Creating an Inbound Rule for RDP using TCP and RDP port 3389.



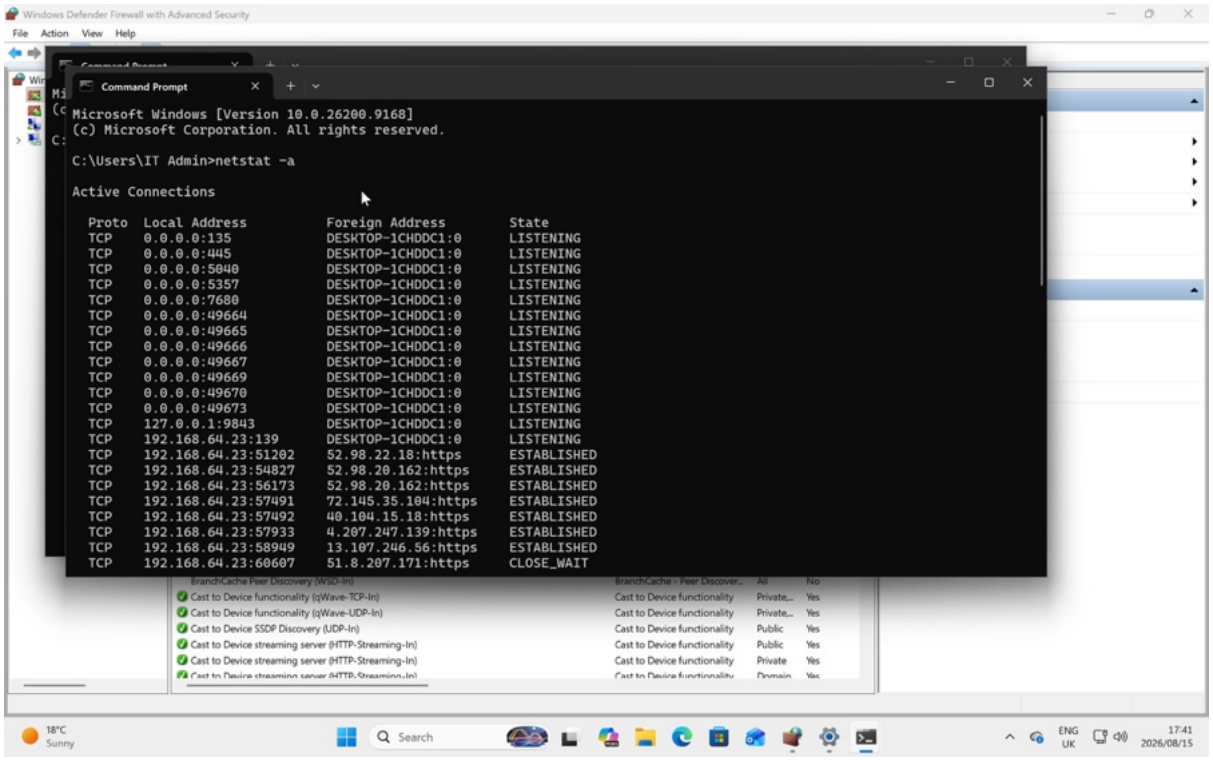
Allowing connection for RDP.



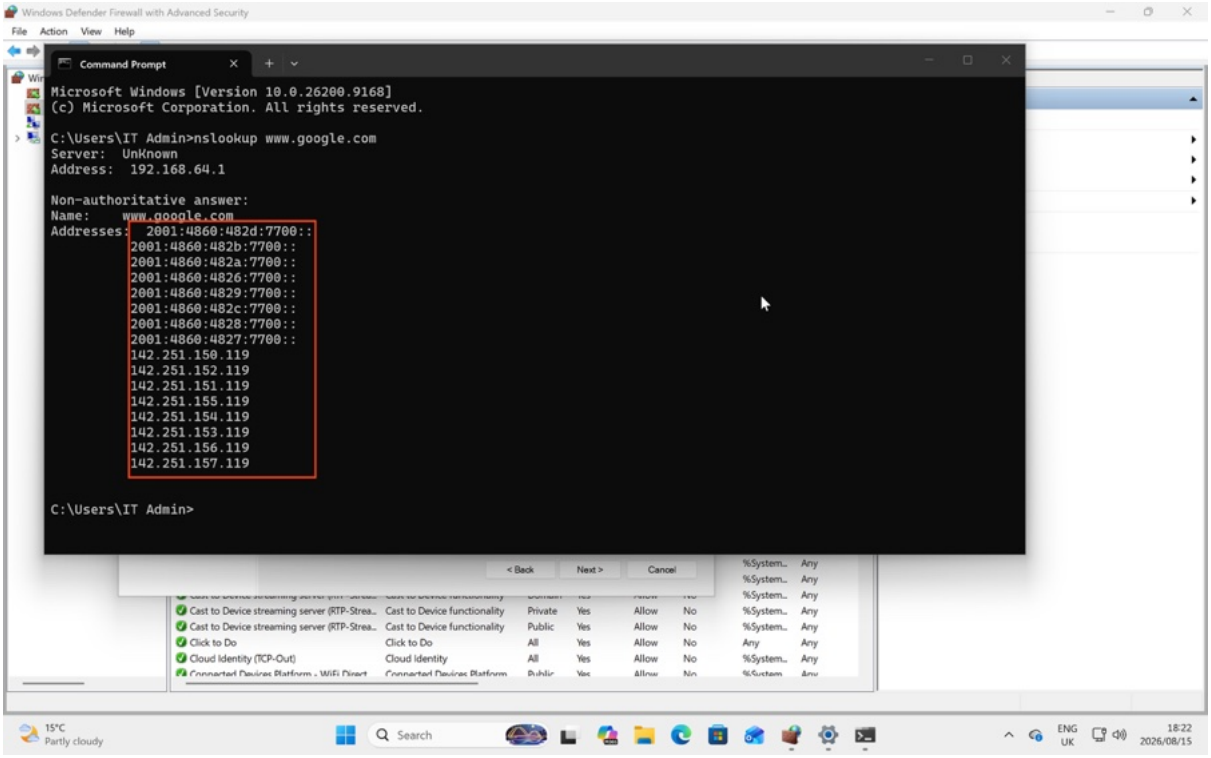
Naming RDP rule : Allow RDP Connection



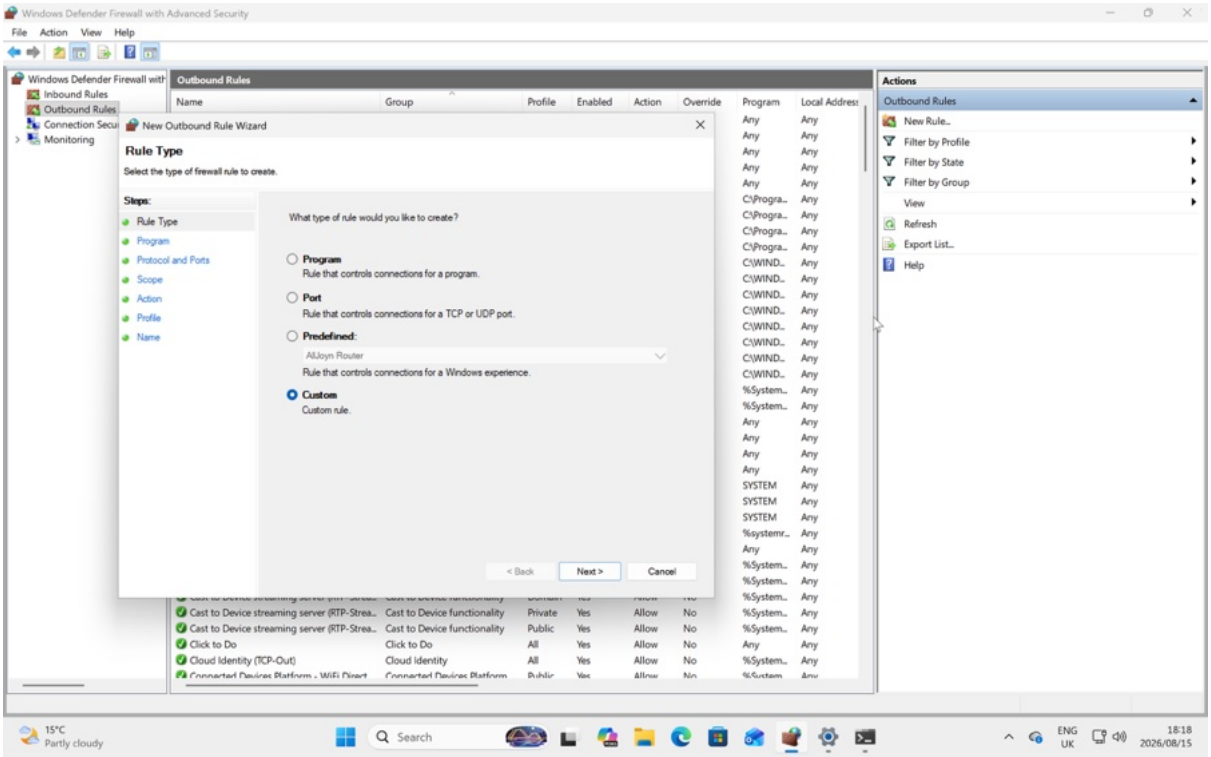
RDP allow rule now appears in Inbound Rules. And confirming that allow RDP rule remains enabled.



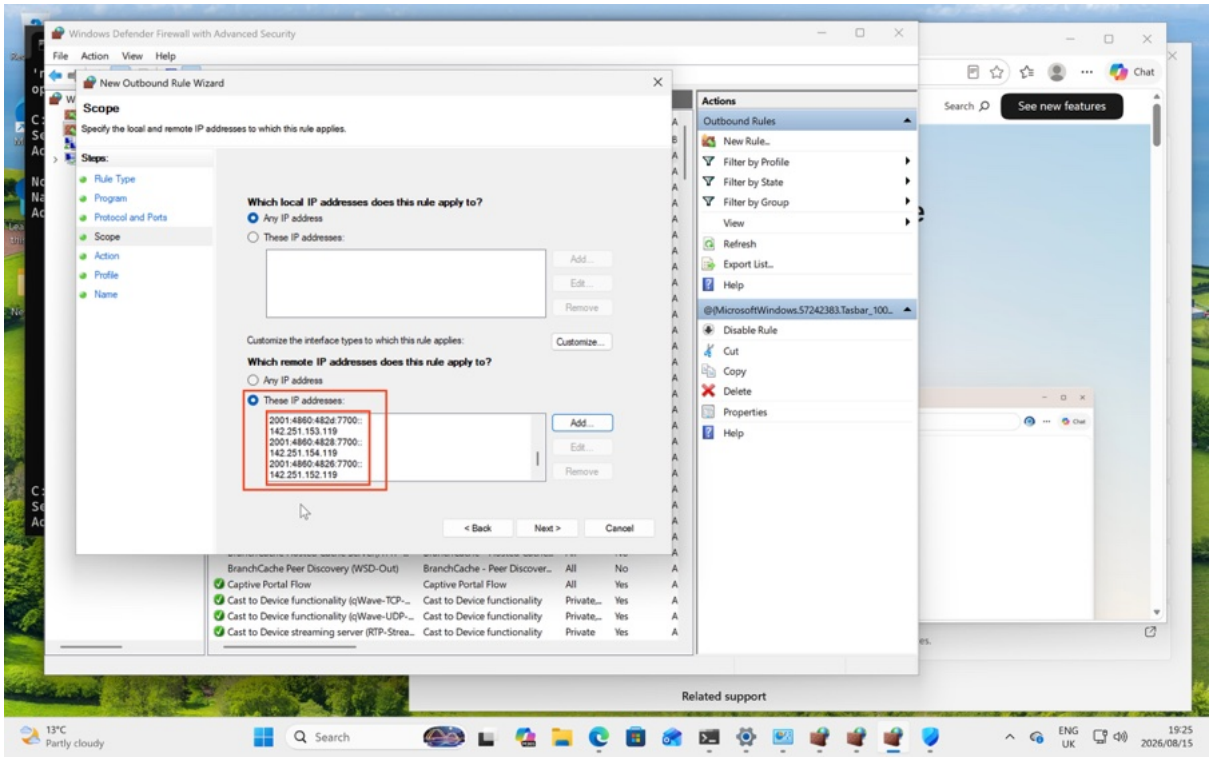
Computer listening but RDP is not yet configured.



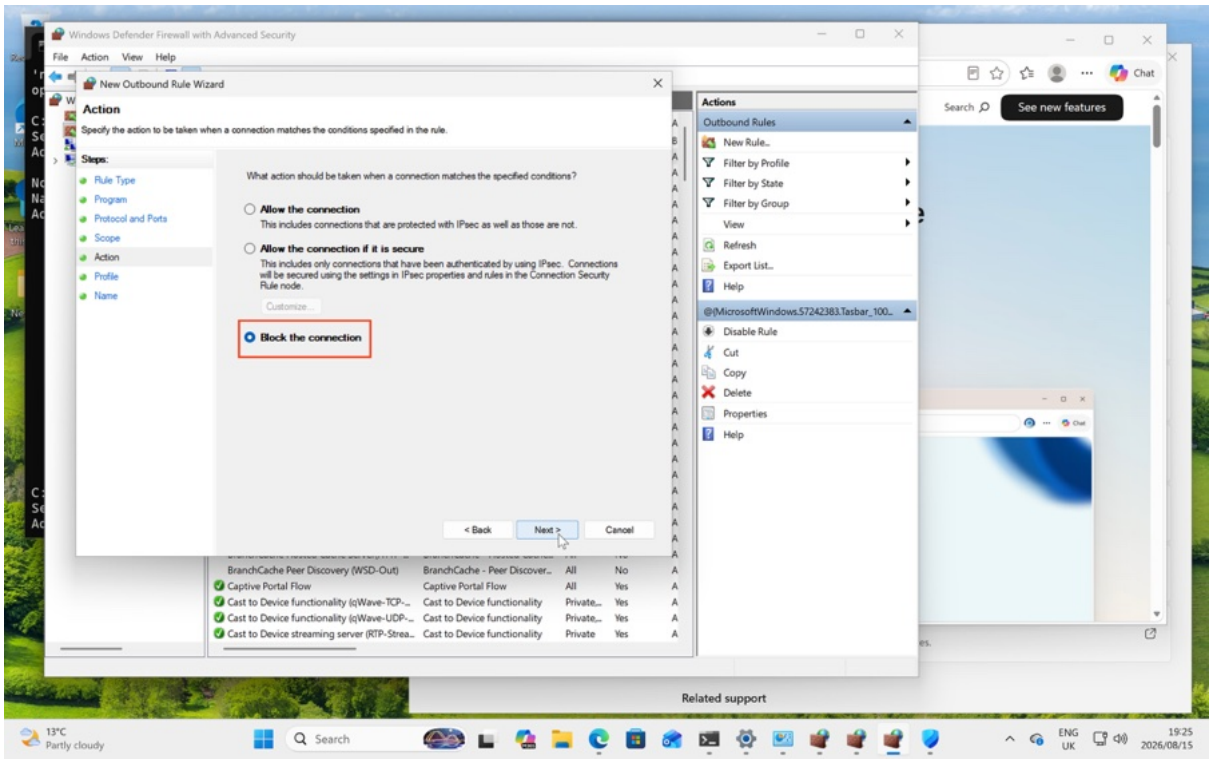
Checking website www.google.com targeted for blockage IP Addresses.



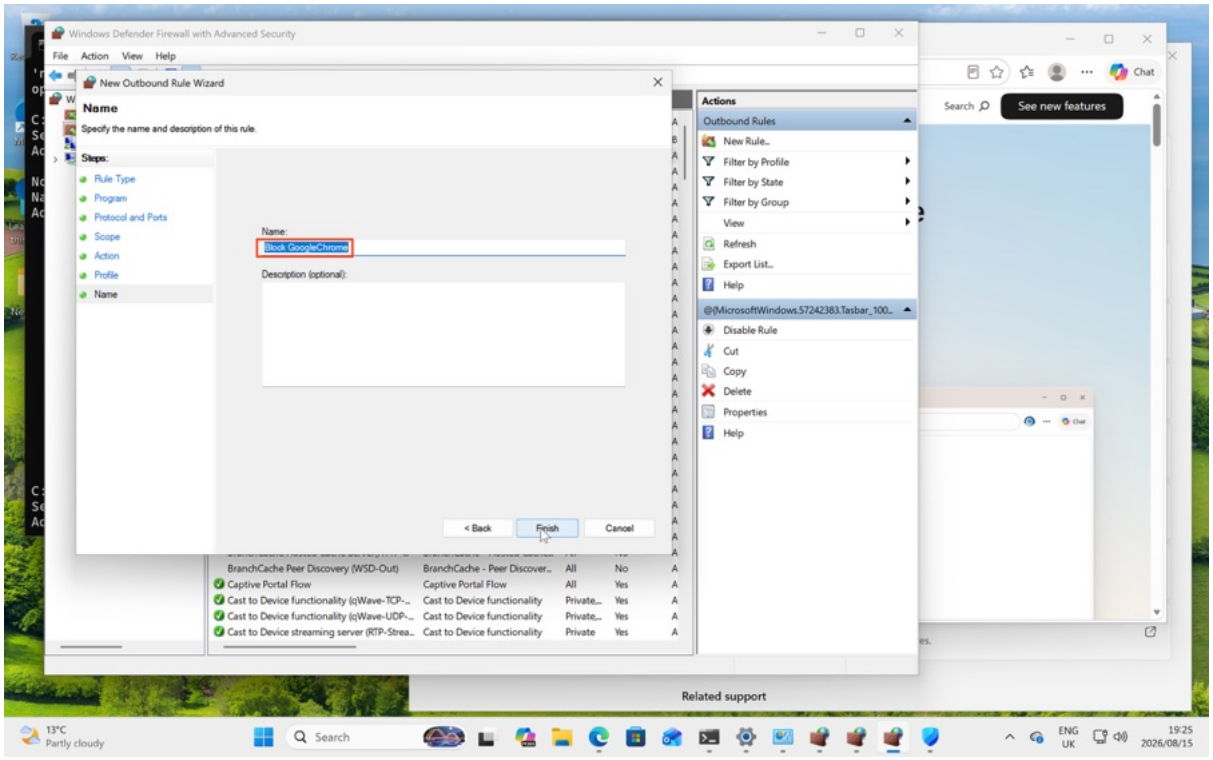
Creating custom Outbound rule for www.google.com



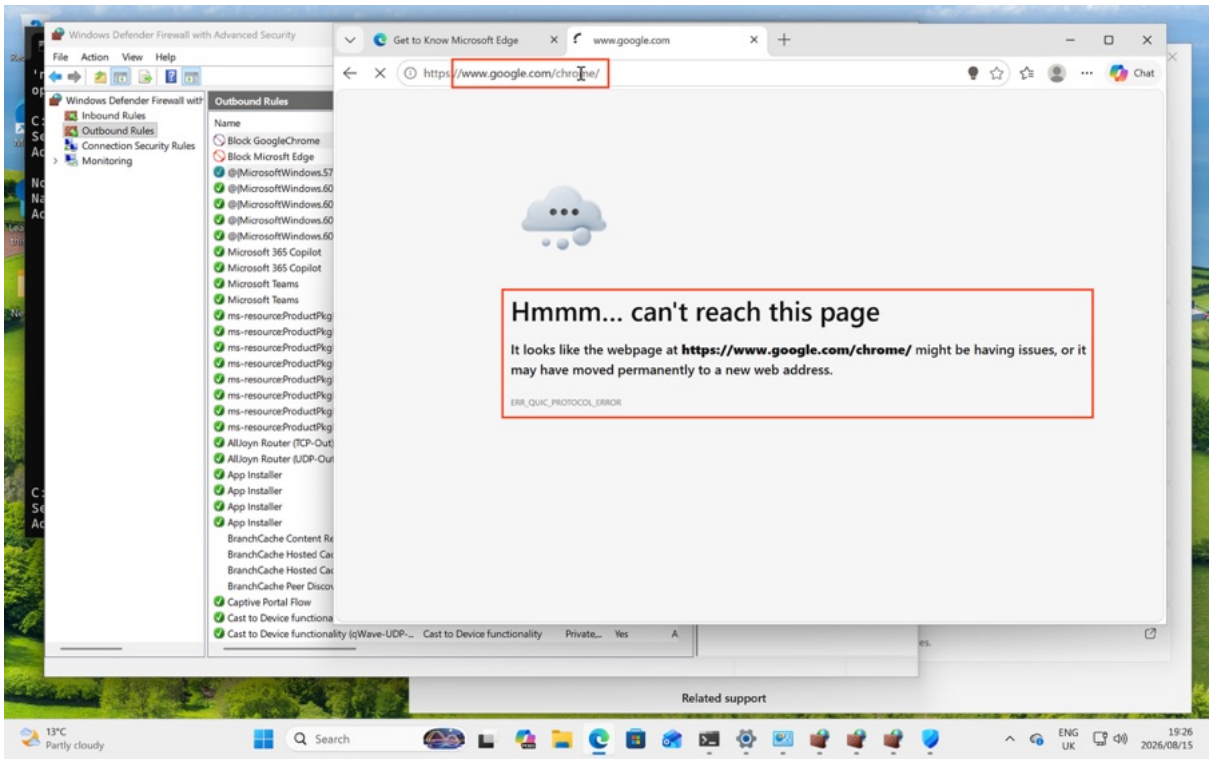
Inserting scope of all IP addresses belonging to www.google.com



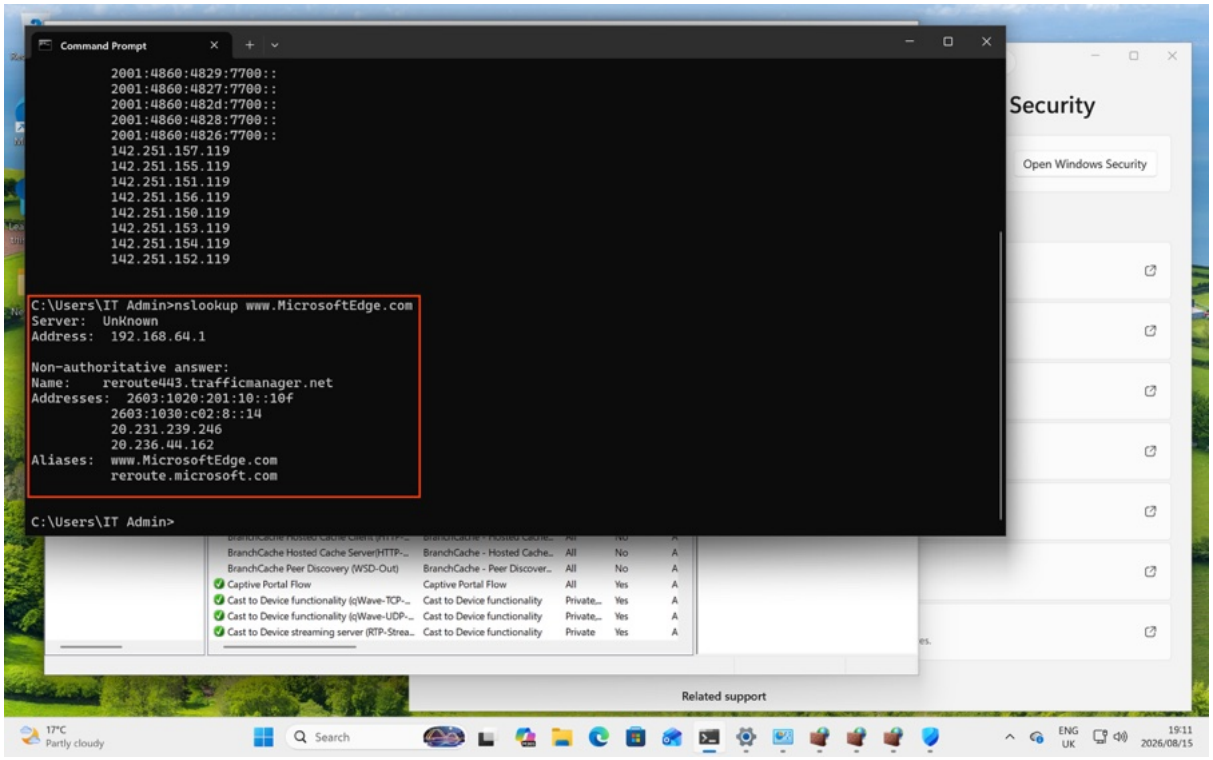
Blocking connection to www.google.com



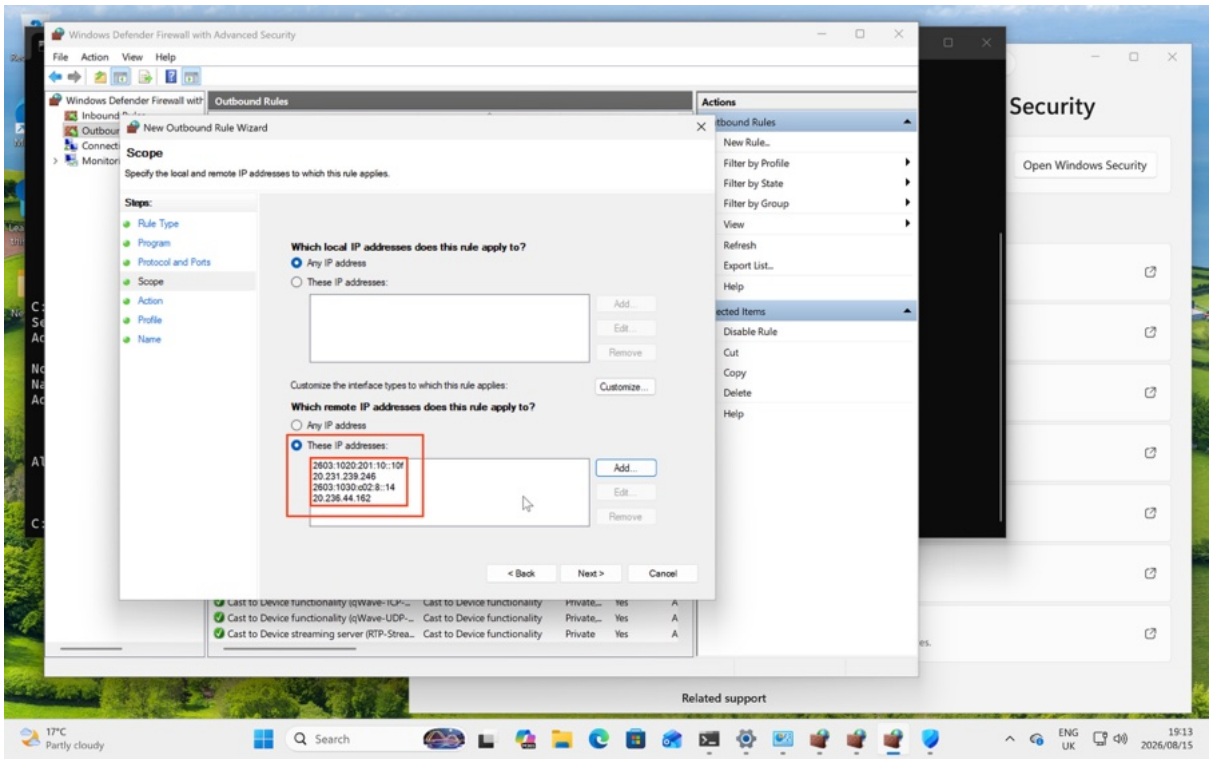
Naming the rule Block GoogleChrome



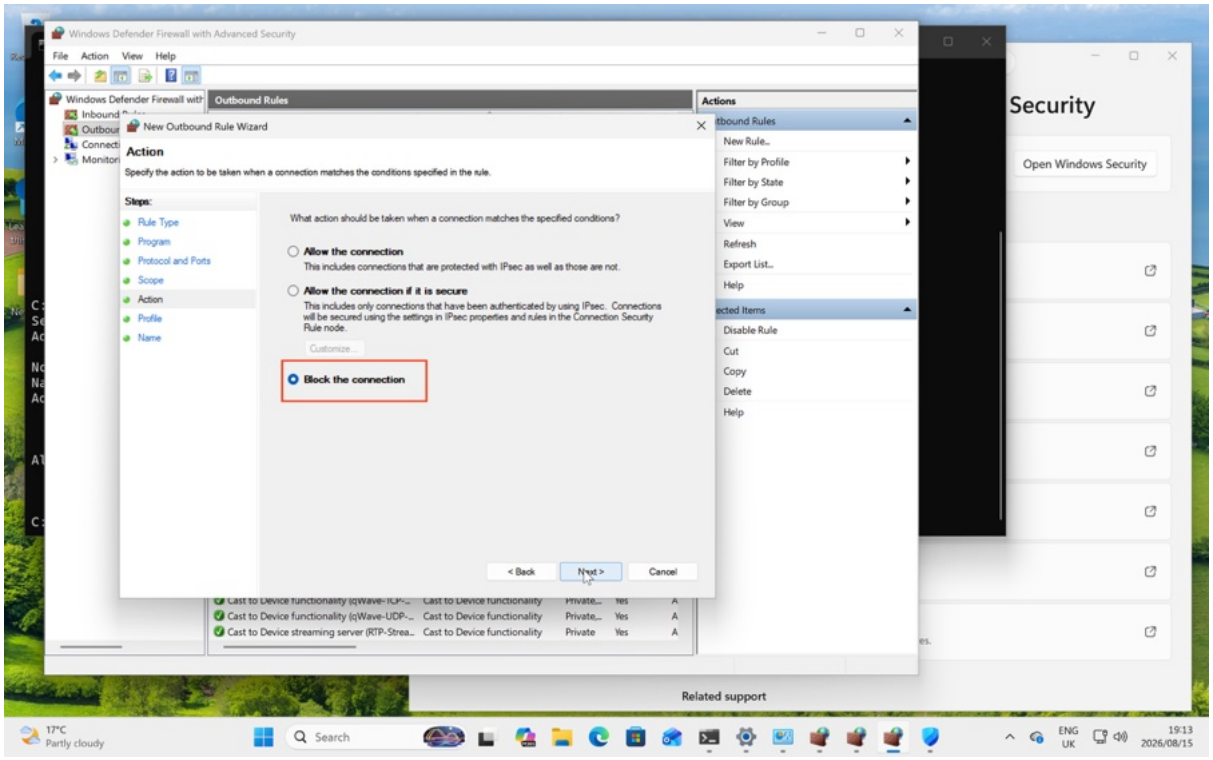
www.google.com has successfully been blocked and added to the Outbound Rules list.



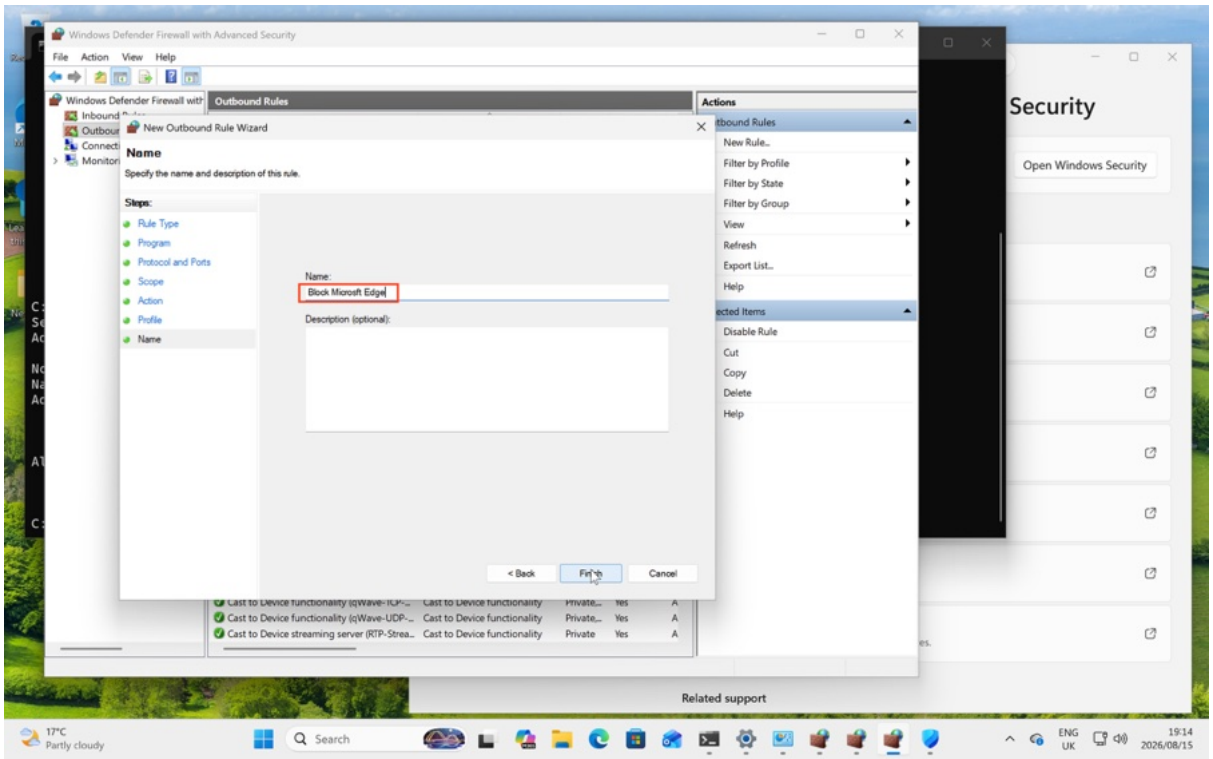
Checking website www.MicrosoftEdge.com targeted for blockage IP Addresses



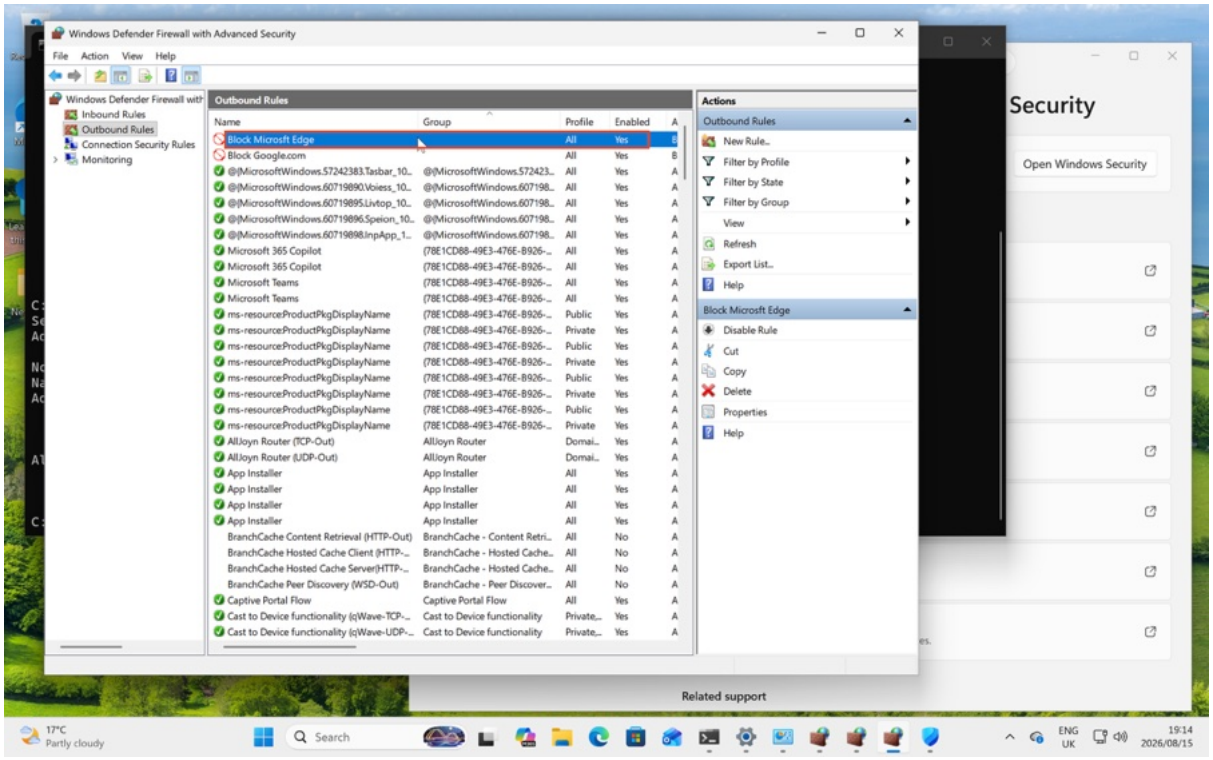
Inserting scope of all IP addresses belonging to www.MicrosoftEdge.com



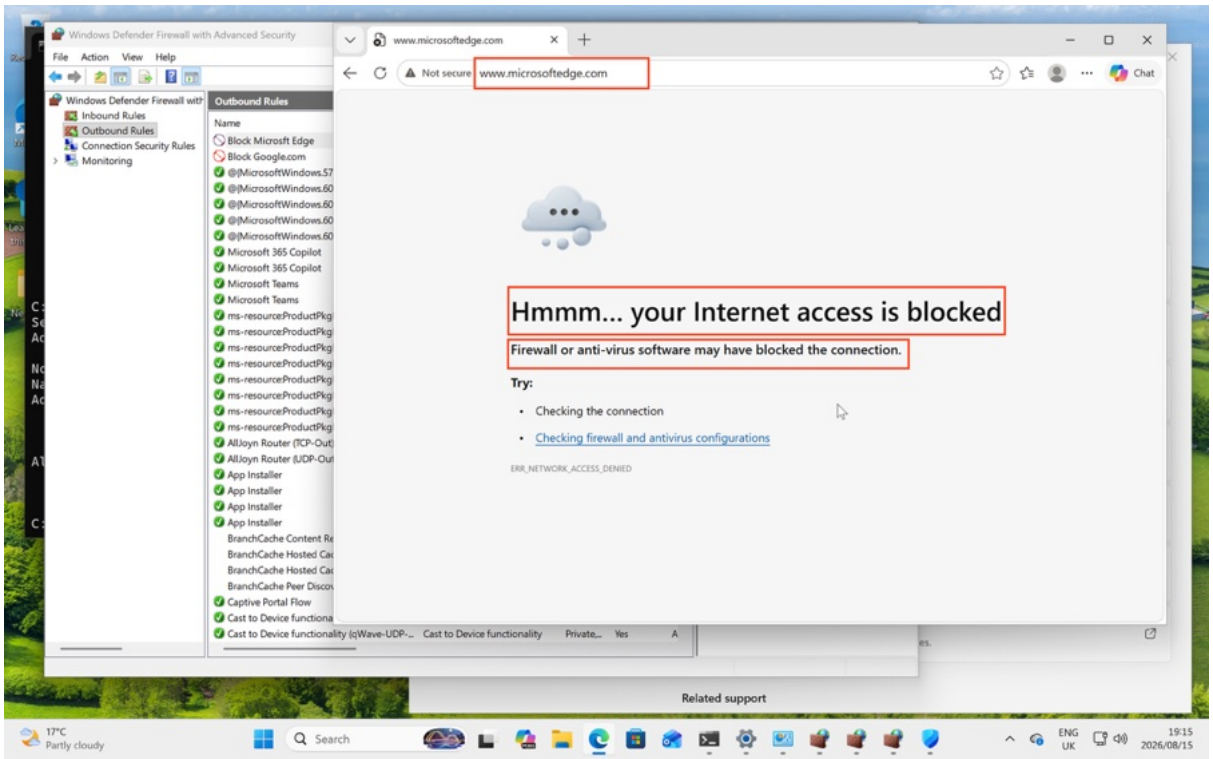
Blocking connection to www.MicrosoftEdge.com



Naming the rule Block Microsoft Edge



Block Microsoft Edge rule has been added to the Outbound Rules list



www.MicrosoftEdge.com has successfully been blocked.

The best practice recommendation for managing windows firewall in an organisation like Eduvos is using centralized control like Group Policy Management and Policy Configuration Profile.

With Group Policy Management allowing lockdown local settings and disabling rule merging.

And Policy Configuration Profile hardening and enabling least privilege keeping profiles active, Enforcing default deny/allow, tailoring profile, limiting remote address scope, documenting every exception and restricting administrative access.